

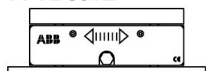
Web Tension Systems with Tension Electronics PFEA111/112

User Manual

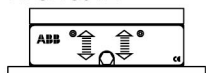
3BSE029380R0001 Rev B

Web Tension measurement

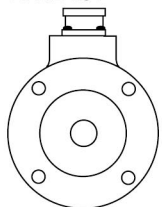
PFTL 301E



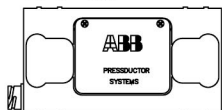
PFCL 301E



PFRL 101



PFTL 101



Use of **DANGER**, **WARNING**, **CAUTION**, and **NOTE**

This publication includes, **DANGER**, **WARNING**, **CAUTION** and **NOTE** information where appropriate to point out safety related or other important information.

DANGER	Hazards which could result in severe personal injury or death
WARNING	Hazards which could result in personal injury
CAUTION	Hazards which could result in equipment or property damage
NOTE	Alerts user to pertinent facts and conditions

Although **DANGER** and **WARNING** hazards are related to personal injury, and **CAUTION** hazards are associated with equipment or property damage, it should be understood that operation of damaged equipment could, under certain operational conditions, result in degraded process performance leading to personal injury or death. Therefore, comply fully with all **DANGER**, **WARNING**, and **CAUTION** notices.

TRADEMARKS

Pressductor[®] is a registered trademark of ABB Automation Technologies AB.

NOTICE

The information in this document is subject to change without notice and should not be construed as a commitment by ABB Automation Technologies AB. ABB Automation Technologies AB assumes no responsibility for any errors that may appear in this document.

In no event shall ABB Automation Technologies AB be liable for direct, indirect, special, incidental or consequential damages of any nature or kind arising from the use of this document, nor shall ABB Automation Technologies AB be liable for incidental or consequential damages arising from use of any software or hardware described in this document.

This document and parts thereof must not be reproduced or copied without ABB Automation Technologies AB's written permission, and the contents thereof must not be imparted to a third party nor be used for any unauthorized purpose.

The software described in this document is furnished under a license and may be used, copied, or disclosed only in accordance with the terms of such license.

CE-MARKING

The Tension Electronics PFEA111/112 fulfills the requirements as stated in the EMC Directive 89/336/EEC and the Low Voltage Directive 73/23/EEC, provided that the installation is carried out in accordance with the installation instructions given in [Chapter 2 Installation](#), included in this User Manual.



The Tension Electronics PFEA111-20, PFEA111-65 and PFEA112-20 fulfill the requirements of safety approval for USA and Canada according to the standard UL61010C-1 for Process Control Equipment and CSA C22.2 No. 1010-1 Safety Requirements for Measurement, Control and Laboratory Use, Part 1: General Requirements Certificate No. 170304-E240621 and No. 240504-E240621, provided that the installation is carried out in accordance with the installation instructions given in [Chapter 2 Installation](#), included in this User Manual.

TABLE OF CONTENTS

Chapter 1 - Introduction

1.1	About this Manual	1-1
1.2	How to Use this Manual	1-1
1.2.1	Getting Started	1-1
1.2.2	Saving Actual Data and Settings at Commissioning	1-1
1.3	About this System	1-2
1.4	Safety Instructions	1-3
1.4.1	Personnel Safety	1-3
1.4.2	Equipment Safety	1-3
1.5	The Measuring Technique Based on Pressductor, Technology	1-4

Chapter 2 - Installation

2.1	About this Chapter	2-1
2.2	Safety Instructions	2-1
2.3	Mounting the Load Cells	2-1
2.4	Installing the Electronic Unit	2-2
2.4.1	Selecting and Routing the Cabling	2-2
2.4.1.1	Recommended Cabling	2-2
2.4.1.2	Interference	2-4
2.4.1.3	Synchronization	2-4
2.4.2	Mounting the Tension Electronics PFEA111/112	2-5
2.4.2.1	IP 65-version (NEMA 4)	2-5
2.4.2.2	IP 20-version (Unsealed)	2-7
2.4.3	Earthing	2-8
2.5	Connecting the Load Cells	2-9
2.6	Connecting Optional Units	2-10
2.6.1	Insulation Amplifier PXUB 201 (only for the IP 20-version)	2-10
2.6.2	Power Supply Unit	2-10

Chapter 3 - Commissioning

3.1	About this Chapter	3-1
3.2	Safety Instructions	3-1
3.3	Necessary Equipment and Documentation	3-1
3.4	Using the Panel Buttons	3-2
3.4.1	Navigating and Confirming	3-2
3.4.2	Changing Numerical Values and Parameters	3-2
3.5	Menu Overview	3-3
3.6	Step-by-step Commissioning Guide	3-4
3.7	Performing Basic Settings	3-5

CONTENTS (continued)

3.8	Performing a Fast Setup	3-5
3.8.1	Performing Fast Setup using Hanging Weights	3-6
3.8.2	Performing Fast Setup using Wrap Gain	3-8
3.9	Checking Load Cell Signal Polarity	3-9
3.10	Checking Load Cell Function	3-10
3.11	Performing a Complete Setup	3-11
3.11.1	Overview	3-11
3.12	Complete Setup Sequence	3-12
3.12.1	Presentation Menu	3-12
3.12.1.1	Set Language	3-12
3.12.1.2	Set Unit	3-12
3.12.2	Set Object	3-13
3.12.3	Nominal Load	3-14
3.12.4	Zero Set	3-15
3.12.5	Set Wrap Gain	3-16
3.12.6	Voltage Output	3-18
3.12.7	Current Output	3-20
3.12.8	Miscellaneous Menu	3-22
3.12.8.1	Profibus	3-22
3.12.8.2	Set Factory Default	3-22
3.12.9	Service Menu	3-23
3.12.9.1	Maximum Load / Present Offset	3-24
3.12.9.2	Reset A/B	3-24
3.13	Profibus DP Communication with PFEA112	3-25
3.13.1	General Data about Profibus DP	3-25
3.13.2	Master-slave Communication	3-25
3.13.3	Commands through Profibus	3-26
3.13.4	Handling of Measurement Data through Profibus	3-26
3.13.4.1	Scaling of Profibus Measurement Values	3-26
3.13.4.2	Filtering of Profibus Measuring Values	3-26
3.13.4.3	Input Buffer, Communication Block from PFEA112 to PLC	3-27
3.13.4.4	Output Buffer, Communication Block from PLC to PFEA112	3-27
3.14	Commissioning of Optional Units	3-28
3.14.1	Insulation Amplifier PXUB 201	3-28

CONTENTS (continued)

Chapter 4 - Operation

4.1	About this Chapter	4-1
4.2	Safety Instructions	4-1
4.3	Operating Devices	4-1
4.4	Start-up and shut-down	4-2
4.4.1	Start-up	4-2
4.4.2	Shut-down	4-2
4.5	Normal Run	4-2
4.6	Operator Menus	4-3
4.6.1	Web Tension	4-3
4.6.1.1	Standard Roll (two load cells)	4-3
4.6.1.2	Single Side Measurement (one load cell)	4-3
4.6.2	Error and Warning Messages	4-4

Chapter 5 - Maintenance

5.1	About this Chapter	5-1
5.2	Preventive Maintenance	5-1

Chapter 6 - Fault-tracing

6.1	About this Chapter	6-1
6.2	Safety Instructions	6-1
6.3	Interchangeability	6-2
6.4	Necessary Equipment and Documentation	6-2
6.5	Fault-tracing Procedure	6-3
6.6	Error and Warning Messages in PFEA111/112	6-4
6.6.1	Error Messages	6-4
6.6.2	Warning Messages	6-4
6.7	Fault Symptoms and Measures	6-5
6.8	Warnings and Errors Detected by the Tension Electronics	6-7
6.8.1	Errors	6-7
6.8.1.1	Flash Memory Error	6-7
6.8.1.2	EEPROM Memory Error	6-7
6.8.1.3	Supply Error	6-7
6.8.1.4	Load Cell Excitation Error	6-8
6.8.2	Warnings	6-8
6.8.2.1	Profibus Communication Problem	6-8
6.8.2.2	Synchronization Problem	6-8
6.8.3	Changing to Single Side Measurement If One Load Cell Is Faulty	6-9
6.9	Changing Load Cells	6-10

CONTENTS (continued)

Appendix A - Technical Data for Tension Electronics PFEA111/112

A.1	About this Appendix	A-1
A.2	Definitions used in the Web Tension Systems	A-2
A.2.1	Coordinate System	A-3
A.3	Technical Data	A-4
A.4	Factory Default Settings	A-7
A.5	Optional Units	A-8
A.5.1	Insulation Amplifier PXUB 201	A-8
A.5.2	Power Supply Unit	A-9
A.5.3	Junction Box PFXC 141	A-9
A.6	Profibus DP - GSD File for PFEA112	A-10

Appendix B - PFCL 301E - Designing the Load Cell Installation

B.1	About This Appendix	B-1
B.2	Basic Application Considerations	B-1
B.3	Step-by-Step Guide for Designing the Load Cell Installation	B-2
B.4	Installation Requirements	B-3
B.5	Mounting Alternatives, Calculating Force and Calculating Wrap Gain	B-4
B.5.1	Horizontal Mounting	B-4
B.5.2	Inclined Mounting	B-5
B.6	Force Calculation for Measurement with a Single Load Cell	B-6
B.6.1	The Most Common and Simple Solution	B-6
B.6.2	Force Calculation when the Web is not Centered on the Roll	B-7
B.7	Mounting the Load Cells	B-8
B.7.1	Routing the Load Cell Cable	B-8
B.7.2	Connecting the Load Cell Extension Cable	B-8
B.8	Technical Data	B-9
B.9	Cable Diagram 3BSE028140D0065, page 2/5, Rev. B	B-11
B.10	Mounting Instruction, Cable Connector, 3BSE019064, Rev. A	B-12
B.11	Dimension Drawing, 3BSE015955D0094, Rev. C	B-13
B.12	Assembly Drawing, 3BSE015955D0096, Rev. B	B-14

CONTENTS (continued)

Appendix C - PFTL 301E - Designing the Load Cell Installation

C.1	About This Appendix.....	C-1
C.2	Basic Application Considerations.....	C-1
C.3	Step-by-Step Guide for Designing the Load Cell Installation	C-2
C.4	Installation Requirements	C-3
C.5	Mounting Alternatives, Calculating Force and Calculating Wrap Gain.....	C-4
C.5.1	Horizontal Mounting.....	C-4
C.5.2	Inclined Mounting.....	C-5
C.6	Force Calculation for Measurement with a Single Load Cell	C-6
C.6.1	The Most Common and Simple Solution	C-6
C.6.2	Force Calculation when the Web is not Centered on the Roll	C-7
C.7	Mounting the Load Cells	C-8
C.7.1	Routing the Load Cell Cable	C-8
C.7.2	Connecting the Load Cell Extension Cable.....	C-8
C.8	Technical Data	C-9
C.9	Cable Diagram 3BSE028140D0065, Page 1/5, Rev. B	C-11
C.10	Mounting Instruction, Cable Connector, 3BSE019064, Rev. A	C-12
C.11	Dimension Drawing, 3BSE019040D0094, Rev. B.....	C-13
C.12	Assembly Drawing, 3BSE019040D0096, Rev. B	C-14

Appendix D - PFRL 101 - Designing the Load Cell Installation

D.1	About This Appendix.....	D-1
D.2	Basic Application Considerations.....	D-1
D.3	Step-by-Step Guide for Designing the Load Cell Installation	D-2
D.4	Installation Requirements	D-3
D.5	Load Cell Orientation Depending on Measurement Direction.....	D-4
D.6	Mounting Alternatives, Calculating Force and Calculating Wrap Gain.....	D-5
D.6.1	Horizontal Mounting.....	D-5
D.6.2	Inclined Mounting.....	D-6
D.7	Force Calculation for Measurement with a Single Load Cell	D-7
D.7.1	The Most Common and Simple Solution	D-7
D.7.2	Force Calculation when the Web is not Centered on the Roll	D-8
D.8	Mounting the Load Cells	D-9
D.8.1	Mounting with Brackets.....	D-11
D.8.2	Mounting Screws for the Load Cells	D-12
D.8.3	Routing the Load Cell Cable	D-12
D.9	Technical Data	D-13
D.10	Cable Diagram, 3BSE028140D0065, Page 3/5, Rev. B	D-15
D.11	Cable Diagram, 3BSE028140D0065, Page 4/5, Rev. B	D-16
D.12	Dimension Drawing 3BSE004042D0003, page 1/2, Rev. N.....	D-17
D.13	Dimension Drawing, 3BSE004042D0003, page 2/2, Rev. N.....	D-18
D.14	Dimension Drawing, 3BSE004042D0065, Rev. -	D-19
D.15	Dimension Drawing, 3BSE010457, Rev. A	D-20

CONTENTS (continued)

Appendix E - PFTL 101 - Designing the Load Cell Installation

E.1	About This Appendix	E-1
E.2	Basic Application Considerations	E-1
E.3	Step-by-Step Guide for Designing the Load Cell Installation	E-2
E.4	Installation Requirements.....	E-3
E.5	Mounting Alternatives, Calculating Force and Calculating Wrap Gain	E-4
E.5.1	Horizontal Mounting.....	E-4
E.5.2	Inclined Mounting.....	E-5
E.6	Force Calculation for Measurement with a Single Load Cell.....	E-6
E.6.1	The Most Common and Simple Solution.....	E-6
E.6.2	Force Calculation when the Web is not Centered on the Roll	E-7
E.7	Mounting the Load Cells.....	E-8
E.7.1	Routing the Load Cell Cable.....	E-9
E.8	Technical Data.....	E-10
E.9	Cable Diagram 3BSE028140D0065, Page 3/5, Rev. B	E-12
E.10	Cable Diagram 3BSE028140D0065, Page 4/5, Rev. B	E-13
E.11	Cable Diagram 3BSE028140D0065, Page 5/5, Rev. B	E-14
E.12	Dimension Drawing, 3BSE004171, Rev. B	E-15
E.13	Dimension Drawing, 3BSE004995, Rev. C	E-16
E.14	Dimension Drawing, 3BSE023301D0064, Rev. B	E-17
E.15	Dimension Drawing, 3BSE004196, Rev. C	E-18
E.16	Dimension Drawing, 3BSE004999, Rev. C	E-19
E.17	Dimension Drawing, 3BSE023223D0064, Rev. B	E-20
E.18	Dimension Drawing, 3BSE012173, Rev. C	E-21
E.19	Dimension Drawing, 3BSE012172, Rev. C	E-22
E.20	Dimension Drawing, 3BSE012171, Rev. C	E-23
E.21	Dimension Drawing, 3BSE012170, Rev. C	E-24

Appendix F - Actual Data and Settings at Commissioning

F.1	Document the Commissioning in this Form	F-1
-----	---	-----

Chapter 1 Introduction

1.1 About this Manual

This User Manual describes your new Measure^{IT} Web Tension System. When you have read the manual, you will have the necessary knowledge for mechanical and electrical installation, commissioning, operation, preventive maintenance and basic fault tracing of your measurement system.

To get the best reliability and precision out of your measurement system, study this User Manual first.

1.2 How to Use this Manual

This user manual comprises two main parts.

1. Information about the Tension Electronics:

- System and safety information (Chapter 1)
- Installation, commissioning, maintenance, operation and fault tracing (Chapters 2-6)
- Technical data (Appendix A)

2. Information about Designing the Load Cell Installation:

- Vertical-force sensing load cell PFCL 301E (Appendix B)
- Horizontal-force sensing load cell PFTL 301E (Appendix C)
- Radial-force Tensiometer PFRL 101 (Appendix D)
- Horizontal-force sensing load cell PFTL 101 (Appendix E)

Each appendix contains detailed information about one of the above load cell types when used in web tension systems with Tension Electronics PFEA111/112.

1.2.1 Getting Started

You can use the Fast Setup sequence to set up your system for basic measurement.

The Fast Setup guides you through a minimum number of steps to set up the tension electronics. Perform the actions in the following sections:

- [Section 3.6 Step-by-step Commissioning Guide](#)
- [Section 3.7 Performing Basic Settings](#)
- [Section 3.8 Performing a Fast Setup](#)

For extended functionality, use “Performing a Complete Setup”.

See [Section 3.11 Performing a Complete Setup](#).

1.2.2 Saving Actual Data and Settings at Commissioning

When commissioning is completed you can use the form in Appendix F, where actual commissioning data and settings can be filled in and saved for future use.

1.3 About this System

Your system for tension measurement consists of:

- Tension Electronics PFEA111 or PFEA112
 - **PFEA111** is a cost effective compact and user friendly tension electronics providing an accurate and reliable fast analog SUM signal from two load cells for control and/or monitoring. The display can show the SUM, individual A & B and difference signal. The small size and DIN-rail mount make this unit very easy to integrate into many types of electrical cabinets
 - **PFEA112** provides the same functionality and user friendliness as the PFEA111 with the addition of fieldbus communication via Profibus-DP.

Covering a wide range of applications the Tension Electronics comes in three versions (The PFEA 113 is described in a separate manual), with different levels of performance and functionality. All three versions have multi-language digital display and configuration keys. The configuration keys being used for setting different parameters and to check the status of the tension system. The 2 x 16 character display can present sum, difference or individual load cell signals. All three versions are available in both DIN-rail version (IP 20-version, unsealed) and enclosed IP 65-version (NEMA 4) for mounting in more severe environments.

- Load cells of type PFCL 301E, PFTL 301E, PFRL 101 and PFTL 101.

The equipment is intended for use in a wide range of manufacturing processes where a web of any kind of material, e.g. paper, plastic or textile, is transported in a machine. The only requirement is that the web is wrapped over a roll. The force on the roll is proportional to the web tension. The resulting force is transferred through the bearing housings into the load cells. The load cells create a signal that is proportional to the force acting in the measuring direction of the load cells. This signal is processed and amplified in the tension electronics and can be used as an input signal for process control, presentation on a display or for registration.

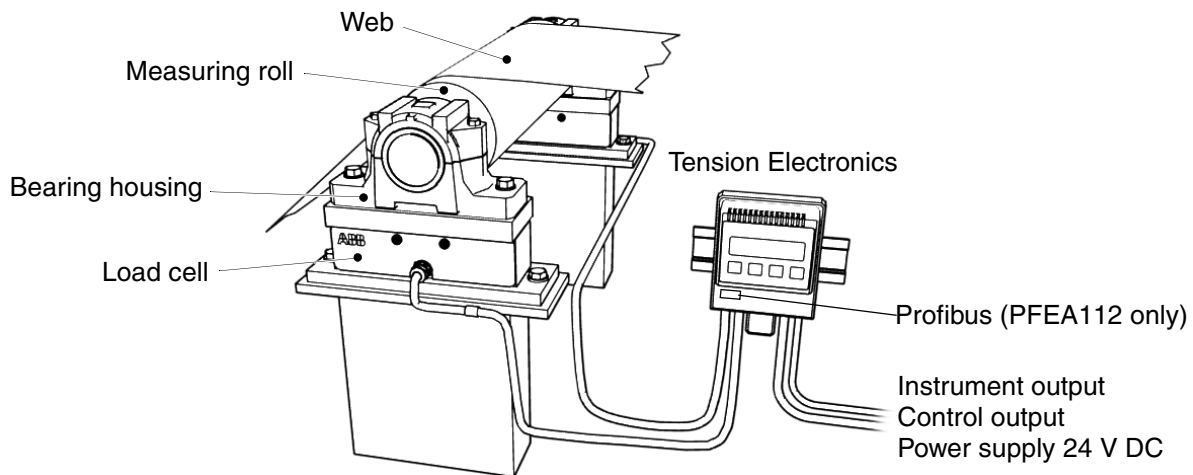


Figure 1-1. Typical Tension Measurement System with Tension Electronics PFEA111/112 (IP 20-version)

1.4 Safety Instructions

Read and follow the safety instructions given in this section before starting any work. However, local statutory regulations, if stricter, are to take precedence.

The tension measurement system contains no moving parts. However, the load cells are mounted close to a rotating roll over which a web is running.

1.4.1 Personnel Safety



WARNING

Never work with the load cells, or nearby, when the production line is in service. Before starting any work, switch off and lock the operating switch of the drive section for the measuring roll.



DANGER

Switch off and lock the mains operating switch to the tension electronics before performing any work on the tension electronics. When the work is completed, check that there are no loose wires, and that all units are properly secured.

NOTE

All personnel working with the installation must know the location of the main power supply switch to the measurement system and how it is operated.

1.4.2 Equipment Safety

CAUTION

Always switch off the mains supply voltage to the measurement system before replacing a unit.



CAUTION

Handle the electronic unit carefully to reduce the risk of damage from Electric Static Discharge (ESD). Note the warning label on the circuit boards.

1.5 The Measuring Technique Based on Pressductor® Technology

The operating principle of a force transducer has a great effect on how well it will perform. It also affects how stiff and vibration-free the entire load cell will be, as well its robustness and tolerance to overload. All of these factors impact on the design, operation, and maintenance of the web processing machinery.

ABB's Pressductor® transducer technology produces a signal as a result of changes in an electromagnetic field when the load cell is subjected to mechanical force. It is an operating principle that has its origin in a metallurgical phenomenon according to which mechanical forces alter the ability of some steels to convey a magnetic field. Unlike other types of load cell technologies, physical movement such as compression, bending or stretching is not required for signal generation.

A Pressductor® transducer (the sensor inside the load cell) is a simple and elegant design. Essentially, two perpendicular windings of copper wire around a steel core combine to provide a measurement signal.

An electromagnetic field is created by continuously feeding an alternating current to one of the windings. The field is positioned in such a way that, since the windings are at right angles to each other, there is no magnetic coupling between them when the load cell is unstressed.

However, when the transducer is subjected to a force, as shown in the figure, the magnetic field pattern changes. A portion of the field couples with the second winding and induces an AC voltage that reflects the tension exerted by the web on the measurement roll. This voltage - a comparatively strong transducer signal - is converted by the load cell system's tension electronics into a system output.

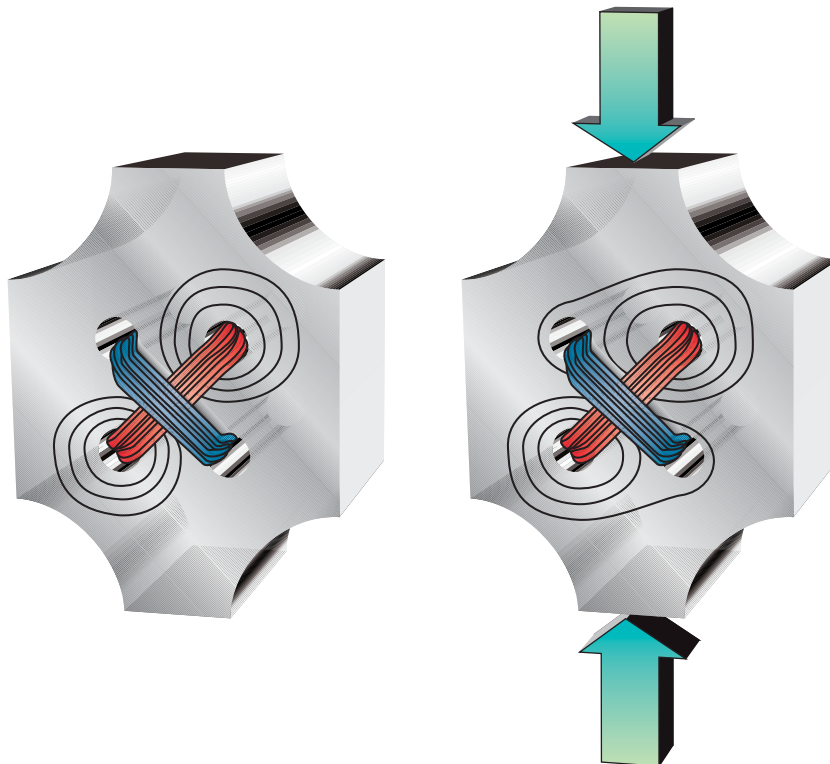


Figure 1-2. The Sensor Based on Pressductor® Technology

Chapter 2 Installation

2.1 About this Chapter

The way you install your system has more influence on its functionality, accuracy and reliability than you might think. The more accurate the installation, the better the measurement system. By following the instructions in this chapter, you will fulfill the most important requirements for proper mechanical and electrical installation.

The equipment is a precision instrument which, although intended for severe operating conditions, must be handled with care.

2.2 Safety Instructions

Read and follow the safety instructions given in [Chapter 1 Introduction](#), before starting any installation work. However, local statutory regulations, if stricter, are to take precedence.

2.3 Mounting the Load Cells

Installation requirements and mounting instructions are given in:

- [Appendix B PFCL 301E - Designing the Load Cell Installation](#)
- [Appendix C PFTL 301E - Designing the Load Cell Installation](#)
- [Appendix D PFRL 101 - Designing the Load Cell Installation](#)
- [Appendix E PFTL 101 - Designing the Load Cell Installation](#)

2.4 Installing the Electronic Unit

2.4.1 Selecting and Routing the Cabling

2.4.1.1 Recommended Cabling

The cabling between the load cells and the electronic unit and electrical connections must be carefully carried out in accordance with connection diagram 3BSE028140D0065 (See the appendix for your type of load cell) or according to order-specific documentation.

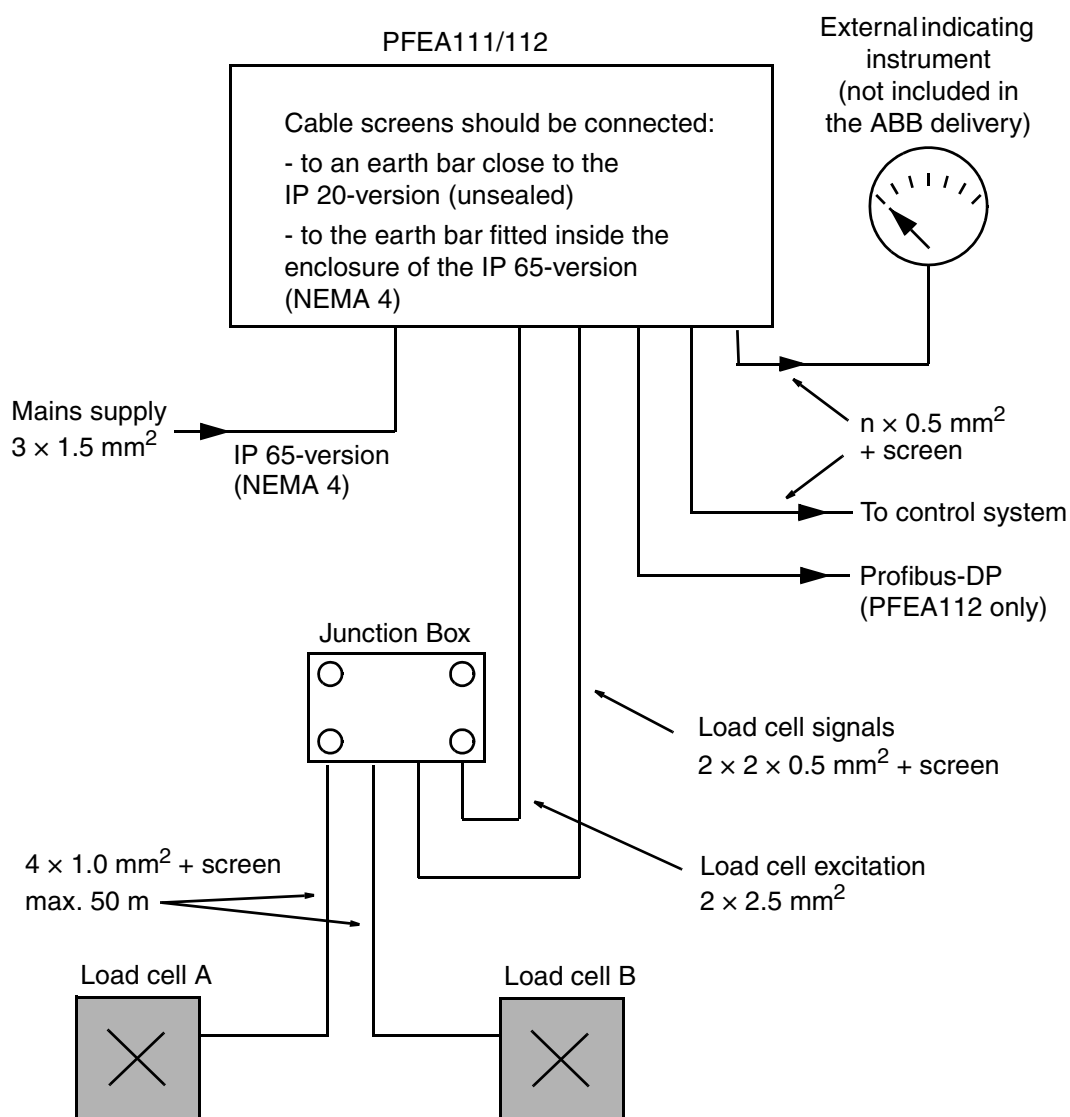


Figure 2-1. Recommended Cabling

- The maximum permitted cable resistance in the excitation circuit is shown in [Table 2-1](#). Before commissioning, check cable resistance in the load cell excitation circuit.

Table 2-1. Maximum Permitted Cable Resistance

Load cell	Max. permitted cable resistance
PFCL 301E	5 Ω
PFTL 301E	5 Ω
PFRL 101	5 Ω
PFTL 101	5 Ω

- Solid conductors should not be connected to terminals. Pins should not be crimped to stranded cores.
- The cable from the load cell **must be a robust four core cable**, see [Figure 2-2](#). Diagonal pairs must be used for the signal circuits and excitation circuits.

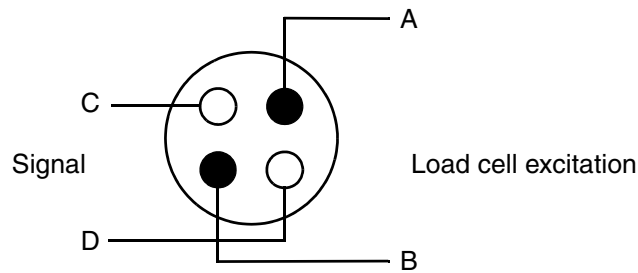


Figure 2-2. Core Arrangement in Load Cell Cable

- Between the junction box and the tension electronics, signal and excitation must be routed in separate cables. For example: a $2 \times 2.5 \text{ mm}^2$ cable for the excitation and a shielded $2 \times 2 \times 0.5 \text{ mm}^2$ cable with twisted pair cores for the load cell signals.
- Cable for synchronization of two or more tension electronics must be screened or a twisted pair.
- The signal cable between the tension electronics and instruments, or process equipment, must be a screened 0.5 mm^2 cable.
- Cable screens must be connected to the copper earth bar. The screen connection maximum length is 50 mm.
- The protective earth conductor of the incoming mains supply must be connected to the copper earth bar in the cubicle.

2.4.1.2 Interference

For immunity to interference, the load cell cables should be separated as far as possible from noisy power supply cables. A minimum distance of 0.3 m (12 inches) is recommended. Where the measuring system cables meet noisy cables, they must cross at right angles.

2.4.1.3 Synchronization

Synchronization is not required for the wall mounted IP 65-version (NEMA 4) of the tension electronics.

If two or more IP 20-version (unsealed) tension electronics are mounted in the same cabinet they must be synchronized.

Synchronization is done by interconnecting the "SYNC" terminals, screw terminal X1:14, of all the units and interconnecting screw terminal X:15 of all units. A twisted pair cable or a screened cable must be used.

If one unit is turned off or removed the remaining units are still synchronized.

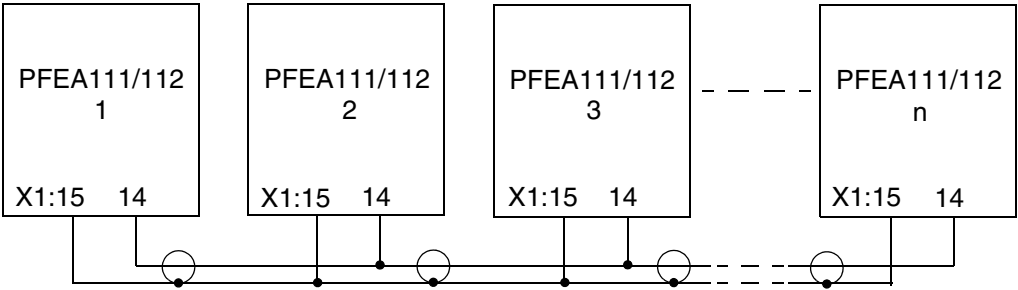


Figure 2-3. Connection for Synchronization

2.4.2 Mounting the Tension Electronics PFEA111/112

2.4.2.1 IP 65-version (NEMA 4)

The electronic unit is delivered in an enclosure intended for wall mounting.

When choosing a place for installation, check that there will be space to open the enclosure lid fully. Also check that there is enough working space in front of the enclosure.

The enclosure is fitted with cable glands (PFEA 111 five and PFEA 112 six cable glands).

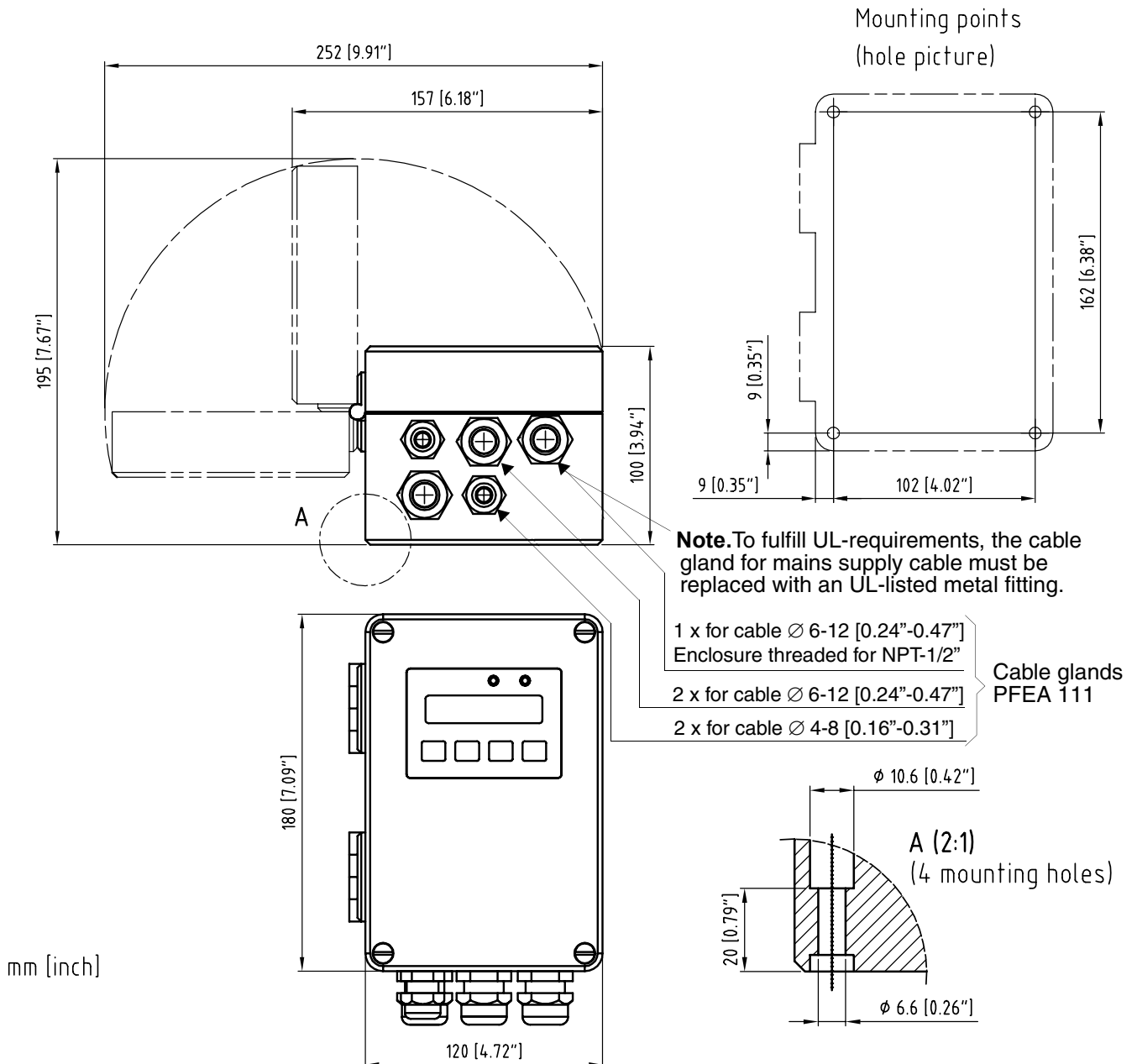


Figure 2-4. Installation Dimensions for PFEA 111/112

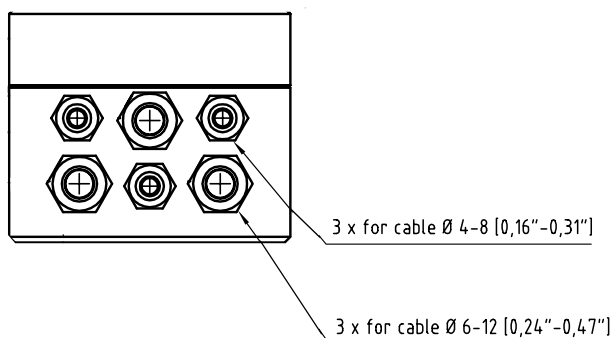


Figure 2-5. Cable Glands PFEA 112

Connect the cables to terminals according cable diagrams in Appendix (B, C, D, or E) depending on installed load cell type.

NOTE

Do not connect solid conductors to terminals. Do not crimp pins to stranded cores.

NOTE

The incoming mains voltage must be provided with fuses and a means of disconnection outside the tension electronics.

2.4.2.2 IP 20-version (Unsealed)

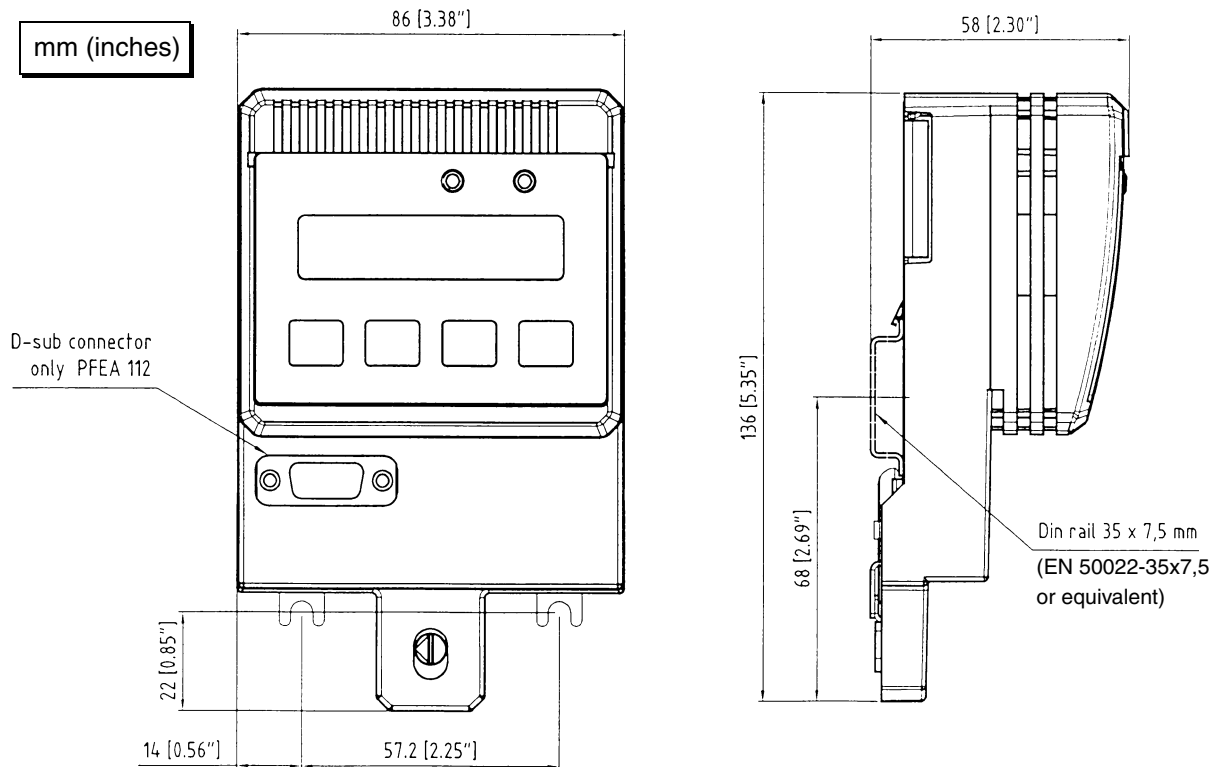


Figure 2-6. Installation Dimensions

Connect the cables to terminals according cable diagrams in Appendix (B, C, D, or E) depending on installed load cell type.

NOTE

Do not connect solid conductors to terminals. Do not crimp pins to stranded cores.

Grounding

The metal bottom of PFEA111-20 and PFEA112-20 connects to the metallic DIN-rail which serves as the Tension Electronics ground connector.

This is to ensure a good ground connection both for internal logic and for the EMI immunity and RF emission of the electronics.

The DIN-rail must have a good connection to the PE (Protective Earth) of the cabinet.

To achieve the best possible corrosion resistance, DIN-rails should be chromium plated, for instance, yellow chromium treated. Use star washers with each screw used to fasten the DIN-rail to the mounting plate.

To fasten the DIN-rail onto the mounting plate, the minimum screw diameter is 5 mm and the maximum distance between screws is 100 mm.

2.4.3 Earthing

For trouble free operation, the earthing must be carried out carefully. Note the following:

- If the free (unscreened) length exceeds 0.1 m (4 in.) the individual pairs of power and signal conductors must be twisted separately
- The external protective earth (PE) cable must be attached to one of the earthbar screw clamps.
- All the cable screens have to be connected to the earth bar and the length of the screen connection must be less than 50 mm (2 in.).

NOTE

The cable screens must be earthed at one end only.

- Since the signal earth of the measurement system is connected to the chassis earth of the tension electronics, the input of a superior system connected to the control system must not be earthed. The best ways of interconnecting the measurement system and a superior system to achieve optimal function are shown in [Figure 2-7](#) and [Figure 2-8](#).

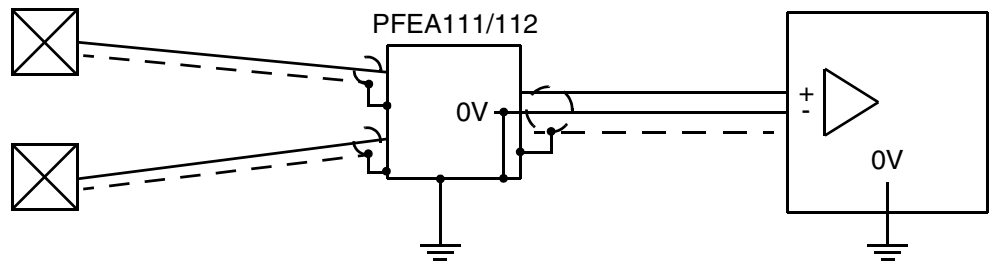


Figure 2-7. Connection to a superior system with an insulated or differential input

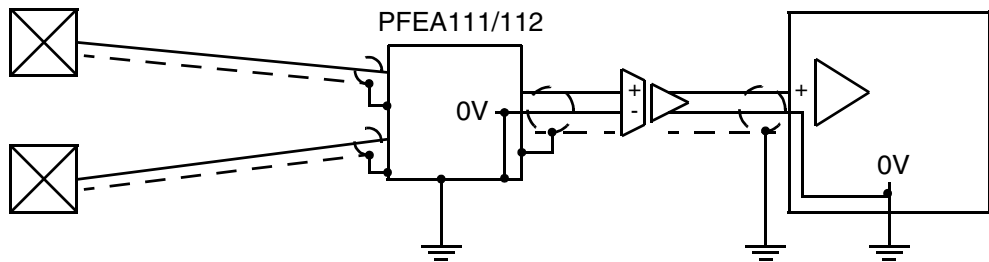


Figure 2-8. Connection to a superior system through a separate insulation amplifier

2.5 Connecting the Load Cells

Information for connecting the load cells is given in the appendix for each load cell type, see table below.

Type of load cell	Cable diagrams in Appendix
PFCL 301E	B
PFTL 301E	C
PFRL 101	D
PFTL 101	E

2.6 Connecting Optional Units

2.6.1 Insulation Amplifier PXUB 201 (only for the IP 20-version)

Insulation amplifier PXUB 201 is used when galvanic insulation between input and output, or between power supply and input/output, is required. See [Section A.5.1 Insulation Amplifier PXUB 201](#).

Insulation amplifier PXUB 201 is intended to be installed on a DIN-rail. The PXUB 201 is connected by screw terminals.

PXUB 201 is normally supplied from the same +24 V DC supply that supplies the tension electronics.

If PXUB 201 is mounted close to the terminal group, it is not necessary for the cable between the tension electronics and PXUB 201 to be screened.

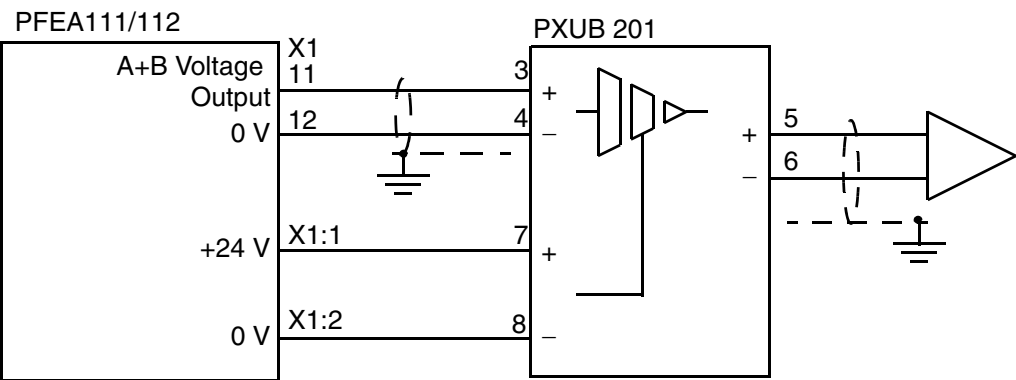


Figure 2-9. Typical Connection of Insulation Amplifier PXUB 201

2.6.2 Power Supply Unit

If no 24 V is available, the power supply units SD 821, SD 822 and SD 823 can be used as power supply for the IP 20-versions.

The main supply voltage for all the three power supply units are:

- 115 V AC (85 - 132 V), 100 V -15% to 120 V + 10%
(Mains voltage selection switch in position 115)
- 230 V AC (176 - 264 V), 207 V -15% to 240 V + 10%
(Mains voltage selection switch in position 230)

Table 2-2. Number of PFEA111/112 that can be supplied

Power Supply Unit	PFEA111	PFEA112
SD 821 (2.5 A)	6	6
SD 822 (5 A)	12	12
SD 823 (10 A)	24	24

Chapter 3 Commissioning

3.1 About this Chapter

This chapter contains necessary information for commissioning your Web Tension System.

It is assumed that the Web Tension System has been installed according to the instructions given in [Chapter 2 Installation](#) and [Appendix \(B, C, D or E\)](#) depending on installed load cell type.

You must know the following data before starting commissioning:

1. Load cell type and nominal load, see Appendix for installed load cell type
2. Object type, see [Section 3.12.2](#)
 - Standard roll (two load cells)
 - Single side measurement (one load cell)
3. Maximum web tension
4. Desired output data at given web tension
5. Communication data, see [Section 3.13](#)

3.2 Safety Instructions

Read and follow the safety instructions stated in [Chapter 1 Introduction](#), before starting any commissioning work. However, local statutory regulations, if stricter, are to take precedence.

3.3 Necessary Equipment and Documentation

The following items are required:

- Cable diagram
- Service tools

3.4 Using the Panel Buttons

3.4.1 Navigating and Confirming

Display	Button	How to use it
		Steps back to previous menu. Sometimes you must press this button two or more times to get back to the desired menu.
		Steps up in a list.
		Steps down in a list. Steps to next main menu.
		OK (acknowledgment) button. Confirms a selection or a parameter setting.

3.4.2 Changing Numerical Values and Parameters

SetTensionAt10V XXXXXX N	NominalLoad ZZ kN ZZ lbs
-----------------------------	-----------------------------

- X indicates a numerical value.
- Z indicates that a parameter can be chosen from a list.

SetTensionAt10V [XXXXXX] N	NominalLoad [ZZ kN ZZ lbs]
-------------------------------	-------------------------------

To change a numerical value, **X**, or parameter, **Z**, press . The numerical value or parameter is then placed within brackets [XXXXXX] or [ZZ] to show that it can be changed.

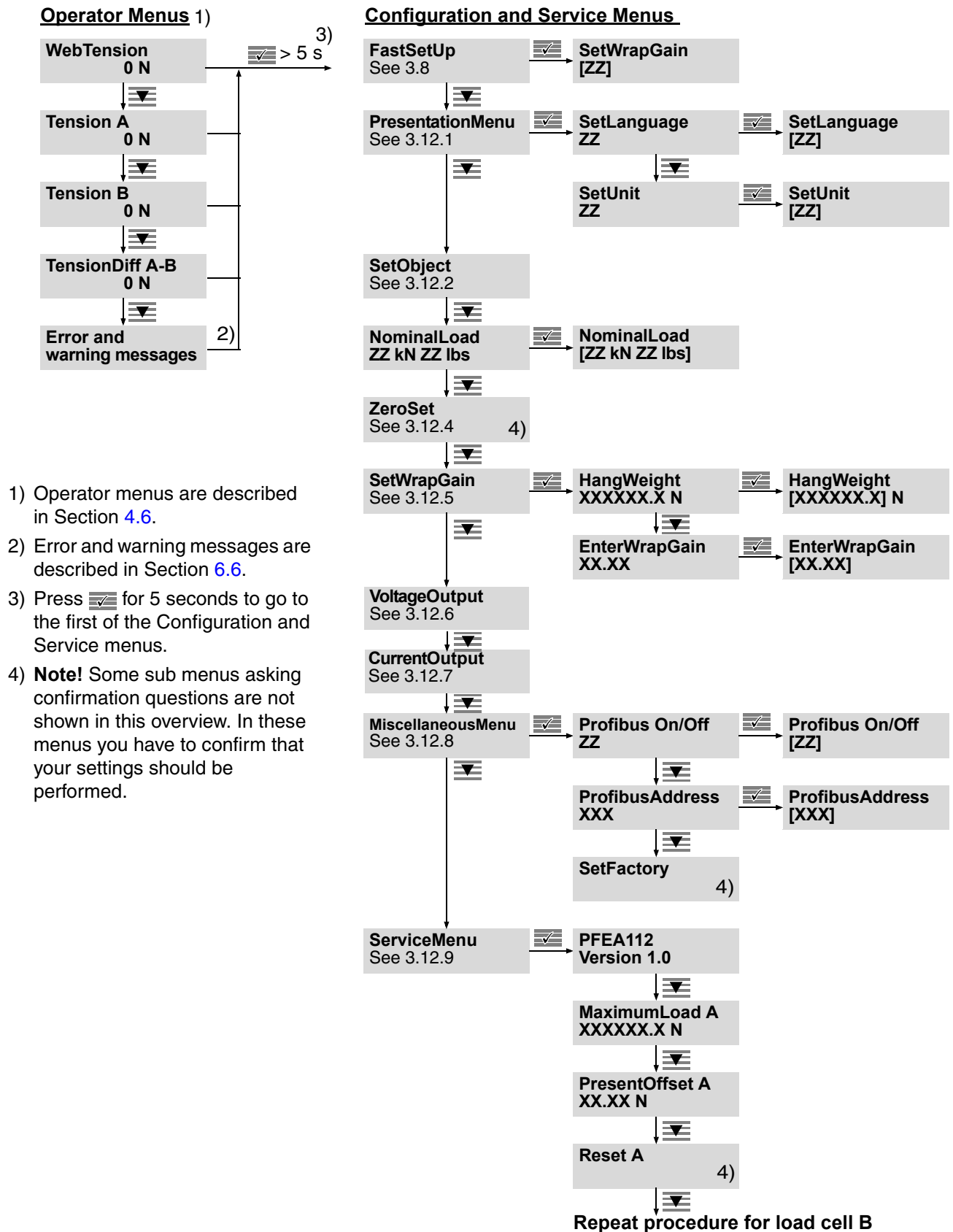
If it is a “**Z**” parameter, use  and  to go up or down in the list. When the desired value is shown on the display press . When  is pressed the new parameter value is saved and the brackets around the value disappear.

If you have pressed  so that the parameter is placed within brackets, you can cancel the entering mode by pressing . The selections you have done with  and  will not be stored. If  is pressed the old value is displayed without brackets.

To change a numerical value, press  so that the value is placed within brackets. Then the first digit can be changed with  and . When the first digit has the desired value press , then the second digit can be changed with  and . When pressing  after the last digit has been set, the new value is saved and displayed without brackets.

Using  when entering a numerical value means returning to the previous digit. By pressing  a sufficient number of times you will leave the entering mode and the old value will be displayed without brackets.

3.5 Menu Overview

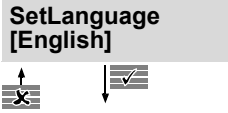
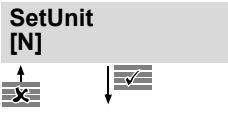


3.6 Step-by-step Commissioning Guide

Step	Measurement	See section
1	Check that the mains supply voltage is switched off.	
2	Check all cabling according to the cable diagrams.	Appendix B, C, D or E
3	Check the supply voltage <u>DIN-rail mounted IP20-unit (unsealed)</u> Nominal 24 V DC, Working range 18 - 36 V DC, X1:1-2 <u>Wall mounted IP65-unit (NEMA 4)</u> 85 - 264 V AC (100 V - 15% to 240 V + 10%), 45-65 Hz, X9:1-2 Nominal 24 V DC, Working range 18 - 36 V DC, X1:1-2	3.7
4	Perform basic settings (if necessary)	3.7
5	Perform setup: Fast setup Complete setup	3.8 3.11
6	Check load cell signal polarity	3.9
7	Check load cell function	3.10

3.7 Performing Basic Settings

When the tension electronics has been powered on for the first time after delivery, you are asked to **SetLanguage** and then to **SetUnit**. These two settings must be performed to be able to proceed with the rest of the setup. Language and unit can be changed later on if desired.

1		Select desired language from the list with  and . English is set as default. Confirm with .
2		Select desired presentation unit from the list with  and . N (Newton) is set as default. Confirm with .
3		Press  to start the fast setup sequence. See Section 3.8. If you want to make a complete setup, go to the separate setup menus by pressing . See Section 3.11.

3.8 Performing a Fast Setup

The fast setup guides you through a minimum number of steps to set up the tension electronics. You are asked to answer some questions and to enter desired values. These selections and parameter settings must be made to get the tension electronics ready for measurement.

Only a limited number of selections and parameter settings are set in the fast setup. All other parameters are set as factory default values. See [Appendix A.4 Factory Default Settings](#).

The fast setup can be made in two ways, depending on how the wrap gain is set.

The wrap gain can be set by selecting “HangWeight” or “EnterWrapGain”.

- Using Hanging Weights, see [Section 3.8.1](#).
- Using Wrap Gain, see [Section 3.8.2](#).

Hanging Weights and Wrap Gain are explained in [Section 3.12.5](#).

3.8.1 Performing Fast Setup using Hanging Weights

The simplest method to set the wrap gain in this tension system is to use a known weight that loads the center of the roll with a rope that follows the web path exactly.

All rolls must be free turning idlers. To keep friction losses low, only use the closest rolls to define web path.

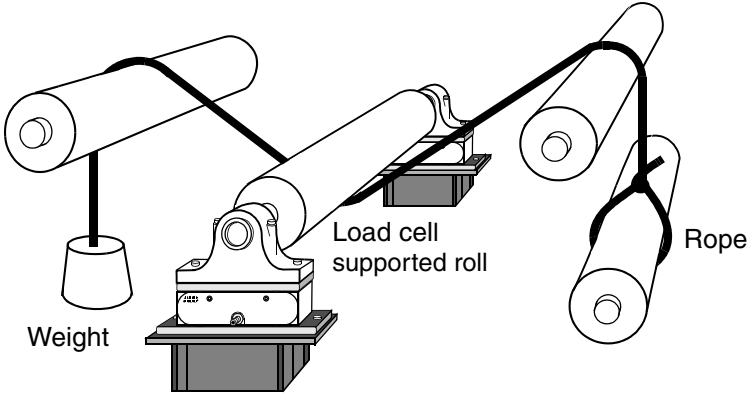
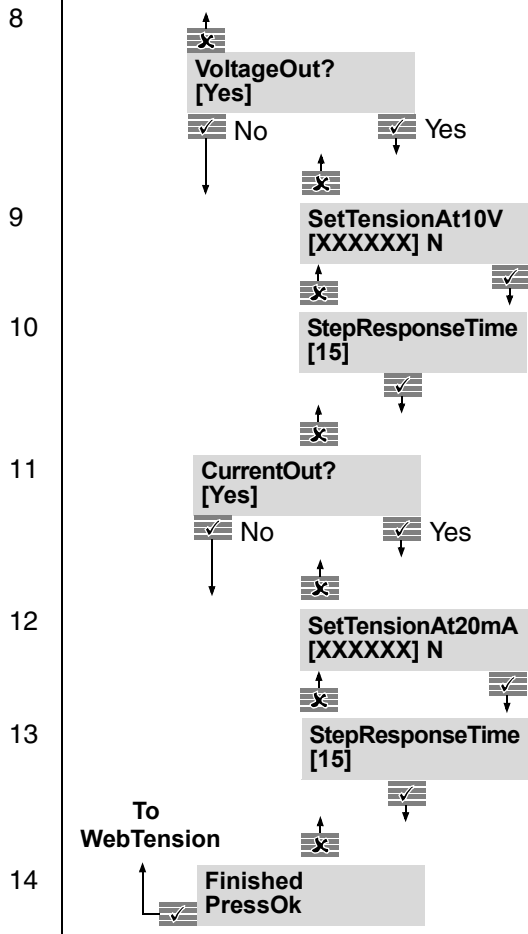


Figure 3-1. Setting Wrap Gain with Hanging Weights (example of installation)

Follow the steps below to run a fast setup using hanging weights.

1	WebTension > 5 seconds	Press for 5 seconds to go to the FastSetUp menu.
2	FastSetUp	Press to start the fast setup sequence.
3	SetWrapGain [HangingWeights]	Select HangingWeights from the list with and .
		Confirm with .
4	CellsOnRoll [2]	Select the number of load cells that support the roll (1 or 2) from the list with or . Confirm with .
5	NominalLoad [1 kN 225 lbs]	Check the nominal load on the load cell name plate. Select nominal load from the list with or . Confirm with .
6	WithNoTension PressOkToZeroSet	1. Check that no load is applied on the roll. 2. Press to zero set. "ActionDone" is shown on the display for one second to confirm the zero setting action.
7	HangWeight [XXXXXX] N	1. Apply a known weight on the roll (see Figure 3-1). 2. Enter the value of the known weight. Confirm with .



Select **Yes** and press to set up the voltage output.

Enter the tension value corresponding to 10 V.
Confirm with .

Select the step response time (15, 75, 250 or 750 ms) from the list with or . Confirm with .

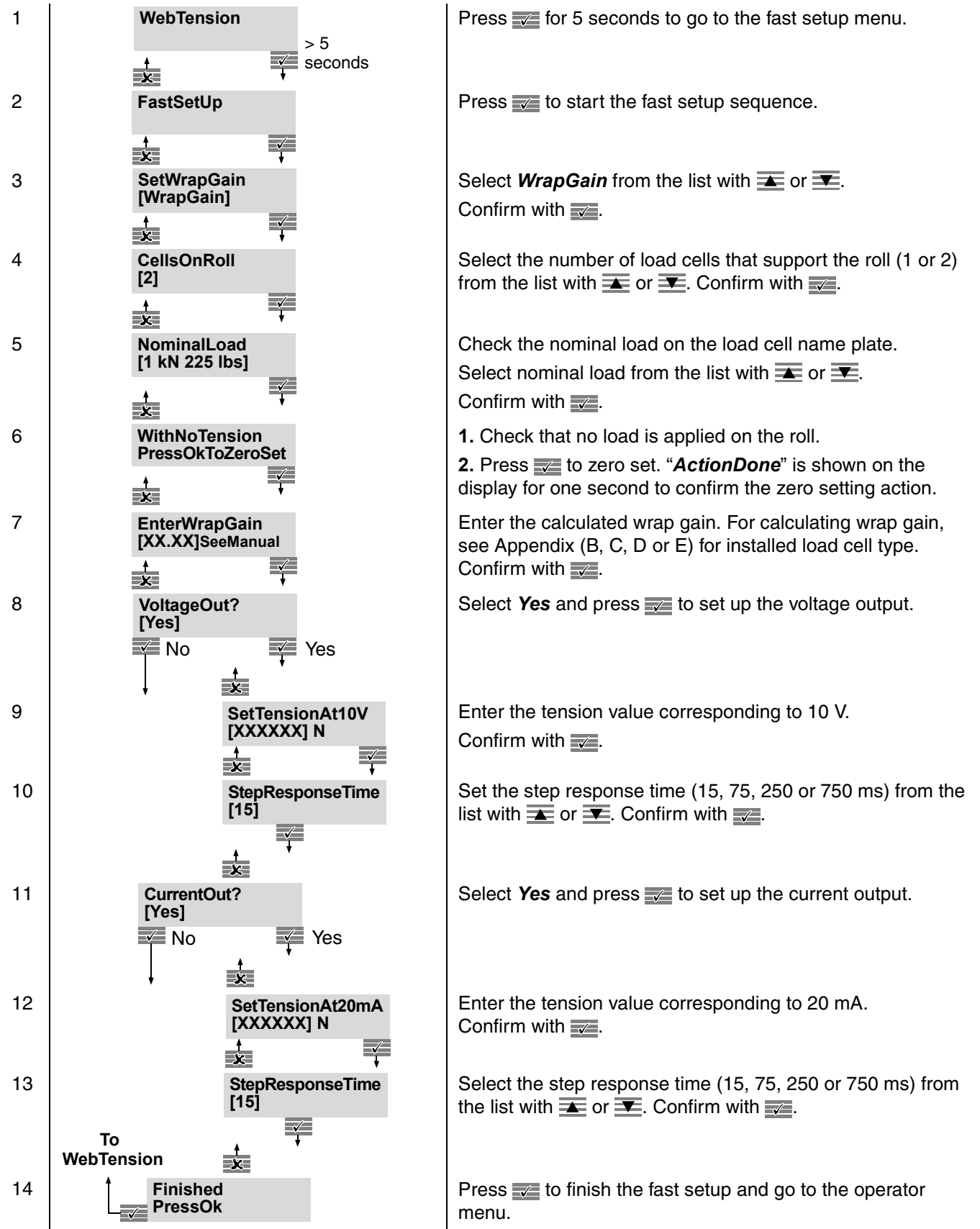
Select **Yes** and press to set up the current output.
Confirm with .

Enter the tension value corresponding to 20 mA.
Confirm with .

Select the step response time (15, 75, 250 or 750 ms) from the list with or . Confirm with .

Press to finish the fast setup and go to the operator menu.

3.8.2 Performing Fast Setup using Wrap Gain



3.9 Checking Load Cell Signal Polarity

This is a simple method to check that the load cells are connected to give a positive output signal change from the tension electronics for increased web tension.

1. Push with your hand to apply a force corresponding to increased web tension on one load cell at a time (as close to the load cell as possible) and check if the display reading is positive. If the display reading is negative, invert the load cell signal connection to the tension electronics.

NOTE

If you don't know in which direction the force is acting, connect load cell A and B with the same force direction.

To change the polarity of load cell A, invert X1:5 and 6 (In A+ and In A-).
To change the polarity of load cell B, invert X1:9 and 10 (In B+ and In B-).

2. After changing load cell polarity, check that the display reading is positive for increased web tension.

3.10 Checking Load Cell Function

The “Hanging Weight” procedure can also be used as a function test on load cells, see [Section 3.8.1](#).

The rope should then be placed in the web path but as close as possible to one of the load cells. The output signal should be noted, and the rope moved close to the other load cell. Check that the difference in output signal is small.

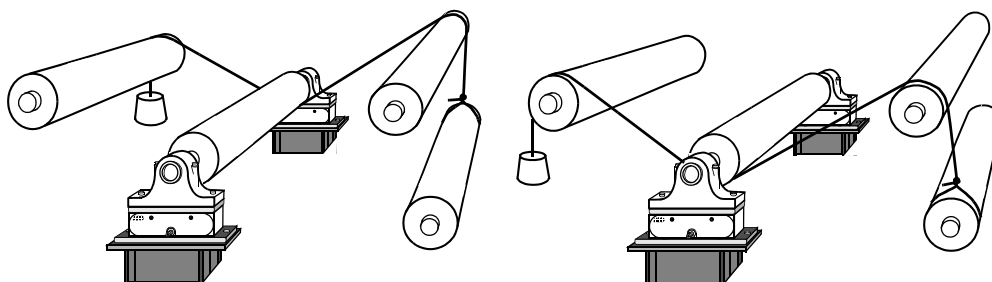


Figure 3-2. Load Cell Function Test

3.11 Performing a Complete Setup

3.11.1 Overview

The complete setup is built up by a number of main and sub menus. The table below shows the main menus in order of appearance when stepping through the complete setup. The table also gives an overview of the selections and parameter settings you can do below in each main menu.

The complete setup sequence is described in [Section 3.12](#).

Main menus	Selections and parameter settings	See details in section...
PresentationMenu ↓ ▾	Set language Set unit	3.12.1
SetObject ↓ ▾	Set object type - Standard roll (load cells A and B) or - Single side measurement (load cell A)	3.12.2
NominalLoad 1000 N 225 lbs ↓ ▾	Set nominal load	3.12.3
ZeroSet ↓ ▾	Zero set load cells	3.12.4
SetWrapGain ↓ ▾	Set hanging weights (actual force) or Set wrap gain (calculated value)	3.12.5
VoltageOutput ↓ ▾	Set step response time Set high tension value and high output voltage Set low tension value and low output voltage Set high and low output voltage limit	3.12.6
CurrentOutput ↓ ▾	Set step response time Set high tension value and high output current Set low tension value and low output current Set high and low output current limit	3.12.7
Miscellaneous Menu ↓ ▾	Set Profibus address Reset all values to factory default	3.12.8
ServiceMenu	Read service information Reset maximum load for load cell A Reset maximum load for load cell B	3.12.9

3.12 Complete Setup Sequence

This section gives a step-by-step description with detailed information of all available setup menus with related parameters, data and settings.

3.12.1 Presentation Menu

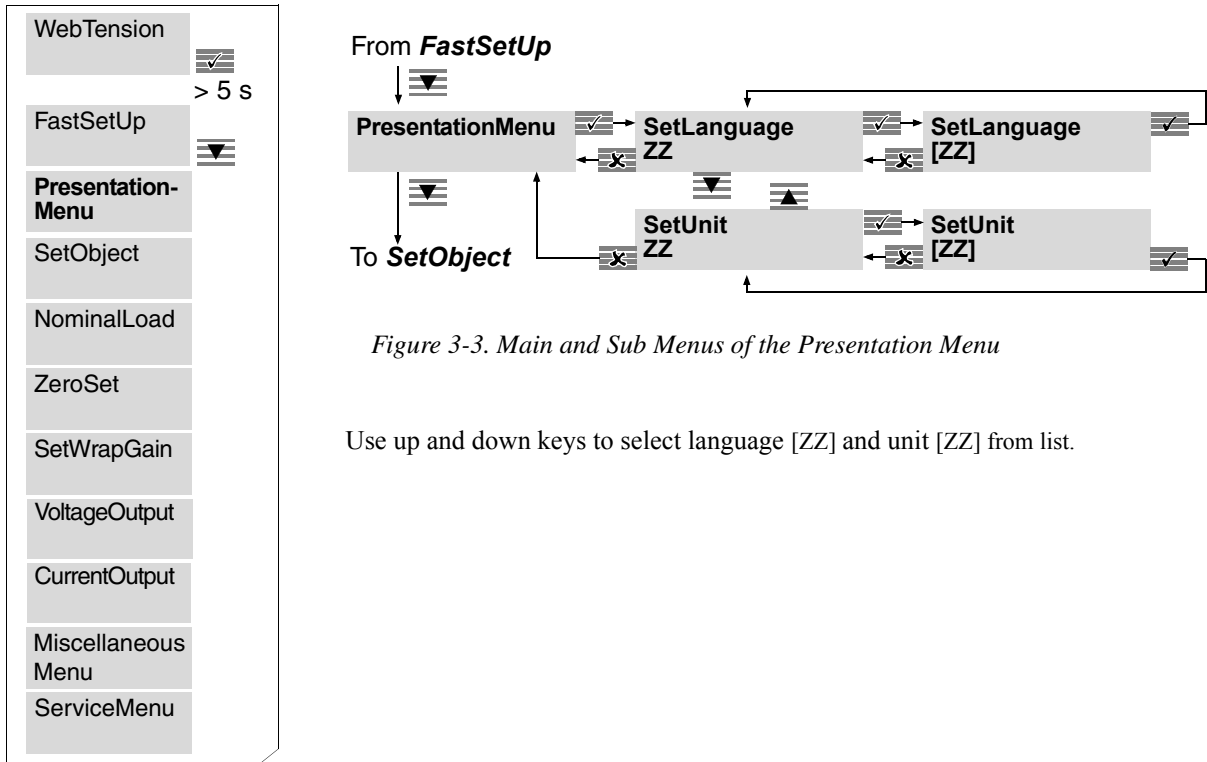


Figure 3-3. Main and Sub Menus of the Presentation Menu

Use up and down keys to select language [ZZ] and unit [ZZ] from list.

3.12.1.1 Set Language

The following languages are available:

- English
- German
- Italian
- French
- Portuguese
- Japanese

3.12.1.2 Set Unit

The following units can be set:

- N (Newton)
- kN (kiloNewton)
- kg (kilogram)
- lbs (US pounds)

3.12.2 Set Object

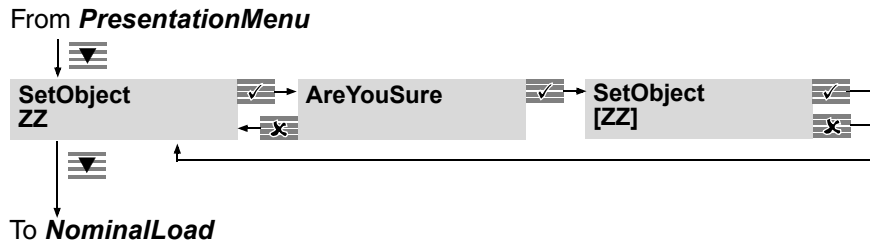
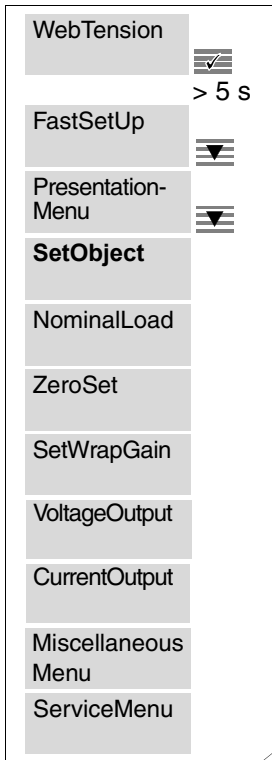


Figure 3-4. Set Object Menus

Use up and down keys to select type of object [ZZ] from list.

There are two object types that can be selected.

- Standard roll (if both load cell A and B are connected to the roll)

Standard roll

Two load cells with separate signals

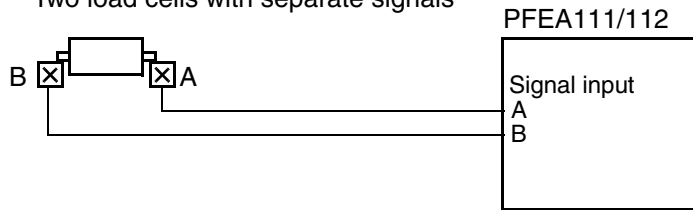


Figure 3-5. Object Type Standard Roll

- Single side measurement (if only load cell A is connected to the roll)

Single side measurement

One load cell signal



Figure 3-6. Object Type Single Side Measurement

When single side measurement is selected the measured signal is multiplied by two and presented as web tension on display and analog output.

3.12.3 Nominal Load

WebTension

> 5 s

FastSetUp

Presentation-Menu

SetObject

NominalLoad

ZeroSet

SetWrapGain

VoltageOutput

CurrentOutput

Miscellaneous Menu

ServiceMenu

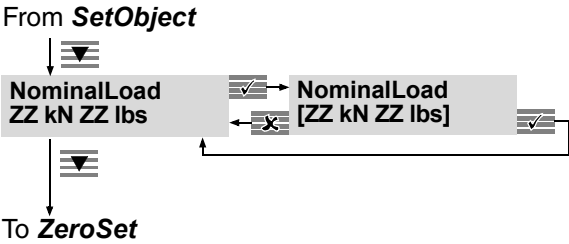


Figure 3-7. Nominal Load Menus

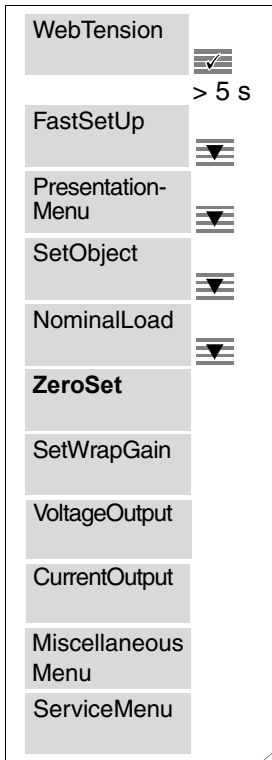
The nominal load is selected from the list below and must be the same as the nominal load on the load cell name plate. The load cell nominal load is displayed in kN and lbs on the same row.

The following nominal loads can be selected:

Table 3-1. Nominal Loads

[kN]	[lbs]
0.1	22
0.2	45
0.5	112
1.0	225
2.0	450
5.0	1125
10	2250
20	4500
50	11250
100	22500
200	45000

3.12.4 Zero Set



Zero set is used to compensate for load cell zero signal and tare weight.

The zero setting range is $\pm 2 \times F_{nom}$ (load cell nominal load).

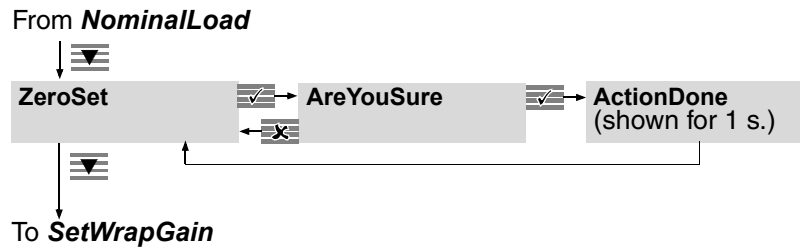


Figure 3-8. Zero Set Menus

NOTE

Zero set must be done with no tension applied to the roll.

3.12.5 Set Wrap Gain

To be able to present actual web tension on the display, the ratio between web tension and measured force on the load cell must be determined.

This ratio is a scaling factor called Wrap Gain.

The Wrap Gain depends on the web's wrap angle on the measuring roll and the orientation of the load cells. Therefore, Wrap Gain depends on the actual installation.

This gives:

$$T \text{ (tension)} = \text{Wrap Gain} \times F_R \text{ (force of web tension in load cell measuring direction)}$$

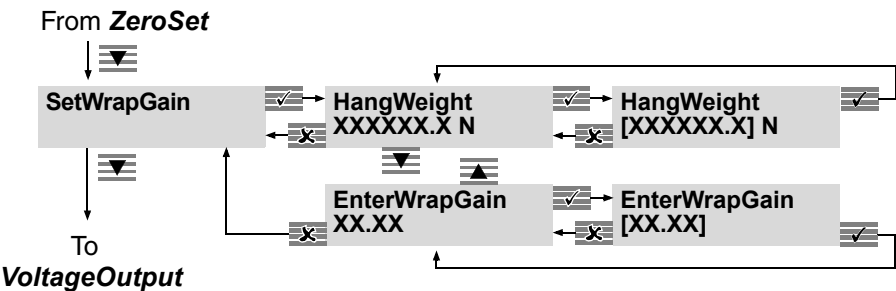
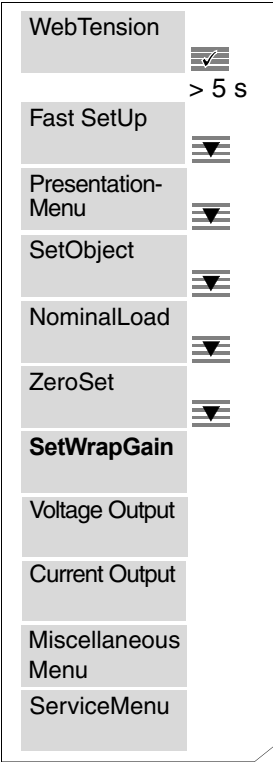


Figure 3-9. Wrap Gain Menus

There are two ways to find out the ratio between web tension and measured force on the load cells, by Hanging weights or by Calculation.

- **With hanging weights** (Menu *HangWeight*)

String a rope that follows the web path exactly and apply a known weight.

The applied known weight simulates the actual web tension and the electronics measures the resulting force on the load cells caused by the applied weight.

When both web tension (T) and the corresponding measured force (F_R) are known the tension electronics calculates the ratio T / F_R and stores the value as Wrap Gain.

When web tension is applied to the roll the tension electronics calculates web tension by multiplying the measured force on the load cells with the Wrap Gain.

After the hanging weight procedure the wrap gain calculated by the tension electronics can be found in the EnterWrapGain menu.

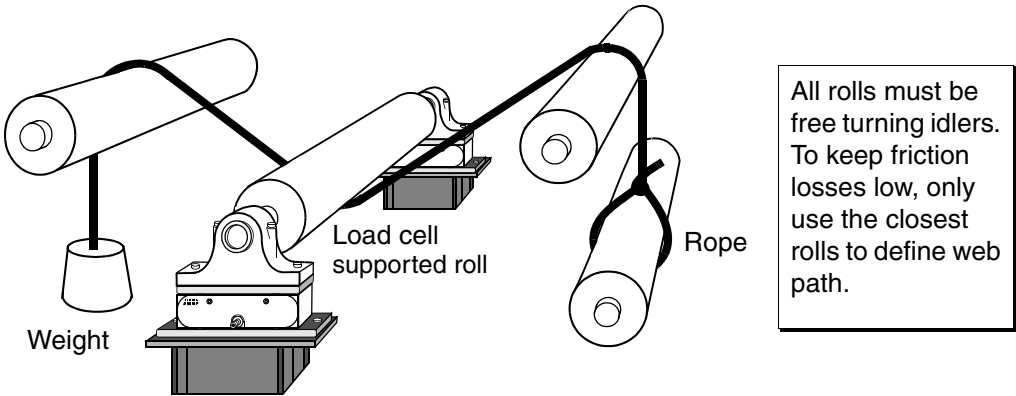


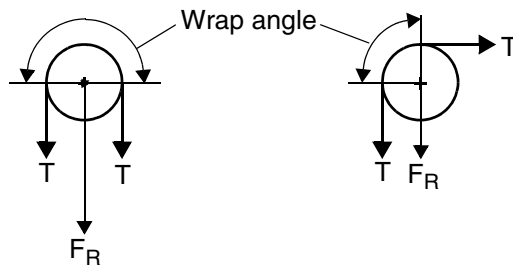
Figure 3-10. Setting Wrap Gain with Hanging Weights (example of installation)

- **By calculation** (Menu *EnterWrapGain*)

Wrap Gain is a scaling factor which corresponds to the ratio between Web Tension (T) and the force component (F_R) from web tension that is acting in the load cell measuring direction.

The Wrap Gain range is 0.5 - 10. If you try to set the wrap gain outside this range, the message “**OutOfRange**” will be shown on the display. The wrap gain can be set with a resolution of 0.01.

Examples describing the principle of calculating Wrap Gain:



$$F_R = 2T$$

$$\text{Wrap gain} = \frac{T}{F_R}$$

$$\text{Wrap gain} = \frac{T}{2T} = 0.50$$

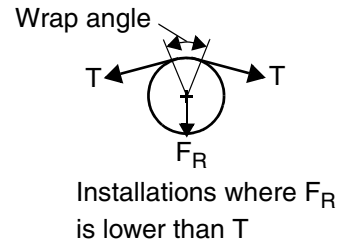
Wrap gain = 0.50
(Min. value of Wrap Gain)

$$F_R = T$$

$$\text{Wrap gain} = \frac{T}{F_R}$$

$$\text{Wrap gain} = \frac{T}{T} = 1.00$$

Wrap gain = 1.00



$$F_R < T$$

$$\text{Wrap gain} = \frac{T}{F_R}$$

Wrap gain = > 1
(Max. permitted value of Wrap Gain is 10)

See calculation of Wrap Gain in Appendix (B, C, D or E) for installed load cell type.

3.12.6 Voltage Output

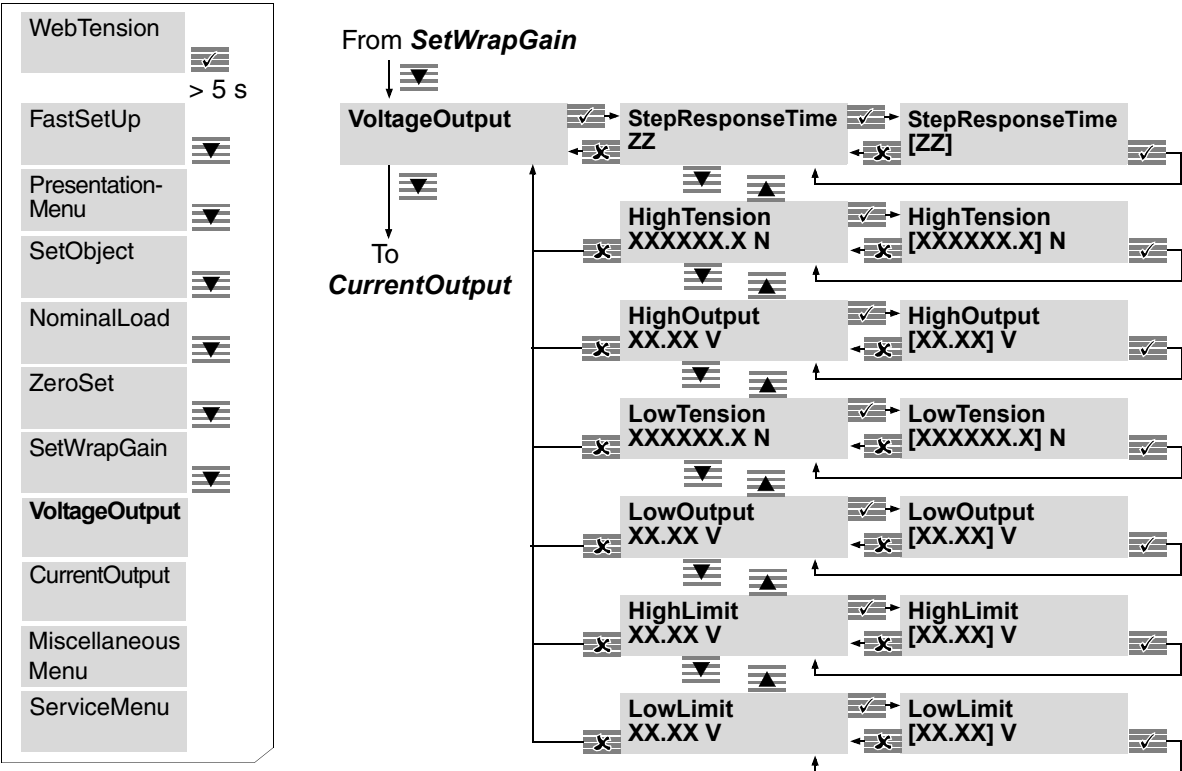


Figure 3-11. Voltage Output Menus

The following parameters can be set:

- Step response time (filtering)
See [Table 3-2](#).
- High Tension (N, kN, kg, lbs), (factory default = 2000 N)
- High Output, (factory default = +10 V)
- Low Tension (N, kN, kg, lbs), (factory default = 0 N)
- Low Output, (factory default = 0 V)
- High Limit, (factory default = +11 V)
- Low Limit, (factory default = -5 V)

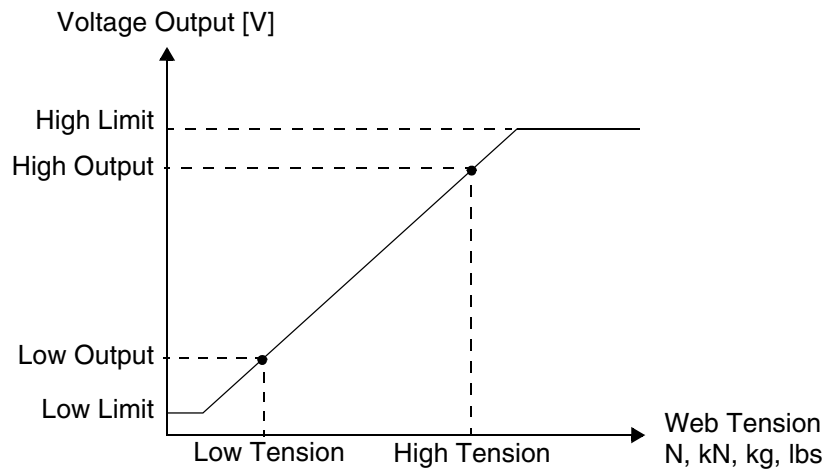


Figure 3-12. Parameter Definitions

Filtering may be used if the voltage output signal is too fast or to compensate roll imbalance. The filters are of type linear phase, maximum flat, 20 dB/decade.

Table 3-2. Filtering

Step response time 0 - 90%	Cut-off frequency -3dB
15 ms	35 Hz
75 ms	5 Hz
250 ms	1.5 Hz
750 ms	0.5 Hz

3.12.7 Current Output

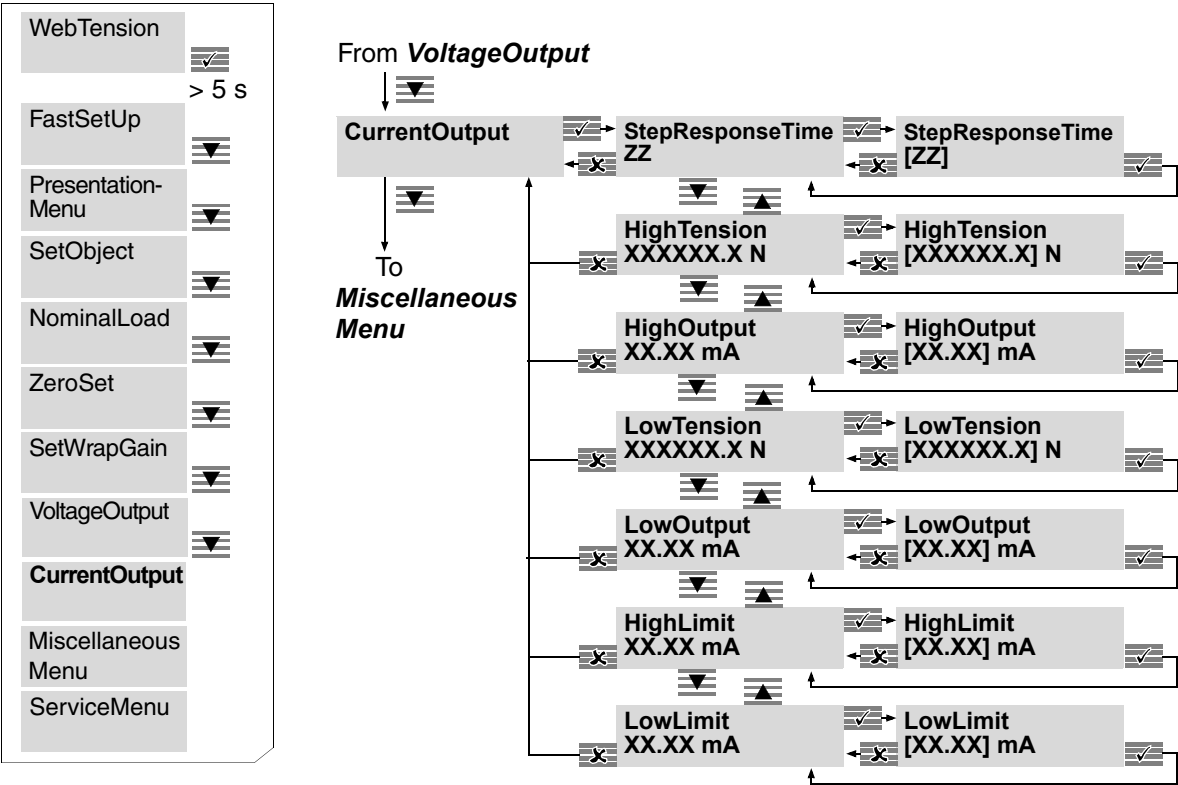


Figure 3-13. Current Output Menus

The following parameters can be set:

- Step response time (filtering)
See [Table 3-3](#).
- High Tension (N, kN, kg, lbs), (factory default = 2000 N)
- High Output, (factory default = 20 mA)
- Low Tension (N, kN, kg, lbs), (factory default = 0 N)
- Low Output, (factory default = 4 mA)
- High Limit, (factory default = 21 mA)
- Low Limit, (factory default = 0 mA)

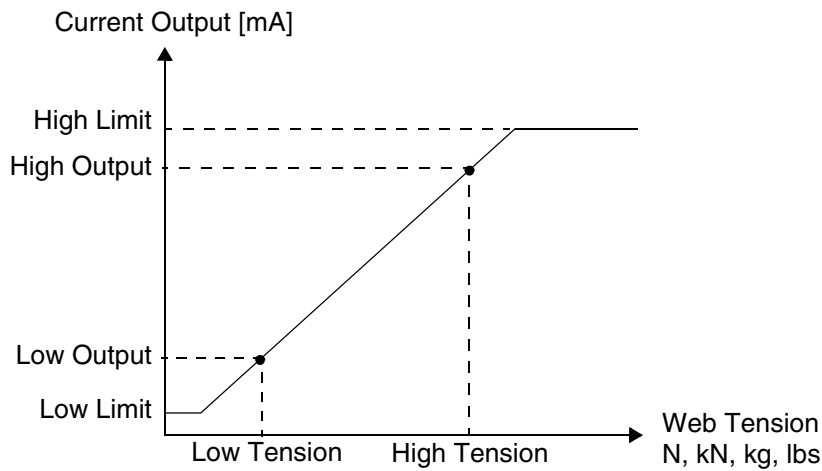


Figure 3-14. Parameter Definitions

Filtering may be used if the current output signal is too fast or to compensate roll imbalance. The filters are of type linear phase, maximum flat, 20 dB/decade.

Table 3-3. Filtering

Step response time 0 - 90%	Cut-off frequency -3dB
15 ms	35 Hz
75 ms	5 Hz
250 ms	1.5 Hz
750 ms	0.5 Hz

3.12.8 Miscellaneous Menu

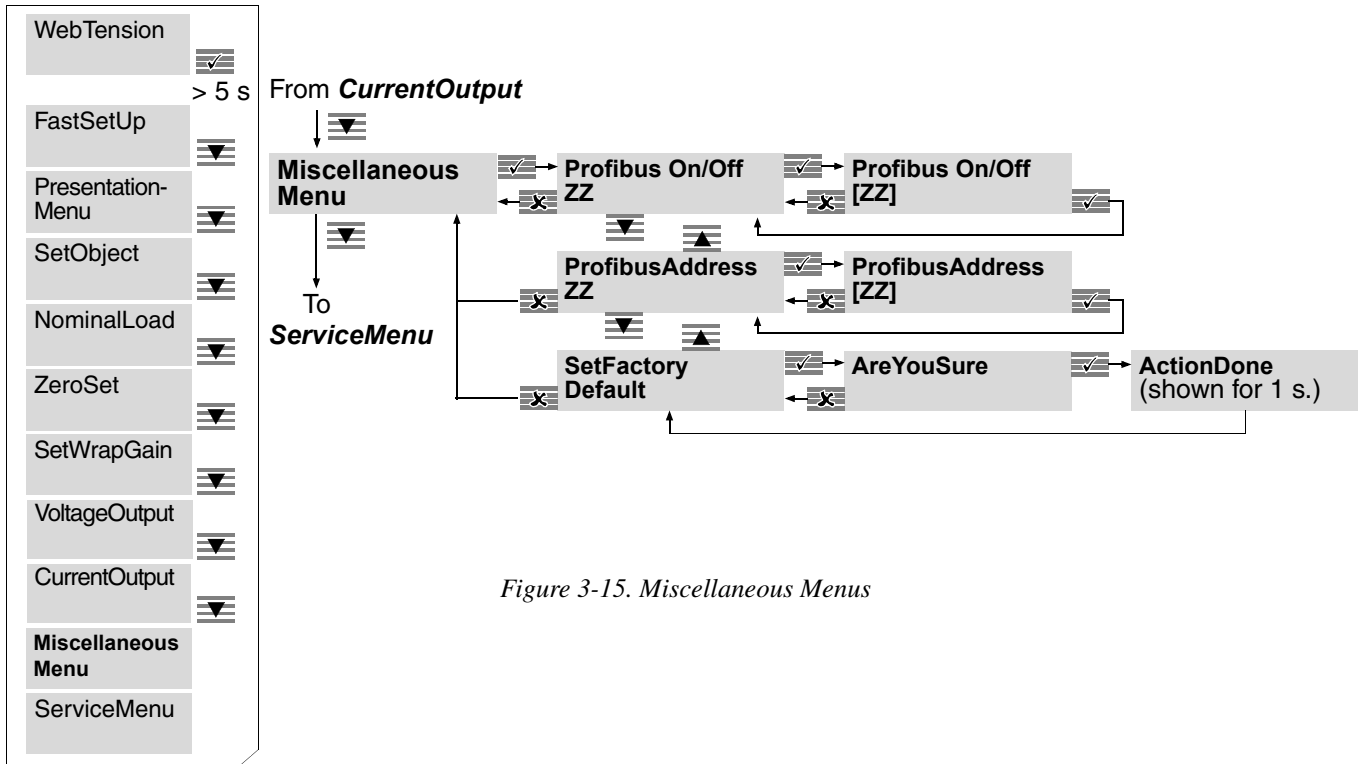


Figure 3-15. Miscellaneous Menus

3.12.8.1 Profibus

- Profibus On/Off
The Profibus can be enabled or disabled.
- Profibus address
If the Profibus is enabled, the Profibus address must be set in the range 000 - 125.
Read more about the Profibus in [Section 3.13](#).

3.12.8.2 Set Factory Default

- Set factory default
The parameters are set as on delivery except **Maximum load A** and **Maximum load B**.
Read more in [Appendix A.4 Factory Default Settings](#).

3.12.9 Service Menu

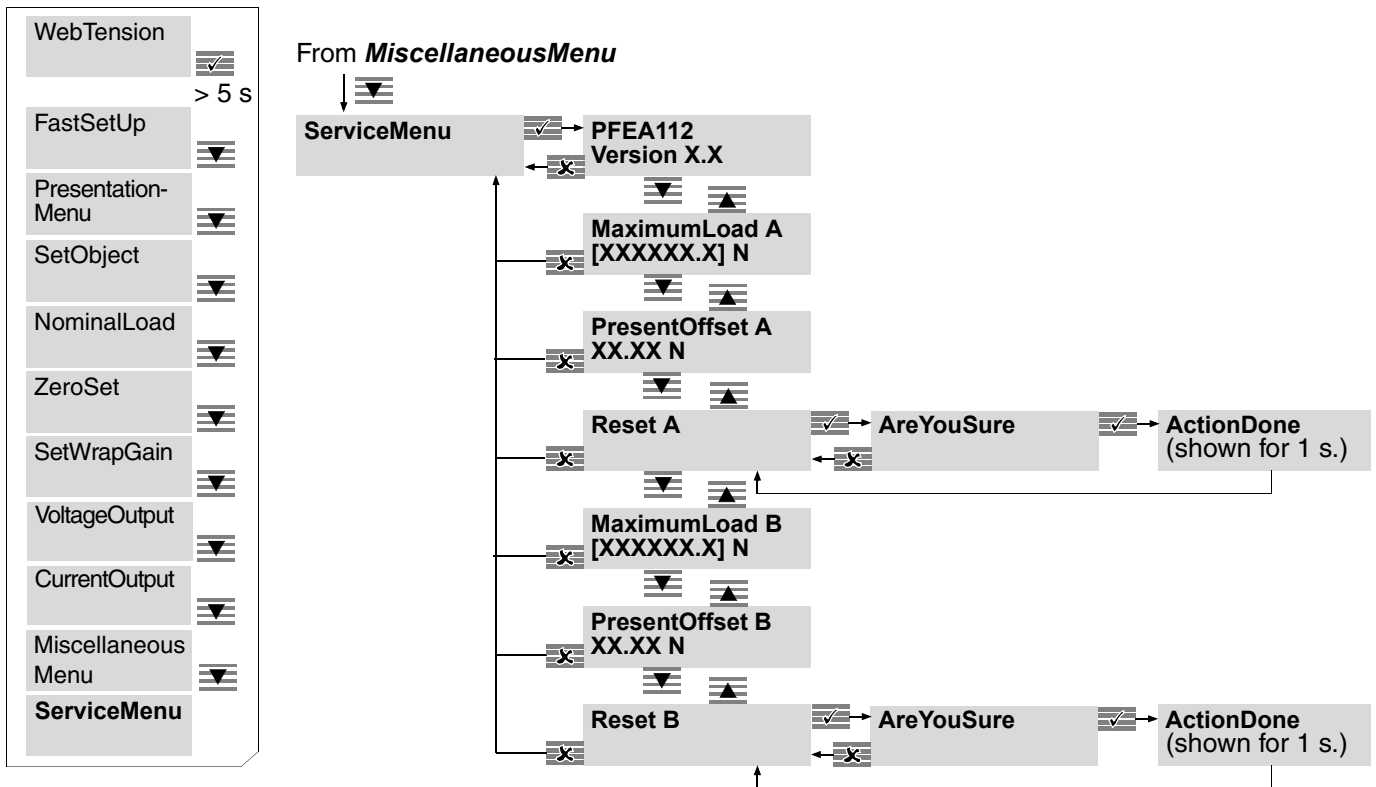


Figure 3-16. Service Menus

The service menu has parameters that can be viewed only and parameters that can be set. If single side measurement is selected in menu **SetObject** input B is not connected.

- Parameters that can be viewed only:
 - Software version number
 - Maximum load A
Displays maximum load since last reset
 - Present offset A
Displays zero offset at last zero set
 - Maximum load B
Displays maximum load since last reset
 - Present offset B
Displays zero offset at last zero set
- Parameters that can be set:
 - Reset A
This action sets “Maximum load A” to zero.
 - Reset B
This action sets “Maximum load B” to zero.

3.12.9.1 Maximum Load / Present Offset

For each load cell connected to the web tension electronics PFEA111/112, a maximum load memory, with the range $\pm 6.5 \times F_{\text{nom}}$, will store the highest load that is applied to the load cell.

The maximum load consists of:

- Load cell zero signal (with no load on the load cell)
- F_{RT} , applied force component of tare in the measuring direction of the load cell
and
- F_{R} , measured force (force component of tension in the measuring direction of the load cell).

The maximum load memory can be reset if a load cell is replaced.

3.12.9.2 Reset A/B

Reset A sets “MaximumLoad A” to zero.

Reset B sets “MaximumLoad B” to zero.

3.13 Profibus DP Communication with PFEA112

3.13.1 General Data about Profibus DP

The purpose of the Profibus DP communication in PFEA112 is to provide a high speed communication link between superior systems and the PFEA112.

Profibus DP is a multidrop communication protocol intended to connect PLCs to sensors (DP means “Distributed Peripherals”).

The physical interface is RS 485 (two wire cable).

The maximum transfer rate is 12 Mbit/s.

The protocol is based on a master-slave principle. The PFEA112 is a slave. A Profibus master polls the slaves all the time, that means the polling is going on with a fixed time interval even when no new data is available from the PFEA112.

Each slave has an address in the range 0 to 125.

Profibus requires that the message format, communication parameters and error codes of slaves are made available in a so called type file, also known as GSD file (See [Appendix A.6 Profibus DP - GSD File for PFEA112](#)). This file is then stored in the Profibus Master.

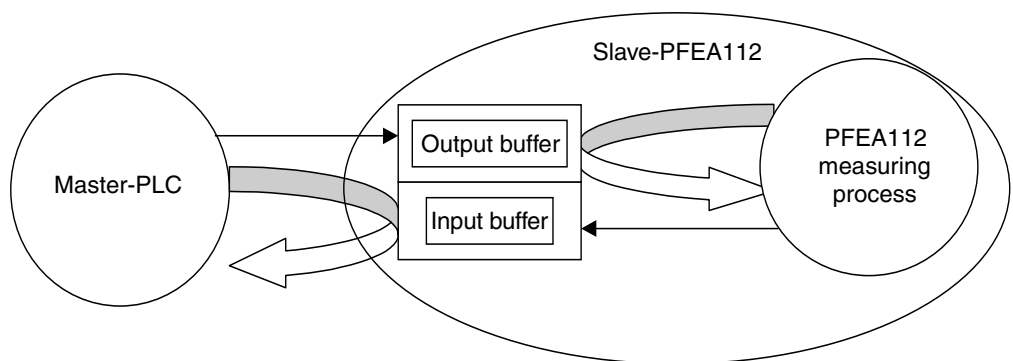
At start-up the Profibus Master verifies that the Slave with the given type file is indeed available on the bus.

3.13.2 Master-slave Communication

The Master and the slave communicate via an output buffer and an input buffer.

The master reads the input buffer and writes to the output buffer once every scan cycle for Profibus.

The slave polls the output buffer and updates values in the input buffer.



3.13.3 Commands through Profibus

Profibus DP is available in PFEA112 (not in PFEA111).
“Zero set” is the only command that can be performed through Profibus in PFEA112.

3.13.4 Handling of Measurement Data through Profibus

Two web tension measurement values are transferred through Profibus:

- Value 1 has the same step response time as "voltage output"
- Value 2 has the same step response time as "current output".

The scaling of "voltage output" and "current output" does not affect the measurement values transferred through Profibus.

If zero set has been performed, the zero set values are transferred through Profibus.

For scaling of Profibus measurement values, see [Section 3.13.4.1](#).

Each measurement value has 16 bit, 2-complement representation (Integer 16).

3.13.4.1 Scaling of Profibus Measurement Values

Depending on the load cell nominal load the least significant bit of the measurement value is as specified in [Table 3-4](#).

Table 3-4. Scaling of Profibus Measurement Values

Number of load cells connected to the analog output channel	Value of least significant bit
1 or 2 load cells	$0.001 \times 2 \times F_{nom}^{(1)}$

(1) F_{nom} = load cell nominal load (see load cell nameplate)

Example for one or two 0.5 kN (500 N) load cells:
The value of the least significant bit is $0.001 \times 2 \times F_{nom} = 0.001 \times 2 \times 500 = 1 \text{ N}$

3.13.4.2 Filtering of Profibus Measuring Values

“Value 1” has the same filtering as the voltage output.
“Value 2” has the same filtering as the current output.

3.13.4.3 Input Buffer, Communication Block from PFEA112 to PLC

This section specifies measurement values and Boolean values in the input buffer communication block.

Data	Byte No.	Bit No.							
		7	6	5	4	3	2	1	0
Value 1	01	MSB							
	02	LSB							
Value 2	03	MSB							
	04	LSB							
Boolean in	05	No. 7	No. 6	No. 5	No. 4	No. 3	No. 2	No. 1	No. 0
	06	Spare for future use							

Data:

Value 1, Web tension

Step response time (filtering) equal to the setting for **voltage** output, 16-bit, 2-complement representation (Integer 16)

Value 2, Web tension

Step response time (filtering) equal to the setting for **current** output, 16-bit, 2-complement representation (Integer 16)

Boolean in:

The error or warning is active when the corresponding bit is set to “1”.

Bit No. 0: Flash memory error

Bit No. 1: EEPROM error

Bit No. 2: Supply error

Bit No. 3: Load cell excitation error

Bit No. 4: Synchronization problem

3.13.4.4 Output Buffer, Communication Block from PLC to PFEA112

This section specifies Boolean values in the output buffer communication block.

Data	Byte No.	Bit No.							
		7	6	5	4	3	2	1	0
Boolean out	01	No. 7	No. 6	No. 5	No. 4	No. 3	No. 2	No. 1	No. 0
	02	Spare for future use							

Bit No. 0: Zero set. Zero set is performed when the bit is changed from “0” to “1”.

3.14 Commissioning of Optional Units

3.14.1 Insulation Amplifier PXUB 201

The insulation amplifier is connected to the tension electronics voltage output.
S1 is normally set for voltage 1:1 ratio.

The output is selected to generate a voltage or current output by means of switch S1 and S2.

Slower response is selected by means of switch S2, position 3.

The switches are located inside the unit.

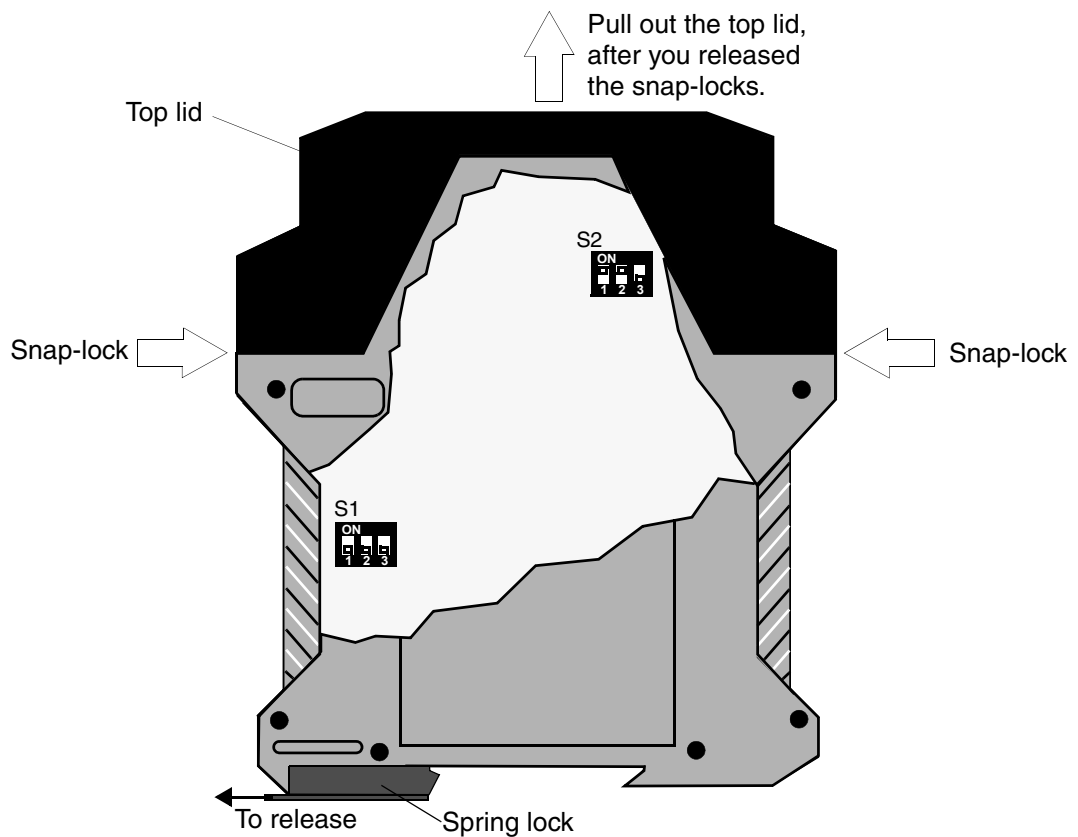


Figure 3-17. Insulation Amplifier PXUB 201

You must open the insulation amplifier to be able to set the switches S1 and S2.

1. Demount the insulation amplifier from the DIN rail.
Use a screw driver to unload the spring at the bottom of the insulation amplifier.
2. Press down the snap-locks on both sides of the insulation amplifier.
3. Pull the top lid open, until you see both the switches S1 and S2.
4. Set the switches S1 and S2.
5. Slide back the top lid to locked position.
6. Remount the insulation amplifier on the DIN rail.

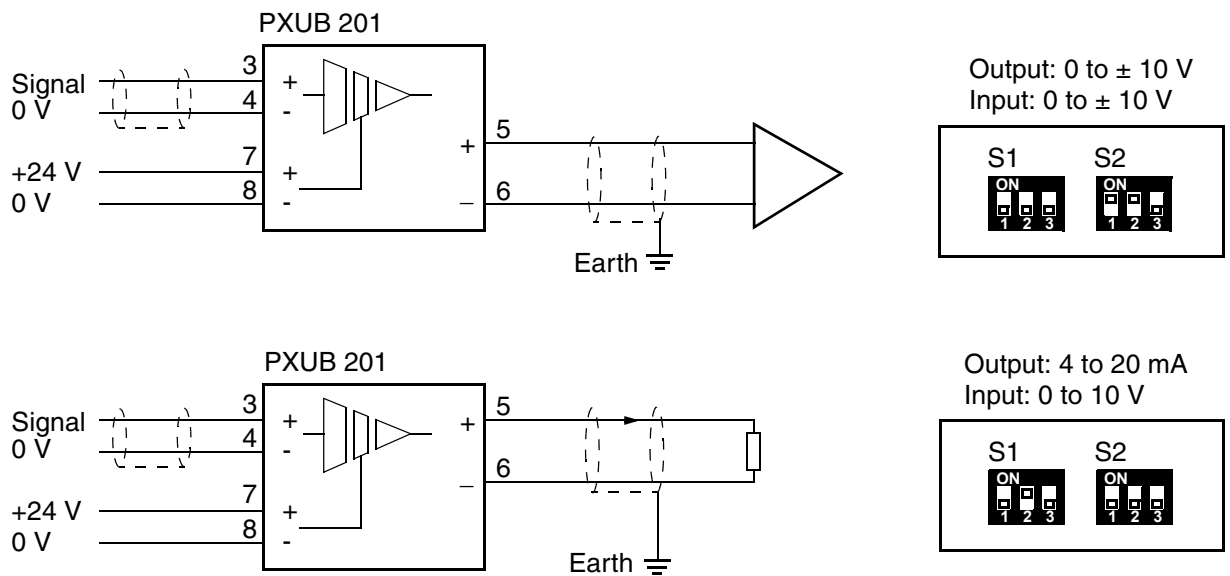


Figure 3-18. Typical Connection of the Insulation Amplifier

Table 3-5. Setting of Input and Output Range

Default	Range		S1			S2		
	Input	Output	1	2	3	1	2	3
×	0 to ± 10 V	0 to ± 10 V				×	×	
	0 to 5 V	4 to 20 mA	×					
	0 to 10 V	4 to 20 mA		×				
	0 to 5 V	0 to 20 mA	×	×				
	0 $\pm$ 10 V	0 $\pm$ 20 mA			×			

Table 3-6. Setting of Bandwidth

Default	Bandwidth	S2, position 3 ($\times$ = ON)
×	10 kHz	
	10 Hz	×

Chapter 4 Operation

4.1 About this Chapter

Your measurement system does not need any attention during normal operation. Measurement runs continuously as long as the system is switched on. However, you need to know how to start and shut down the system, see [Section 4.4 Start-up and shut-down](#).

4.2 Safety Instructions

Read and follow the safety instructions given in [Chapter 1 Introduction](#), before starting any operation work. However, local statutory regulations, if stricter, are to take precedence.

4.3 Operating Devices

The LED-indicators and operator keys are described in [Figure 4-1](#).

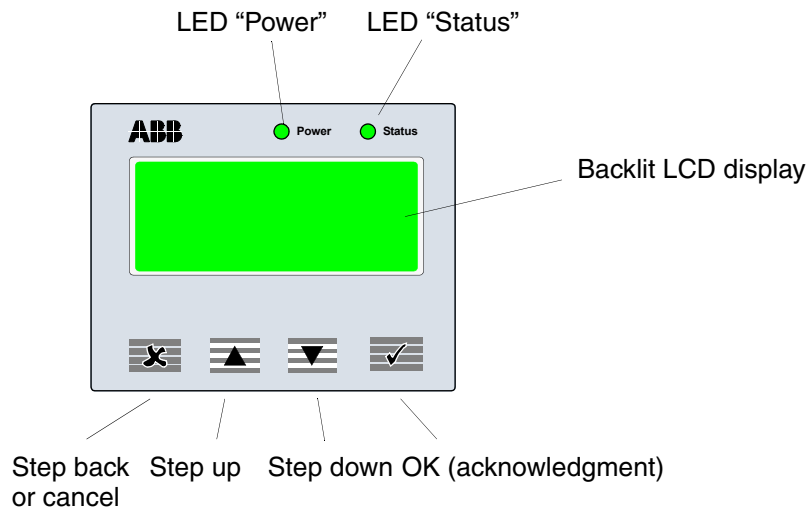


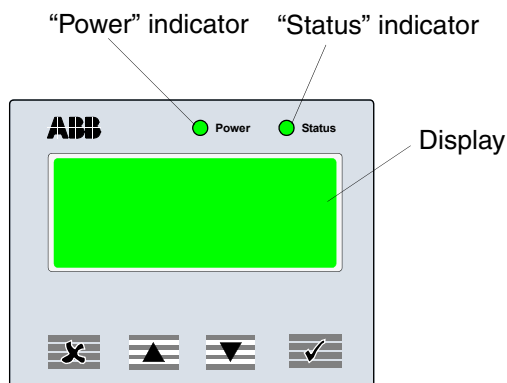
Figure 4-1. Operating Devices

4.4 Start-up and shut-down

4.4.1 Start-up

The tension electronics is started and shut down using an external ON/OFF switch (not supplied by ABB). During normal service no action from the operator is required.

1. Check that the main tension control machinery is ready for normal run.
2. Switch on the tension electronics by setting the external ON/OFF switch to position ON. For the IP 65-version (NEMA 4) also set the internal switch to "ON".
3. Check that:
 - the display is illuminated
 - the "Power" indicator is lit
 - the "Status" indicator is lit (green light). Red light indicates an error.



4.4.2 Shut-down

Shut-down the tension electronics by setting the external ON/OFF switch to position OFF.

4.5 Normal Run

The measurement equipment should be permanently on to achieve best measuring result. This allows the load cells and electronics to operate under even temperature conditions.

The measuring equipment is designed for continuous duty.

4.6 Operator Menus

This section describes the operator menus. The updating time for displayed values is 500 ms. Use up  and  to switch between menus.

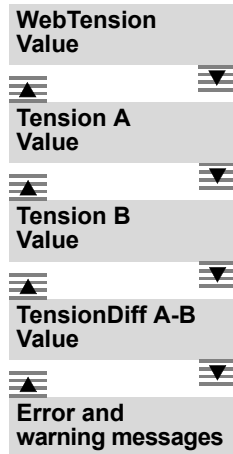


Figure 4-2. Operator Menus

4.6.1 Web Tension

4.6.1.1 Standard Roll (two load cells)

The following menus are available when a standard roll (two load cells) is connected to the tension electronics:

- **WebTension**
Shows the total web tension measured by load cell A and load cell B
- **Tension A**
Shows the part of web tension measured by load cell A
- **Tension B**
Shows the part of web tension measured by load cell B
- **TensionDiff A-B**
Shows the difference between Tension A and Tension B

4.6.1.2 Single Side Measurement (one load cell)

The following menu is shown when only one load cell (single side measurement) is connected to the tension electronics:

- **WebTension**
Web Tension is displayed for single side measurement (only load cell A connected).
Web tension is the tension measured by load cell A multiplied by 2.

4.6.2 Error and Warning Messages

An **ERROR** is something that causes the tension electronics to function incorrectly.

A **WARNING** is something that might affect the accuracy of the measurement.

When a warning or an error occurs, a warning or an error message is displayed on the Operator Panel and the “Status” indication turns from green to red.

When  is pressed the message disappears from the display.

If the problem that activated the warning or error message has disappeared the “Status” indication turns to green.

If the error or warning remains the “Status” indication is red. Use the  to step to the last menu, where you can read the error or warning message.

For handling Error and Warning messages, see [Chapter 6 Fault-tracing](#).

Chapter 5 Maintenance

5.1 About this Chapter

Under normal operating conditions, your system does not require any maintenance. However, we recommend you to perform regular checks. The following preventive measures may be taken depending on the type of environment in which your system operates.

5.2 Preventive Maintenance

Unit	Measures
Load cells	Protect the load cells from prolonged contact with corrosive elements.
	Check the fixing screws and re-tighten if necessary.
	Check the gaps between load cell and adapter plates to ensure that they are not clogged with dirt, which can cause a shunt force into the load cell.
	Clean the gaps with compressed air if necessary.
Tension Electronics	Check that the circuit boards are properly secured and that cables or wires are not damaged.
	Check that all terminal screws and cable glands are properly tightened.
Connection cables	Check that the connection cables between the load cells and the tension electronics are not damaged.
	

Chapter 6 Fault-tracing

6.1 About this Chapter

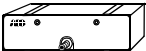
During the working life of your measurement system, events may occur that disturb it and your process. These disturbances may appear in many different ways and the reason for the fault can be difficult to find. However, disturbances similar in character can be grouped together and usually they have the same or similar sources of error.

The fault-tracing instructions in this chapter will help you to quickly find and correct the most common faults.

6.2 Safety Instructions

Read and follow the safety instructions given in [Chapter 1 Introduction](#) when tracing faults. However, local statutory regulations, if stricter, are to take precedence.

6.3 Interchangeability

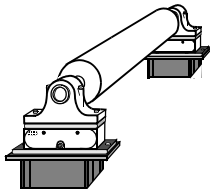
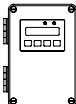
Unit	Measures
Tension Electronics 	The tension electronics PFEA111/112 is interchangeable with a tension electronics of the same type. A new setup is required.
Load cells 	The load cells are directly interchangeable with other load cells of the same type. Zero setting PFEA111/112 and resetting “Maximum Load A” or “Maximum Load B” are required after a load cell has been replaced.

6.4 Necessary Equipment and Documentation

The following items are required to perform fault-tracing and repairs:

- Cable diagrams, see Appendix (B, C, D or E for) for installed load cell type
- Service tools
- Torque wrench
- Multimeter

6.5 Fault-tracing Procedure

Faults in the...	Fault symptoms
Mechanical installation 	Faults in the mechanical installation usually manifest themselves as an unstable zero point or incorrect sensitivity. If a fault is associated with a process parameter, such as the temperature, or can be linked with a particular operation, the fault is likely to emanate from the mechanical part of the installation.
Load cells 	Calibration data for a load cell do not change gradually. A load cell, depending on its size and type, can withstand up to five times⁽¹⁾ the nominal load in the measuring direction. An event in the process line, such as web breakage, may cause an overloading large enough to alter load cell data. Depending on the amount of overload, it may be sufficient to zero set.
Cabling 	Problems such as malfunctions or unstable zero point may arise from faulty cables or wiring. Proximity to noisy cables may cause interference problems. Incorrect installation, such as cable cores connected asymmetrically or screens earthed at both ends instead of only at one end, may show up as an unstable zero point. If the polarity of load cell signals is not correct, the cabling has to be checked.
Tension Electronics 	Intermittent loss of a function is usually due to a fault in the tension electronics. Instability problems seldom originate from the tension electronics. Faults in devices connected to the tension electronics may affect its operation.

(1) Read more about the overload capacity for your load cell type in Appendix B, C, D or E.

6.6 Error and Warning Messages in PFEA111/112

An **ERROR** is something that causes the electronics to function incorrectly.

A **WARNING** is something that might affect the accuracy of the measurement.

When a warning or an error occurs, a warning or an error message is displayed on the Operator Panel and the “Status” indication turns from green to red.

When  is pressed the message disappears from the display.

If the problem that activated the warning or error message has disappeared the “Status” indication turns to green.

If the error or warning remains the “Status” indication is red. Use the  to step to the last operator menu, where you can read the error or warning message.

6.6.1 Error Messages

The following errors can be detected:

- Flash (memory) error
- EEPROM (memory) error
- Supply error
- Load cell excitation error

See [Section 6.8 Warnings and Errors Detected by the Tension Electronics](#).

6.6.2 Warning Messages

The following warnings can be detected:

- Profibus communication problem
- Synchronization problem

See [Section 6.8 Warnings and Errors Detected by the Tension Electronics](#).

6.7 Fault Symptoms and Measures

General remark:

If the free (unscreened) cable length exceeds 0.1 m (4 in.) the individual pairs of power and signal conductors must be twisted.

Free length exceeding 0.1 m can cause unstable zero point or incorrect absolute measurement value.

Table 6-1. Fault Symptoms and Measures

Fault symptom	Measures
Noisy signals	- Check that cable shields are connected to earth according to cable diagram. - Proximity to noisy cables can cause interference problems.
Unstable zero point	- Check that cable screens are not connected in both ends. - Check that cable between load cell and electronic has diagonal pairs, one pair for signal circuit and one pair for excitation circuit, see Figure 2-2. - Check, if a junction box is installed, that load cell signal and load cell excitation between junction box and electronics are routed in separate cables. - Check, if two or more IP 20 units are mounted close to each other in the same cabinet, that they are synchronized (cable for synchronizing the units, see cable diagram and Section 2.4.1.3 Synchronization).
Display and LED indicators are not illuminated	If the Operator panel display is not illuminated and the "Power" and "Status" indications on the Operator Panel are "OFF", check the following: - Check that the cables are correctly connected to the electronics power supply. - Check that the power supply connected to the electronics is correct. - Check that the power switch is "ON" (inside the IP 65-version (NEMA 4) enclosure). - If power supply SD 82x is used as power supply for PFEA111/112 IP 20-version (unsealed), check that the mains voltage selection switch on SD 82x is set to correct position (115 or 230) corresponding to the connected mains supply voltage. - Further tests are described in Section 6.8.1.3 Supply Error.

Table 6-1. Fault Symptoms and Measures

Fault symptom	Measures
No signal when load is applied	1. Check that cables to the electronics are correctly connected. 2. Check that the load cells are connected with correct polarity. If not, the load cell signals cancel out each other. This is displayed on the operator panel as described below: a. The sum signal (A+B) is low b. The difference signal (A-B) is high c. The output signal of the individual load cells have opposite signs (polarity) when a force is applied on the middle of the roll. To check load cell signal polarity, see Section 3.9 Checking Load Cell Signal Polarity. For connecting the load cells to give positive signals for increased web tension, see cable diagram for the installed load cell type. 3. Switch off the tension electronics and measure the cable resistance in the load cell signal circuit between terminals X1:5 and X1:6 and between X1:9 and X1:10. a. The resistance is > 25 ohm: Check cabling and load cells. b. The resistance is < 25 ohm: Check the mechanics.

6.8 Warnings and Errors Detected by the Tension Electronics

6.8.1 Errors

6.8.1.1 Flash Memory Error

- Replace PFEA111/112.

6.8.1.2 EEPROM Memory Error

- Replace PFEA111/112.

6.8.1.3 Supply Error

IP 20-version (unsealed):

When the PFEA111/112 is connected to the 24 VDC power supply the voltage between terminals X1:1 and X1:2 should be 18 - 36 V.

- If the voltage is lower than 18 V:
 - Check the power supplies rating. Rating should be 18-36 V DC.
 - Check that the power supply has sufficient capacity. See power requirements in [Section 2.6.2 Power Supply Unit](#).
- If the power supply has sufficient capacity, check cabling and cable resistance between power supply and PFEA111/112.
- If power supply and cabling are correct, the tension electronics is probably defect.
Replace PFEA111/112.

IP 65-version (NEMA 4):

- Check the mains voltage connected to terminals X9:1 and X9:2.
Mains voltage must be:
85 - 264 V AC (100 V -15% to 240 V +10%)
Frequency range: 45 - 65 Hz

6.8.1.4 Load Cell Excitation Error

- Check that cables are correctly connected to the electronics.
- If single side measurement is used and only load cell A is connected to the electronics, check that a shorting wire is connected between terminals X1:7 and X1:8.
 - Switch off the tension electronics and measure the resistance between terminals X1:3 and X1:8.

The resistance is > 8 ohm:

Check that the total cable resistance between electronics and load cells does not exceed 5 ohm. If cable resistance does not exceed 5 ohm, check cabling and load cells.

The resistance is < 7 ohm:

If cabling is correct, the electronics is probably defect.

Replace PFEA111/112.

6.8.2 Warnings

6.8.2.1 Profibus Communication Problem

Check:

- that the bus is correctly terminated.
- the Profibus address.
- cabling and connectors.

6.8.2.2 Synchronization Problem

Check the cabling and shield.

If the cabling is correct the tension electronics is probably defect.

Replace PFEA111/112.

6.8.3 Changing to Single Side Measurement If One Load Cell Is Faulty

If one load cell is faulty it is possible to change from Standard Roll to Single Side Measurement.

For load cell connections, please refer to cable diagrams in Appendix B, C, D or E for the load cell type used in the installation.

Depending on which load cell that is faulty, do the following:

Load cell A is faulty:

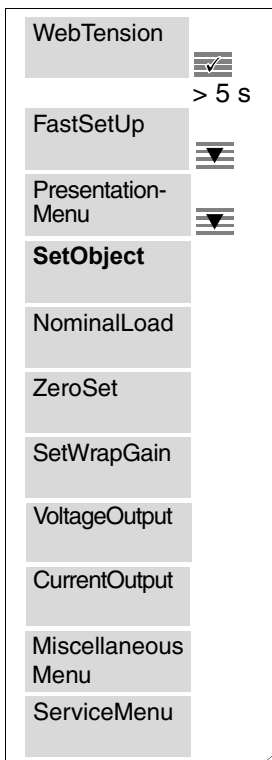
1. Disconnect load cell A from the Tension Electronics.
2. Move connections for load cell B to the terminals where load cell A was connected.
3. Connect a shorting wire for the load cell excitation circuit between X1:7 and X1:8.

Load cell B is faulty:

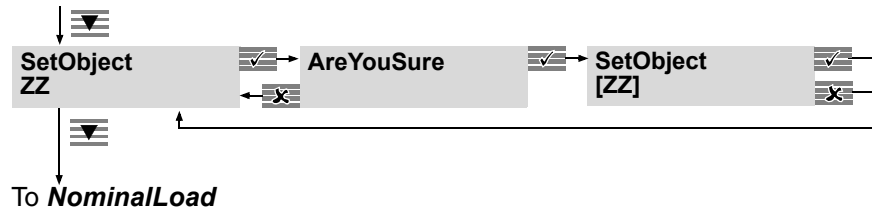
1. Disconnect load cell B from the Tension Electronics.
2. Connect a shorting wire for the load cell excitation circuit between X1:7 and X1:8.

After having changed the load cell connections, one parameter setting in the Tension Electronics must be changed.

Use the menu below to change from *StandardRoll* to *SingleSide*.

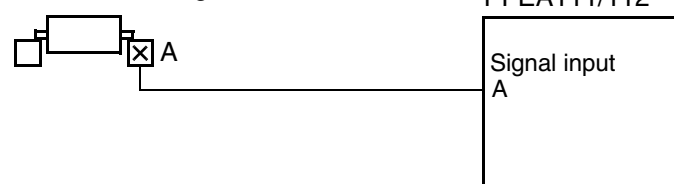


From **PresentationMenu**



Single side measurement

One load cell signal



6.9 Changing Load Cells

1. Before starting work, read the safety instructions stated in [Chapter 1 Introduction](#).
2. For load cells equipped with an extension cable and connector:
Disconnect the connection cable from the load cell and protect the connection cable from dirt and damage.

For load cells equipped with a fixed cable:
Disconnect the load cell connection in the tension electronics or junction box and protect the loose cable ends from dirt and damage.
3. Clean the old load cell before it is unfastened and removed.
4. Unfasten and remove the old load cell.
5. Unfasten and remove the adapter plates from the old load cell.
6. Clean the support structure, adapter plates and other mounting surfaces.
7. For mounting instructions for the new load cell, see:
 - [Appendix B PFCL 301E - Designing the Load Cell Installation](#)
 - [Appendix C PFTL 301E - Designing the Load Cell Installation](#)
 - [Appendix D PFRL 101 - Designing the Load Cell Installation](#)
 - [Appendix E PFTL 101 - Designing the Load Cell Installation](#)
8. Set the zero point, see [Section 3.12.4 Zero Set](#).

Appendix A Technical Data for Tension Electronics PFEA111/112

A.1 About this Appendix

This appendix comprises technical data for tension electronics PFEA111/112.

Data for the load cells are given in:

- [Appendix B PFCL 301E - Designing the Load Cell Installation](#)
- [Appendix C PFTL 301E - Designing the Load Cell Installation](#)
- [Appendix D PFRL 101 - Designing the Load Cell Installation](#)
- [Appendix E PFTL 101 - Designing the Load Cell Installation](#)

The definitions used in the load cell appendices are explained in [Section A.6, Profibus DP - GSD File for PFEA112](#).

A.2 Definitions used in the Web Tension Systems

Table A-1. Definitions

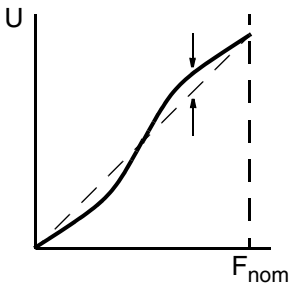
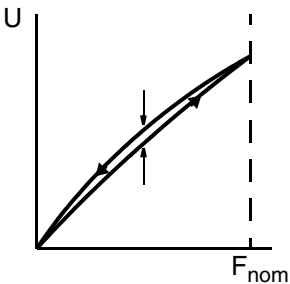
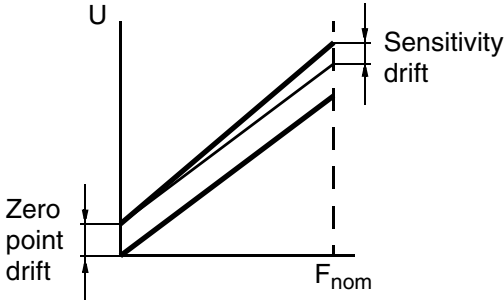
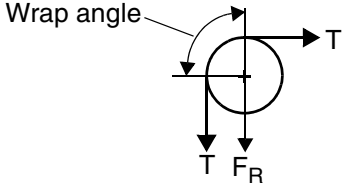
Nominal load, F_{nom}, is the load for which the load cell is dimensioned and calibrated, i.e. the sum of the stationary load and the maximum measured load in the measuring direction.	<u>Linearity deviation</u> 
F_{ext} = Extended range. Between F_{nom} and F_{ext} some decline in measurement accuracy may be experienced.	
Sensitivity is defined as the difference in output signal between nominal load and no load.	
Accuracy class is defined as the maximum deviation, and is expressed as a percentage of the sensitivity at nominal load. This includes linearity deviation, hysteresis and repeatability error.	<u>Hysteresis</u> 
Linearity deviation is the maximum deviation from a straight line drawn between the output values of zero and nominal load, related to the nominal load.	
Hysteresis is the maximum deviation of the output signal at the same load during a cycle from zero to nominal load and back to zero, related to the sensitivity at nominal load. The hysteresis is proportional to the cycle.	
Repeatability error is defined as the maximum deviation between repeated readings under identical conditions. It is expressed as a percentage of the sensitivity at nominal load.	<u>Temperature dependence</u> 
Temperature dependence is the drift in %/K related to the sensitivity at nominal load.	
Zero point drift is defined as the drift in the output signal when there is no load on the load cell.	
Sensitivity drift is defined as the drift in the output signal at nominal load, excluding the zero point drift.	

Table A-1. Definitions

T = Web tension.	Example:
Tare = Force of tare (weight of roll and bearing arrangement mounted on the load cells)	
F_R = Measured force (force component of the web tension in the measuring direction of the load cell).	
F_{RT} = Applied force component of tare in the measuring direction of the load cell.	F_R = T
F_{Rtot} = Total applied force in the measuring direction of the load cell.	Wrap gain = $\frac{T}{F_R}$
Wrap Gain = The ratio between web tension, T , and measured force, F_R .	Wrap gain = $\frac{T}{T} = 1.00$
	Wrap gain = 1.00

A.2.1 Coordinate System

A coordinate system is defined for the load cell. This is used in force calculations to derive force components in the load cell principal directions.

Where direction designations R, V and A are recognized as suffixes for force components, F, this represents the force component in the respective direction. The suffix R may be omitted, when measuring direction is implied by the context.

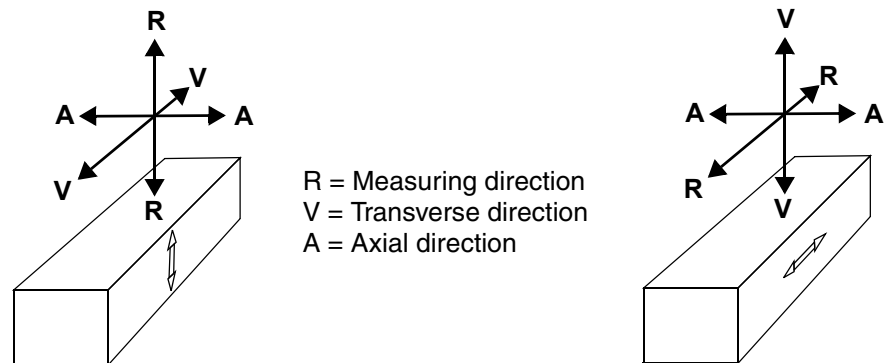


Figure A-1. Coordinate System Defining Directions used in Force Calculations

A.3 Technical Data

Table A-2. Data for Supply Voltage

	Data	Comments
Supply voltage		
IP 20-unit (unsealed)	24 V DC	18 - 36 V DC
IP 65-unit (NEMA 4)	24 V DC	18 - 36 V DC
	85 - 264 V AC	100 V -15% to 240 V +10%
Mains frequency	45 - 65 Hz	100 - 240 V AC, 0.2 - 0.1 A
Power consumption	7.5 W (24 V)	
Fuse		
IP 20-unit (unsealed)	Automatic reset	
IP 65-unit (NEMA 4)	Slow blow, 2 A, 250 V	

Table A-3. Data for Load Cell Excitation

	Data	Comments
Current	0.5 A rms, 330 Hz	Regulated
Max. load	Two load cells + max. 5 Ω cable resistance (1 μ F cable capacitance).	Load cells of type: PFCL 301E, PFTL 301E, PFRL 101 and PFTL 101.

Table A-4. Data for Load Cell Inputs

	Data	Comments
Number of inputs	2	
Input impedance	10 kohm	

Table A-5. Data for Signal Outputs

	Data	Comments
Voltage output	0 - 10 V	Range -5 to +11 V
Max. load	5 mA	
Ripple	<10 mV _{p-p}	Wrap gain = 1
Step response time	15 ms	
Band width	35 Hz	
Current output	4 - 20 mA	Range 0 to 21 mA
Max. load	550 Ω	
Step response time	15 ms	
Band width	35 Hz	
Extra filtering for voltage and current output		
Step response time	75 ms 250 ms 750 ms	
Cut-off frequency	5 Hz 1.5 Hz 0.5 Hz	
Wrap gain adjustment	0.5 - 10	

Table A-6. Measurement Ranges for the Tension Electronics

Type	Range ⁽¹⁾
Zero Setting Range	$\pm 2.0 \times F_{nom}$
Dynamic Measurement Range (including zero set)	$-2.5 \times F_{nom}$ to $+3.5 \times F_{nom}$

(1) F_{nom} = Load cell nominal load

Table A-7. Communication PFEA112

	Data	Comments
Profibus	1	12 Mbit
Communication protocol	Profibus DP slave	According to EN 50 170
Transfer speed	Max. 12 Mbits / s	
Address range	0 - 125	

Table A-8. Environmental Data

	Data	Comments
Temperature dependence		
Zero point drift	< 50 ppm/K (28 ppm/°F)	
Sensitivity drift	< 75 ppm/K (42 ppm/°F)	
Operating temperature		
+5 to +55 °C (32 - 131 °F)		
Outside the IP 20-version (unsealed) and IP 65-version (NEMA 4)		
Non-operating temperature		
-40 to +70 °C (-40 - 158 °F)		
Degree of protection		
DIN-rail version	IP 20 (unsealed)	
Wall mounting unit	IP 65 (NEMA 4)	According to EN 60 529

Table A-9. Dimensions

	Data	Comments
Dimensions		
IP 20-version (unsealed)	86 x 136 x 58	Width x Height x Depth
IP 65-version (NEMA 4)	120 x 180 x 100	Width x Height x Depth
Weight		
IP 20-version (unsealed)	0.3 kg	
IP 65-version (NEMA 4)	1.9 kg	

A.4 Factory Default Settings

Table A-10. Factory Default Settings

	PFEA111	PFEA112
Display language	English	English
Display unit	N	N
Load cells per roll	2	2
Object type	Standard roll	Standard roll
Load cell nominal load	1.0 kN 225 lbs	1.0 kN 225 lbs
Wrap gain	1	1
Current output		
Step response time	15 ms	15 ms
High tension	2000 N	2000 N
High output	20.00 mA	20.00 mA
Low tension	0 N	0 N
Low output	4.00 mA	4.00 mA
High limit	21.00 mA	21.00 mA
Low limit	0.00 mA	0.00 mA
Voltage output		
Step response time	15 ms	15 ms
High tension	2000 N	2000 N
Low tension	0 N	0 N
High output	+10.00 V	+10.00 V
Low output	0.00 V	0.00 V
High limit	+11.00 V	+11.00 V
Low limit	-5.00 V	-5.00 V
Profibus	-	Off
Address	-	126

A.5 Optional Units

A.5.1 Insulation Amplifier PXUB 201

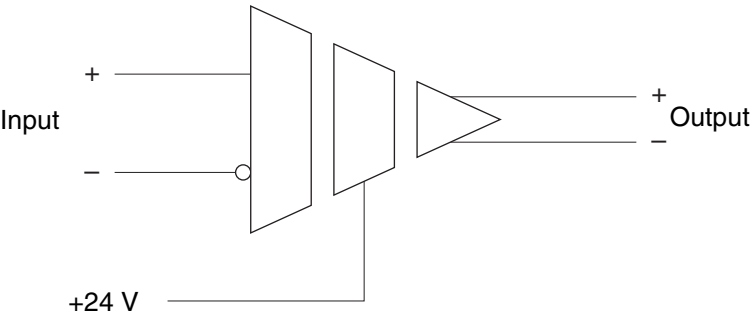


Figure 1-2. Insulation Amplifier PXUB 201

Table 1-11. Data for Insulation Amplifier PXUB 201

Type	Data	
Power supply	20 - 253 V AC/DC	
	AC: 48 - 62 Hz, 2 VA	
	DC: 1 W	
Current consumption	10 mA + external load, at 24 V	
Signal range	Input	Output
	0 ± 10 V	0 ± 10 V
	0 - 10 V	4 - 20 mA
	0 - 5 V	4 - 20 mA
	0 ± 10 V	0 ± 20 mA
	0 - 5 V	0 - 20 mA
Input resistance	1 MΩ at 10 V input	
	500 kΩ at 5 V input	
Max. load	10 mA for voltage output	
	500 Ω for current output	
Rise time	50 μs or 50 ms, selectable	
Ripple	10 mV _{p-p}	
Bandwidth (-3 dB)	10 kHz or 10 Hz	
Rated insulation voltage	600 V, basic insulation	
Insulation test voltage	4 kV	
Dimensions (l × w × d)	99 × 12.5 × 111 mm	
Weight	150 g	

A.5.2 Power Supply Unit

Table A-12. Mains Supply Voltage

	Data	Remarks
Mains supply voltage	115 V AC (85 - 132 V), 100 V -15% to 120 V + 10%	(Mains voltage selection switch in position 115)
	230 V AC (176 - 264 V), 207 V -15% to 240 V + 10%	(Mains voltage selection switch in position 230)

Table A-13. Power Supply Unit

Unit	Dimensions (l x w x d)	Weight
SD821	125 x 50 x 103 mm	0.5 kg
SD822	125 x 65 x 103 mm	0.6 kg
SD823	125 x 120 x 103 mm	1.0 kg

A.5.3 Junction Box PFXC 141

Degree of protection	Dimensions (l x w x d)	Weight
IP 65 (NEMA 4)	220 x 120 x 80 mm	2.0 kg

A.6 Profibus DP - GSD File for PFEA112

```
===== GSD file:ABB_0716.GSD =====  
;  
;  
; DEVICE NAME:          Tension Electronics PFEA112  
; AUTHOR:               M.Sollander  
; REVISION DATE:        January 27, 2003  
;  
=====
```

#Profibus_DP

GSD_Revision = 2

```
===== PRODUCT SPECIFICATION =====
```

Vendor_Name = "ABB Automation Techn. Products"
Model_Name = "Tension Electronics PFEA112"
Ident_Number = 0x0716
Revision = "2.0"
Hardware_Release = "1.0"
Software_Release = "1.0"

```
===== OVERALL PROFIBUS SPECIFICATIONS =====
```

FMS_supp = 0
Protocol_Ident = 0
Station_Type = 0
Slave_Family = 0

```
===== HARDWARE CONFIGURATION =====
```

Implementation_type = "SPC3"
Redundancy = 0
Repeater_Ctrl_Sig = 0
24V_Pins = 0

;===== PROTOCOL CONFIGURATION =====

Set_Slave_Add_supp	= 0
Auto_Baud_supp	= 1
Min_Slave_Intervall	= 1
Freeze_Mode_supp	= 1
Sync_Mode_supp	= 1
Fail_Safe	= 0

;===== SUPPORTED BAUDRATES =====

9.6_supp	= 1
19.2_supp	= 1
45.45_supp	= 1
93.75_supp	= 1
187.5_supp	= 1
500_supp	= 1
1.5M_supp	= 1
3M_supp	= 1
6M_supp	= 1
12M_supp	= 1
MaxTsdrr_9.6	= 60
MaxTsdrr_19.2	= 60
MaxTsdrr_45.45	= 60
MaxTsdrr_93.75	= 60
MaxTsdrr_187.5	= 60
MaxTsdrr_500	= 100
MaxTsdrr_1.5M	= 150
MaxTsdrr_3M	= 250
MaxTsdrr_6M	= 450
MaxTsdrr_12M	= 800

;===== DIAGNOSTIC DEFINITIONS =====

Max_Diag_Data_Len = 6

;===== PARAMETER DEFINITIONS =====

User_Prm_Data_Len = 3

User_Prm_Data = 0, 0, 0

;===== MODULE DEFINITIONS =====

Modular_Station = 0

Module = "PFEA112" 0x51,0x11,0x21

EndModule

;=====

Appendix B PFCL 301E - Designing the Load Cell Installation

B.1 About This Appendix

This appendix describes the procedure for designing the load cell installation.

The following sections are included:

- Basic application considerations
- Designing the load cell installation (step-by-step guide)
- Installation requirements
- Force and wrap gain calculation
 - Horizontal mounting
 - Inclined mounting
 - Single side measurement
- Mounting the load cells
- Technical data
- Drawings
 - Cable diagram(s)
 - Mounting instruction for load cell extension cable
 - Dimension drawing
 - Assembly drawing

B.2 Basic Application Considerations

Each application has its own individual demands that have to be considered; though a few basic considerations tend to repeat themselves.

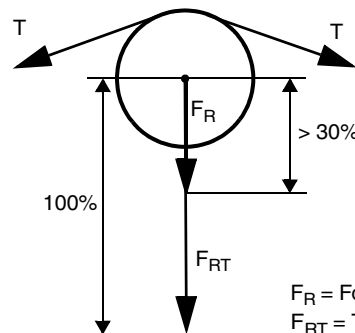
- What type of process is involved (papermaking, converting, etc.)?
Is the environment demanding (temperature, chemicals, etc.)?
- What is the tension measurement purpose; indication or closed loop control?
Are there any specific accuracy demands involved?
- What is the machine design like? Is there the possibility to modify the design,
in order to fit the most suitable load cell, or is the machine design fixed?
- What are the forces acting on the roll like (size and direction)?
Can they be altered by redesign?

If these questions are dealt with thoroughly, the installation is very likely to be successful. However, the extent to which measurement accuracy is needed, defines the requirements when designing a load cell installation.

B.3 Step-by-Step Guide for Designing the Load Cell Installation

The procedure below defines the main considerations involved in designing a load cell installation.

1. Check load cell data so that environmental demands are met.
2. Calculate forces; vertical, horizontal and axial (cross directional).
3. Size and orientate load cell so that the guidelines below are met:
 - a. Try to achieve a measured value no less than 10% of web tension, in the load cell measurement direction!
 - b. Select load cell size so that it is loaded as close as possible to its nominal load! Do not dimension Force component of Tension in measuring direction, F_R , for less than 10% of the load cell nominal load!
 - c. If the span between maximum and minimum tension in the process is large, select load cell so that the maximum tension will be in the load cell extended range (when applicable)!
 - d. The measured force component of web tension is recommended to be at least 30% of tare force component (roll weight) acting in load cell measurement direction. The reason for this recommendation is load cell signal stability, especially when the system operates in a large temperature span.
This means that if $F_{RT} < 1/3$ of F_{nom} , F_R should be at least 10% of F_{nom} .
For larger F_{RT} , lowest F_R is recommended to be at least 30% of F_{RT} .



Rule 1: If $F_{RT} < 1/3$ of F_{nom}
 F_R should be at least 10% of F_{nom}

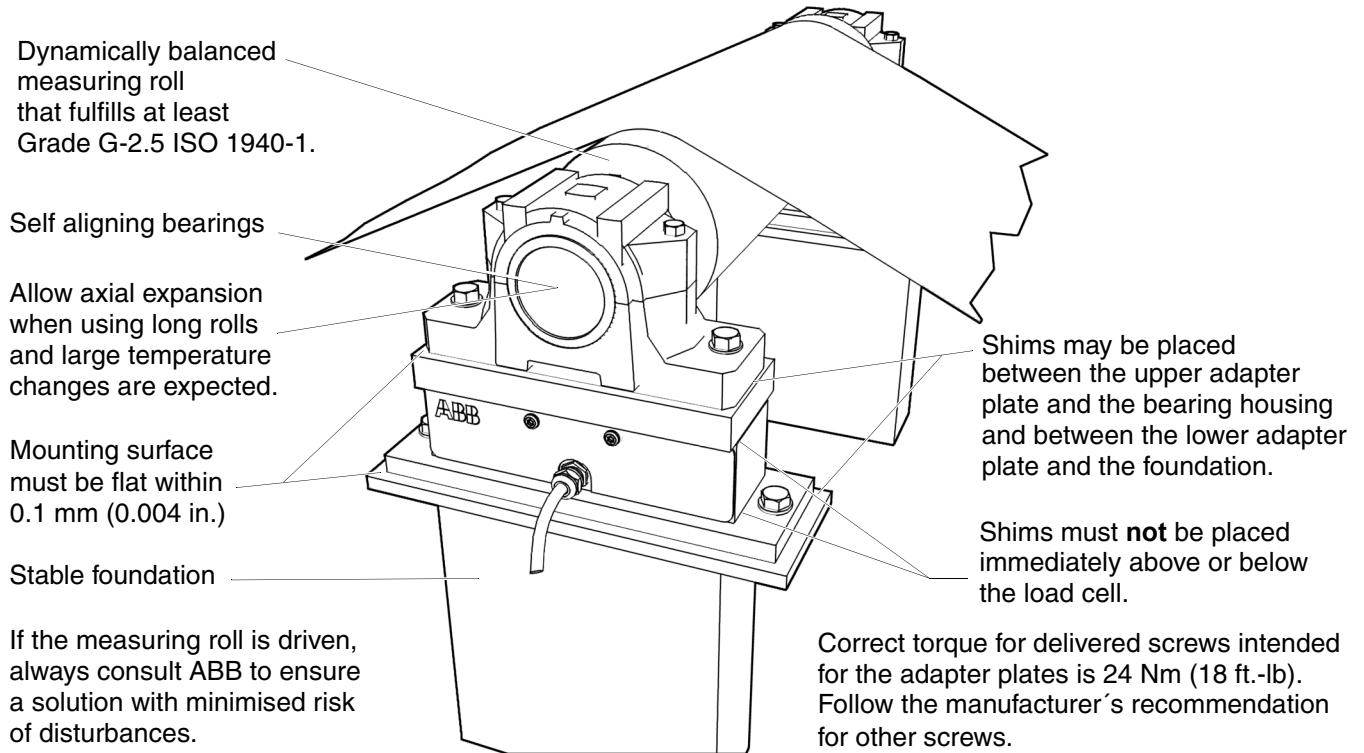
Rule 2: If $F_{RT} > 1/3$ of F_{nom}
 F_R is recommended to be at least 30% of F_{RT}

F_R = Force component of web tension in measuring direction
 F_{RT} = Tare force in measurement direction

- e. Check load cell data so that limits for building height, transverse and axial forces are not exceeded.
4. Design base frame and/or adapter plates.

B.4 Installation Requirements

To achieve the specified accuracy, the best possible reliability and long-term stability, install the load cells in accordance with the requirements below.



Alignment of the load cells

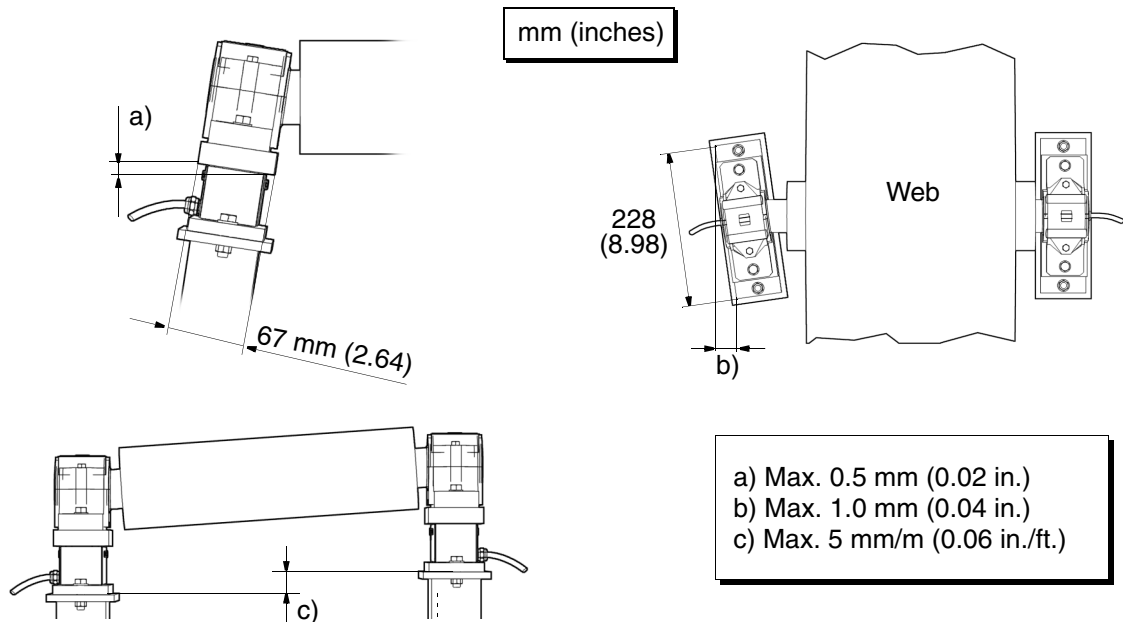
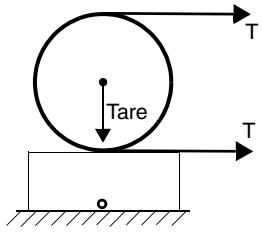
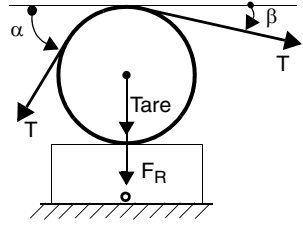
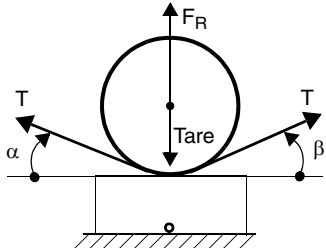


Figure B-1. Installation Requirements

B.5 Mounting Alternatives, Calculating Force and Calculating Wrap Gain

B.5.1 Horizontal Mounting

PFCL 301E  No vertical web tension force applied to the load cell.	In most cases, horizontal mounting is the most obvious and simplest solution. The load cell should thus be mounted horizontally when possible. However, should the machine design require inclined mounting of the load cell or should the web path not give a sufficient vertical force, see figure, the inclined mounting is permitted and the calculations are somewhat more complex, (see Appendix B.5.2).
 $F_R = T \times (\sin \alpha + \sin \beta)$ $F_{RT} = \text{Tare}$ $F_{Rtot} = F_R + F_{RT} = T \times (\sin \alpha + \sin \beta) + \text{Tare}$ $T (\text{Tension}) = \text{Wrap gain} \times F_R$ $\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T(\sin \alpha + \sin \beta)}$ $\text{Wrap gain} = \frac{1}{\sin \alpha + \sin \beta}$	The PFCL 301E load cell measures the vertical forces applied to its top surface. The horizontal applied forces are not measured and do not influence the vertical measurement. There are two sources of vertical forces; the forces from the web tension and the tare weight of the roll. Divide the total vertical force F_{Rtot} by two to get the required capacity of each load cell. Do not oversize an ABB load cell for overload purposes as the load cell has sufficient overload capacity.
 $F_R = T \times (\sin \alpha + \sin \beta)$ $F_{RT} = \text{Tare}$ $F_{Rtot} = F_{RT} - F_R = \text{Tare} - T \times (\sin \alpha + \sin \beta)$ $T (\text{Tension}) = \text{Wrap gain} \times F_R$ $\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T(\sin \alpha + \sin \beta)}$ $\text{Wrap gain} = \frac{1}{\sin \alpha + \sin \beta}$	A PFCL 301E load cell can measure tension as well as compression. If the $T (\sin \alpha + \sin \beta)$ is larger than the tare weight, the load cell will be in tension. To get the capacity of each load cell: 1. Divide $(F_R - \text{Tare})$ by two if F_R is greater than or equal to $(\text{Tare} \times \text{two})$. 2. Divide Tare by two if F_R is smaller than $(\text{Tare} \times \text{two})$.

B.5.2 Inclined Mounting

PFCL 301E

$$F_R = T \times [(\sin(\alpha - \gamma)) + \sin(\beta + \gamma)]$$

$$F_{RT} = Tare \times \cos \gamma$$

$$F_{Rtot} = F_R + F_{RT} =$$

$$T \times [(\sin(\alpha - \gamma)) + \sin(\beta + \gamma)] + Tare \times \cos \gamma$$

$$T \text{ (Tension)} = \text{Wrap gain} \times F_R$$

$$\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T[(\sin(\alpha - \gamma)) + \sin(\beta + \gamma)]}$$

$$\text{Wrap gain} = \frac{1}{\sin(\alpha - \gamma) + \sin(\beta + \gamma)}$$

Sometimes it is necessary to mount the load cell on an incline due to mechanical design constraints of the machine or the need to have a proper force component applied to the load cell.

In this case, the incline angle modifies the tare load and the force components as shown.

B.6 Force Calculation for Measurement with a Single Load Cell

In some cases, it is sufficient to measure the tension with only a single load cell mounted at one end of the roll. The roll should nevertheless be supported at both ends.

B.6.1 The Most Common and Simple Solution

The most obvious and simple solution is horizontal mounting with the web evenly distributed and centered on the roll.

As long as the roll is supported at both ends, the same calculations given in [Section B.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#) are valid.

NOTE

The accuracy of a single load cell measurement is highly dependent on how well the center of force can be determined. Since the cross-directional stress distribution generally is somewhat uneven, this is not easily done. The load cell will, however, produce a stable and repeatable measurement.

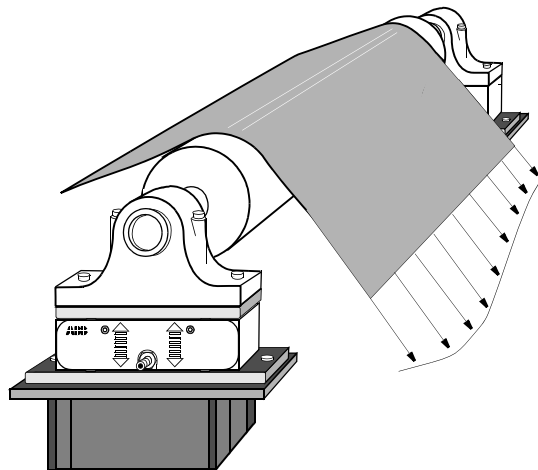
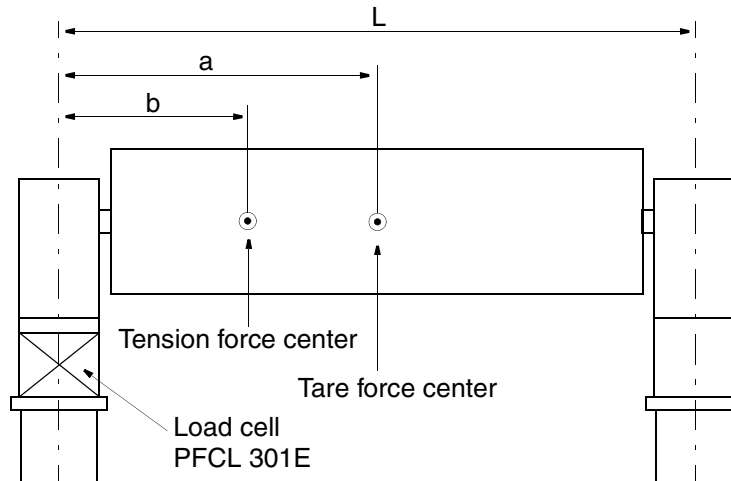


Figure B-2. Cross-directional stress distribution

B.6.2 Force Calculation when the Web is not Centered on the Roll

Use the calculations below for horizontal and inclined mounting when the web is not centered on the roll.

The applied force at the load cell will be proportional to the distance between the tension force center and the load cell centerline.



Calculation procedure:

1. Horizontal or inclined mounting?
2. Calculate F_R and F_{RT} , see [Section B.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#)
3. Use the following equations:

$$F_R \text{ for single load cell} = F_R \times \frac{L-b}{L}$$

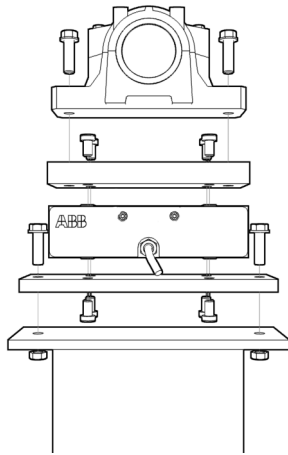
$$F_{RT} \text{ for single load cell} = F_{RT} \times \frac{L-a}{L}$$

$$F_{R_{tot}} \text{ for single load cell} = F_R \text{ for single load cell} + F_{RT} \text{ for single load cell}$$

where:

- L = Distance between load cell centerline and the opposite bearing centerline
 a = Distance between tare force center and load cell centerline
 b = Distance between tension force center and load cell centerline

B.7 Mounting the Load Cells

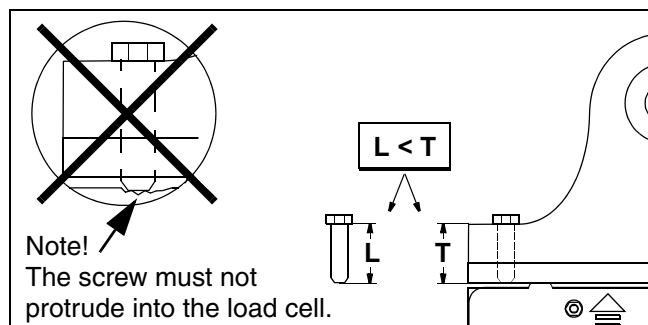


The instructions below apply to a typical mounting arrangement. Variations may be allowed, provided that they comply with the requirements in [Appendix B.4](#).

1. Clean the foundation and other mounting surfaces.
2. Fit the lower adapter plate to the load cell.
Tighten the screws (included in delivery) with a torque wrench to 24 Nm (18 ft.-lb).
3. Fit the load cell and the lower adapter plate to the foundation,
but do not fully tighten the screws.
4. Fit the upper adapter plate to the load cell.
Tighten the screws (included in delivery) with a torque wrench to 24 Nm (18 ft.-lb).
5. Fit the bearing housing and the roll to the upper adapter plate,
but do not fully tighten the screws.

CAUTION

When mounting bearings or other adjacent details to adapter plates, screws must not protrude into the load cell. The load cell can be damaged by an excessive applied force.



6. Adjust the load cells according to the installation requirements.
Tighten the foundation screws.
7. Adjust the roll according to the installation requirements.
Tighten the screws in the upper adapter plate.

B.7.1 Routing the Load Cell Cable

The cable must be supported with clamps and routed to prevent force shunting through the cable.

B.7.2 Connecting the Load Cell Extension Cable

See [Section B.10, Mounting Instruction, Cable Connector, 3BSE019064, Rev. A](#).

B.8 Technical Data

PFCL 301E				Unit
Nominal load				
Nominal load in measuring direction, F_{nom}	0.2 (45)	0.5 (112)	1.0 (225)	kN (lbs)
Permitted transverse load within the accuracy, F_{Vnom} For h = 135 mm (5.3 inch)	0.05 (11)	0.125 (28)	0.25 (56)	
Permitted axial load within the accuracy, F_{Anom} For h = 135 mm (5.3 inch)	0.05 (11)	0.125 (28)	0.25 (56)	
Extended load in measuring direction with accuracy class, compressive force $\pm 2\%$, F_{ext}	0.3 (67)	0.75 (169)	1.5 (337)	
Overload capacity				
Max. load in measuring direction without permanent change of data, $F_{max}^{1)}$	0.6 (135)	1.5 (337)	3 (674)	kN (lbs)
Max. load in transverse direction without permanent change of data, $F_{Vmax}^{(1)}$. For h = 135 mm (5.3 inch)	0.3 (67)	0.75 (169)	1.5 (337)	
Spring constant	9 (52)	22 (124)	34 (197)	kN/mm (1000 lbs/inch)
Accuracy				
Accuracy class, compressive force	± 1.0			%
Linearity deviation	$< \pm 0.5$			
Repeatability error	$< \pm 0.1$			
Hysteresis	$< \pm 0.3$			
Mechanical data				
Weight without adapter plates	approx. 2.5 (approx. 5.5)			kg (lbs)
Weight including adapter plates	approx. 5.4 (approx. 11.9)			
Length, width and height are given in Section B.11, Dimension Drawing, 3BSE015955D0094, Rev. C .				
Material				
Load cell	SS 2387 stainless steel, DIN X4CrNiMo 165. Corrosion resistance properties similar to AISI 304.			
Adapter plates	SS 1312, finished by black chromating. ASTM A 238-79 Grade C.			

(1) F_{max} and F_{Vmax} are allowed at the same time.

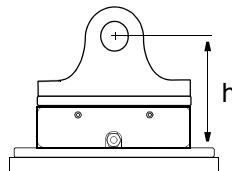


Figure 2-3. Building height

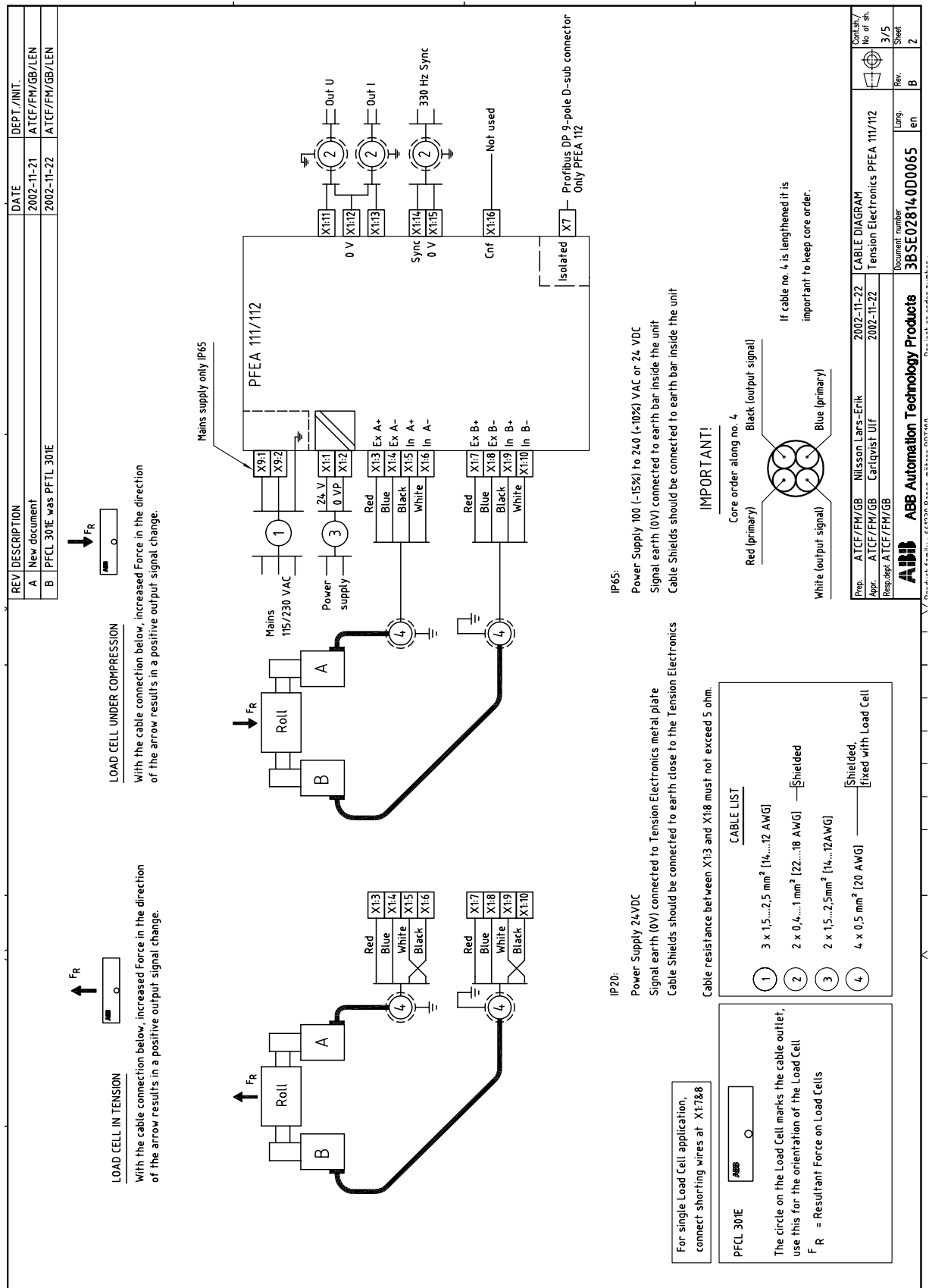
Table B-1. Environmental data for load cell PFCL 301E

PFCL 301E		Unit
Compensated temperature range	+20 - +60 (68 - 140)	°C (°F)
Zero point drift	< ± 150 (83)	ppm/K (ppm/F)
Sensitivity drift	< ± 250 (139)	
Working temperature range	-10 - +80 (14 - 176)	°C (°F)
Zero point drift	< ± 250 (139)	ppm/K (ppm/F)
Sensitivity drift	< ± 350 (194)	
Storage temperature range	-40 - +90 (-40 - 194)	°C (°F)
Degree of protection	IP 66 acc. to EN 60 529	

Table B-2. Mounting screws

Type of screws	Strength class	Dimension	Tightening torque
Electro-zincplated steel screws, lubricated with oil or emulsion. Strength class according to ISO 898/1.	8.8	M8	24 Nm (18 ft-lb)

B.9 Cable Diagram 3BSE028140D0065, page 2/5, Rev. B



B.10 Mounting Instruction, Cable Connector, 3BSE019064, Rev. A

REV	DESCRIPTION	DATE	DEPT./INIT.
-	New Document	1999.07.07	SEAPR / AGB / JRK
A	Core order along cable added.	00-02-25	SEAPR / AGB / JK

Side view of the cable connector assembly. A dimension of 23 is indicated for the length of the connector housing. Section line A-A is shown passing through the center of the connector.

Section A - A: Cross-section of the cable connector. It shows four internal conductors: Blue (top), Black (right), Red (bottom), and White (left). The central area is labeled 'Screen'. The conductors are numbered 1, 2, 3, and 4.

Side view of the cable connector assembly from the opposite side. A dimension of 23 is indicated. Section line B-B is shown passing through the center of the connector.

Section B - B: Cross-section of the cable connector. It shows four internal conductors: Blue (top), White (right), Red (bottom), and Black (left). The central area is labeled 'Screen'. The conductors are numbered 1, 2, 3, and 4.

IMPORTANT!

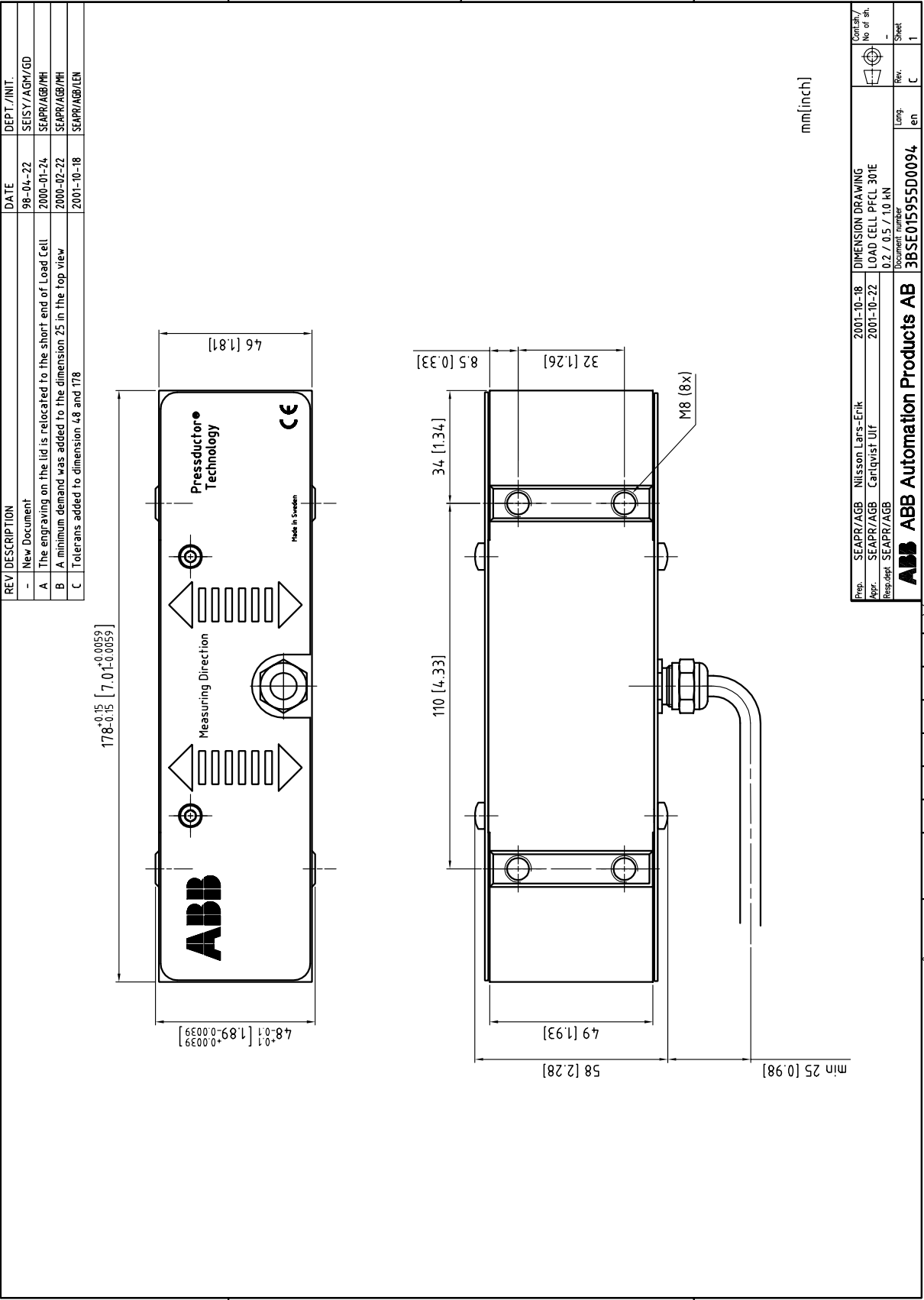
Core order along cable

Diagram showing the core order along the cable: Blue, Black, White, and Red.

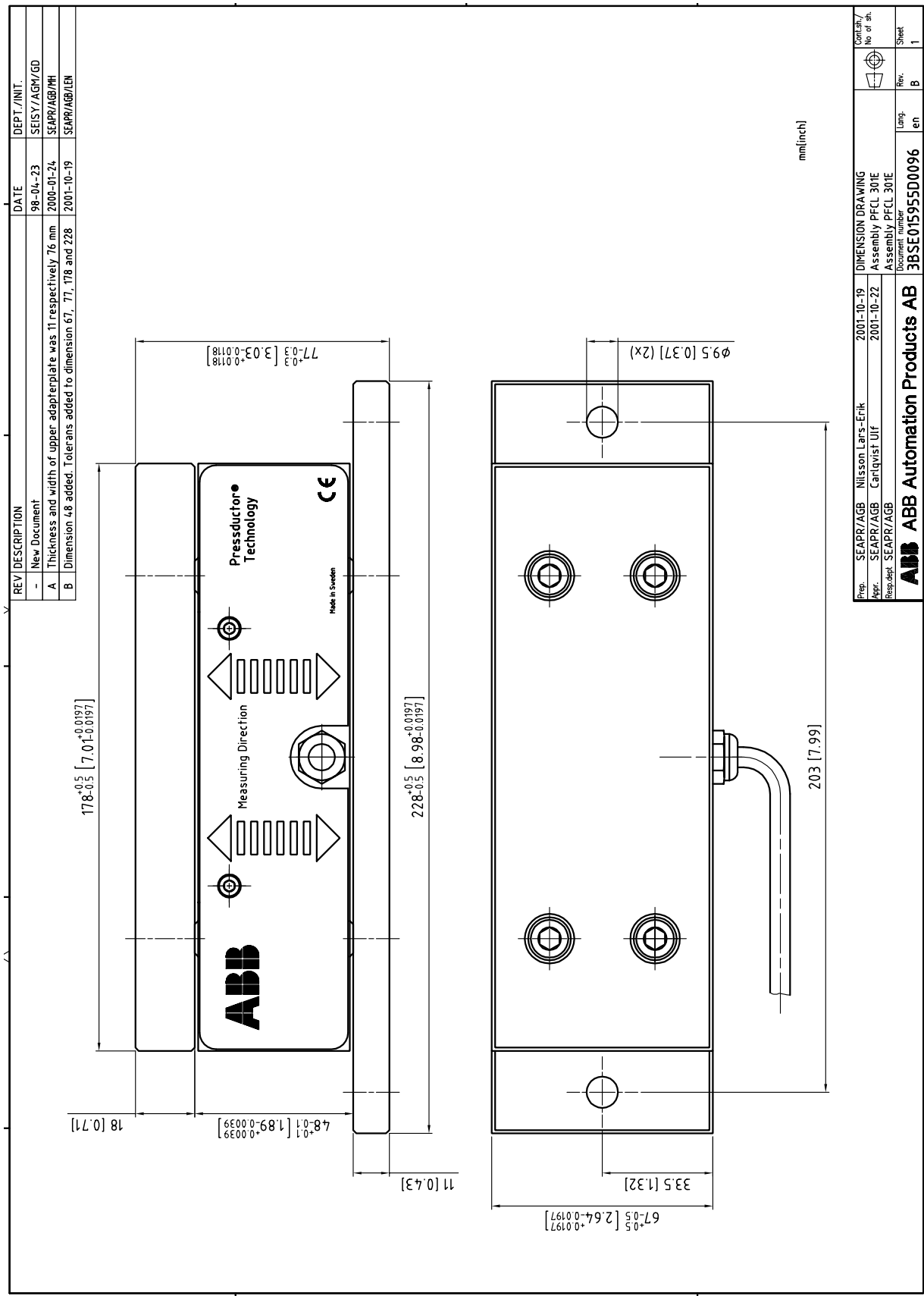
Prep. SEAPR/AGB	Hugosson Mattias	2000-02-25	INSTRUCTION		Cont.sh./ No of sh.	
Appr. SEAPR/AGB	Carlqvist Ulf	2000-02-29	Mounting instr. for cable connector			
Resp.dept SEAPR/AGB			Monteringsinstruktion för kontakt			
ABB ABB Automation Products AB			Document number 3BSE019064	Lang. en	Rev. A	Sheet 1

Product family: 661730 Base, mätare DDT/MV/DDT Product or order number:

B.11 Dimension Drawing, 3BSE015955D0094, Rev. C



B.12 Assembly Drawing, 3BSE015955D0096, Rev. B



Appendix C PFTL 301E - Designing the Load Cell Installation

C.1 About This Appendix

This appendix describes the procedure for designing the load cell installation.

The following sections are included:

- Basic application considerations
- Designing the load cell installation (step-by-step guide)
- Installation requirements
- Force and wrap gain calculation
 - Horizontal mounting
 - Inclined mounting
 - Single side measurement
- Mounting the load cells
- Technical data
- Drawings
 - Cable diagram(s)
 - Mounting instruction for load cell extension cable
 - Dimension drawing
 - Assembly drawing

C.2 Basic Application Considerations

Each application has its own individual demands that have to be considered; though a few basic considerations tend to repeat themselves.

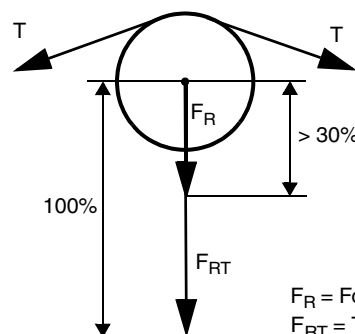
- What type of process is involved (papermaking, converting, etc.)?
Is the environment demanding (temperature, chemicals, etc.)?
- What is the tension measurement purpose; indication or closed loop control?
Are there any specific accuracy demands involved?
- What is the machine design like? Is there the possibility to modify the design,
in order to fit the most suitable load cell, or is the machine design fixed?
- What are the forces acting on the roll like (size and direction)?
Can they be altered by redesign?

If these questions are dealt with thoroughly, the installation is very likely to be successful. However, the extent to which measurement accuracy is needed, defines the requirements when designing a load cell installation.

C.3 Step-by-Step Guide for Designing the Load Cell Installation

The procedure below defines the main considerations involved in designing a load cell installation.

1. Check load cell data so that environmental demands are met.
2. Calculate forces; vertical, horizontal and axial (cross directional).
3. Size and orient load cell so that the guidelines below are met:
 - a. Try to achieve a measured value no less than 10% of web tension, in the load cell measurement direction!
 - b. Select load cell size so that it is loaded as close as possible to its nominal load! Do not dimension Force component of Tension in measuring direction, F_R , for less than 10% of the load cell nominal load!
 - c. If the span between maximum and minimum tension in the process is large, select load cell so that the maximum tension will be in the load cell extended range (when applicable)!
 - d. The measured force component of web tension is recommended to be at least 30% of tare force component (roll weight) acting in load cell measurement direction. The reason for this recommendation is load cell signal stability, especially when the system operates in a large temperature span.
This means that if $F_{RT} < 1/3$ of F_{nom} , F_R should be at least 10% of F_{nom} .
For larger F_{RT} , lowest F_R is recommended to be at least 30% of F_{RT} .



Rule 1: If $F_{RT} < 1/3$ of F_{nom}
 F_R should be at least 10% of F_{nom}

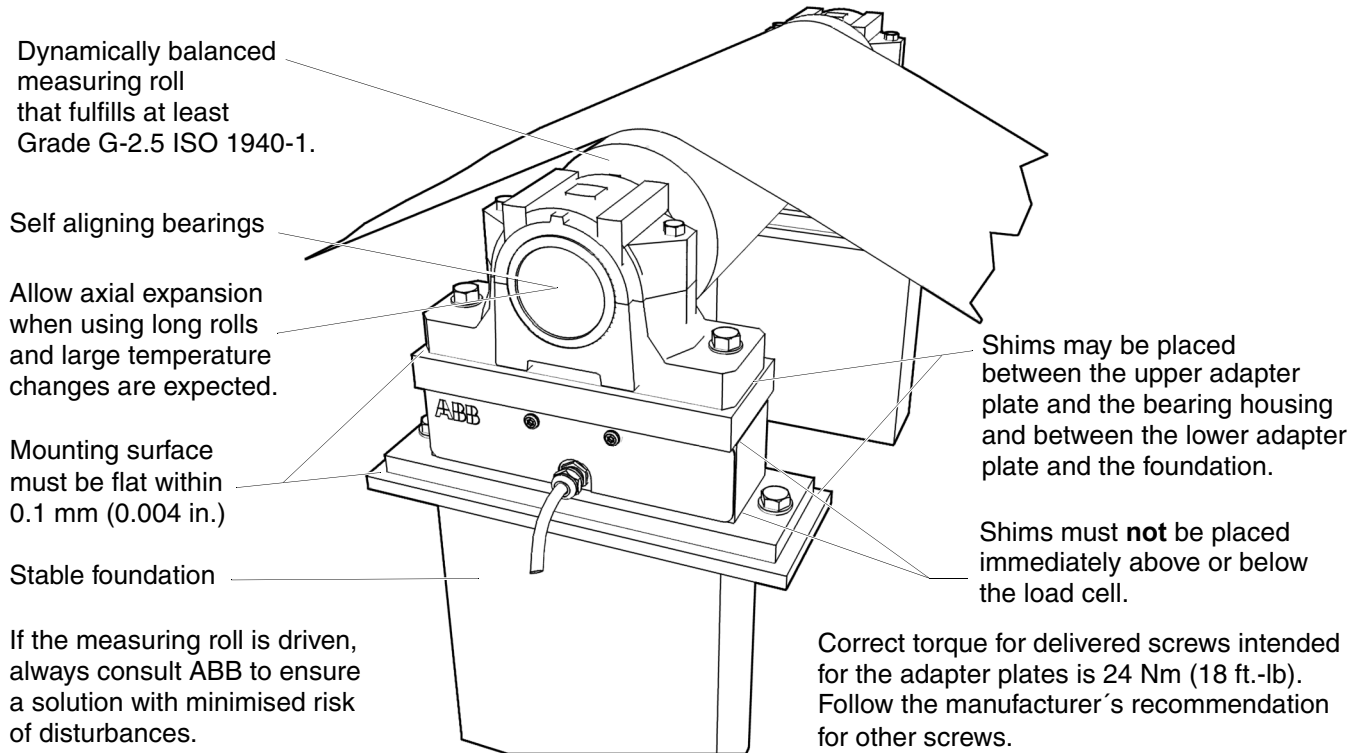
Rule 2: If $F_{RT} > 1/3$ of F_{nom}
 F_R is recommended to be at least 30% of F_{RT}

F_R = Force component of web tension in measuring direction
 F_{RT} = Tare force in measurement direction

- e. Check load cell data so that limits for building height, transverse and axial forces are not exceeded.
4. Design base frame and/or adapter plates.

C.4 Installation Requirements

To achieve the specified accuracy, the best possible reliability and long-term stability, install the load cells in accordance with the requirements below.



Alignment of the load cells

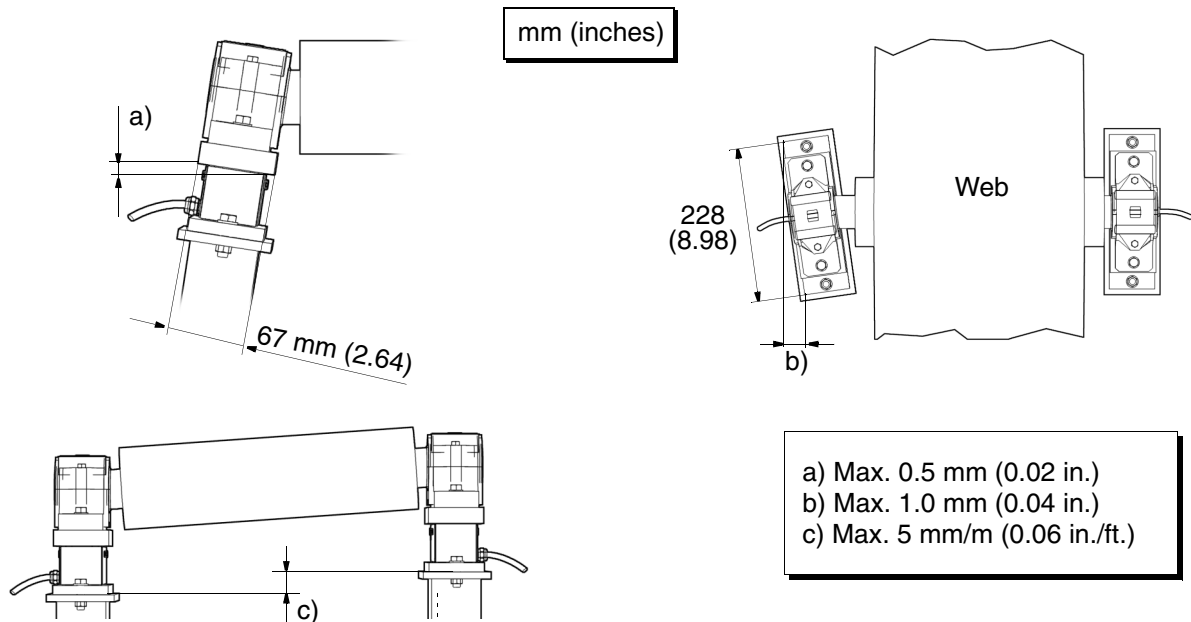
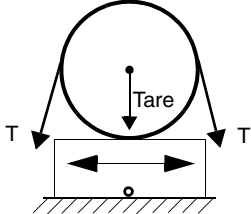


Figure C-1. Installation Requirements

C.5 Mounting Alternatives, Calculating Force and Calculating Wrap Gain

C.5.1 Horizontal Mounting

PFTL 301E



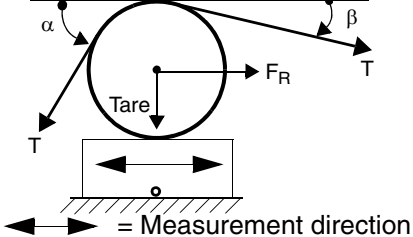
No horizontal web tension force applied to the load cell.

← → = Measurement direction

In most cases, horizontal mounting is the most obvious and simplest solution. The load cell should thus be mounted horizontally when possible.

However, should the machine design require inclined mounting of the load cell or should the web path not give a sufficient horizontal force, see figure, the inclined mounting is permitted and the calculations are somewhat more complex, (see [Section C.5.2, Inclined Mounting](#)).

PFTL 301E



← → = Measurement direction

$$F_R = T \times (\cos \beta - \cos \alpha)$$
$$F_{RT} = 0 \text{ (Tare force is not measured)}$$
$$F_{Rtot} = F_R + F_{RT} = T \times (\cos \beta - \cos \alpha)$$

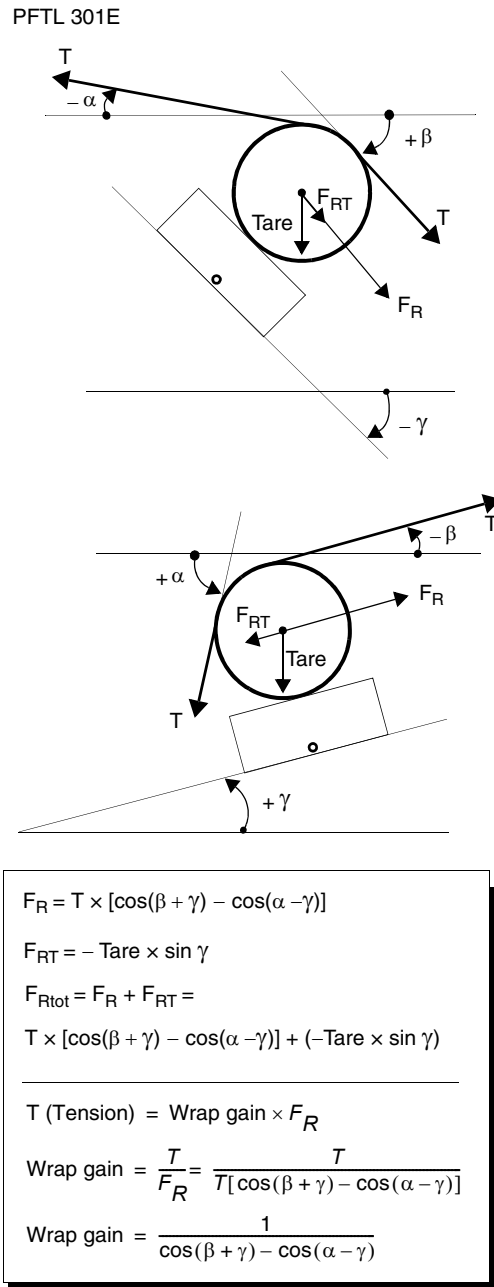
$$T \text{ (Tension)} = \text{Wrap gain} \times F_R$$
$$\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T(\cos \beta - \cos \alpha)}$$
$$\text{Wrap gain} = \frac{1}{\cos \beta - \cos \alpha}$$

The PFTL 301E load cell measures the horizontal forces applied to its top surface. The load cell can measure in both directions. The vertical applied forces are not measured and do not influence the horizontal measurement. There is one source of horizontal forces; the force from the web tension (the tare weight has no force component in measuring direction). See the force calculations in the figure.

Divide the total horizontal force F_{Rtot} by two to get the required capacity of each load cell.

Do not oversize an ABB load cell for overload purposes as the load cell has sufficient overload capacity.

C.5.2 Inclined Mounting



Sometimes it is necessary to mount the load cell on an incline due to mechanical design constraints of the machine or the need to have a proper force component applied to the load cell.

Inclined mounting adds a component of tare force in the measuring direction and modifies the force components as shown.

NOTE

When calculating, it is important that the angles are set into the equations with the correct signs in relation to the horizontal plane.

C.6 Force Calculation for Measurement with a Single Load Cell

In some cases, it is sufficient to measure the tension with only a single load cell mounted at one end of the roll. The roll should nevertheless be supported at both ends.

C.6.1 The Most Common and Simple Solution

The most obvious and simple solution is horizontal mounting with the web evenly distributed and centered on the roll.

As long as the roll is supported at both ends, the same calculations given in [Section C.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#) are valid.

NOTE

The accuracy of a single load cell measurement is highly dependent on how well the center of force can be determined. Since the cross-directional stress distribution generally is somewhat uneven, this is not easily done. The load cell will, however, produce a stable and repeatable measurement.

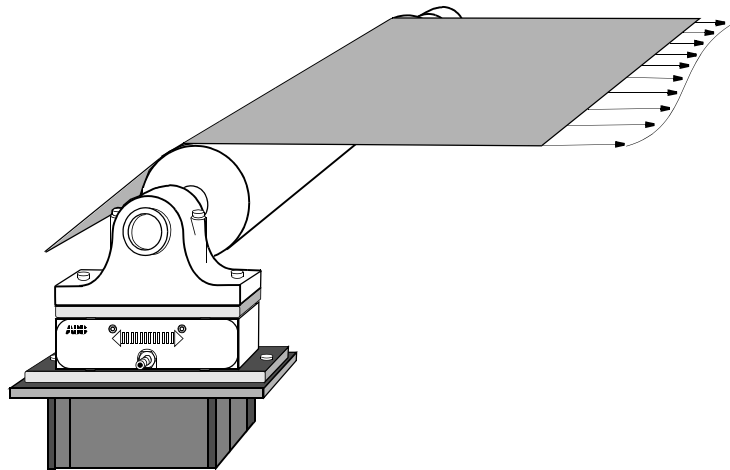
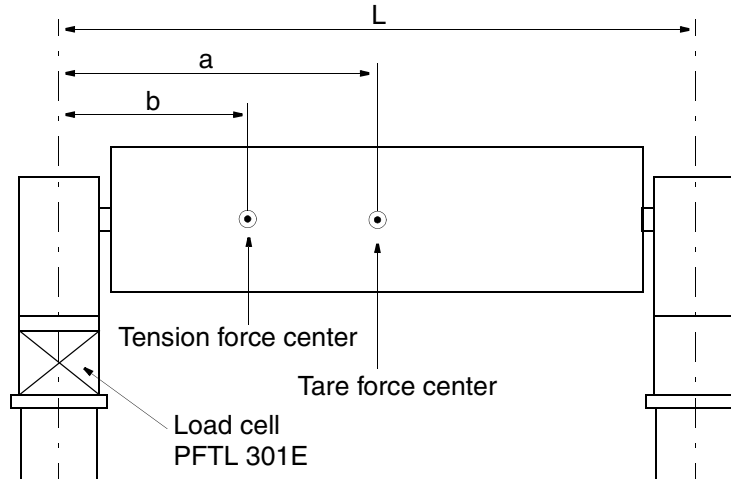


Figure C-2. Cross-directional stress distribution

C.6.2 Force Calculation when the Web is not Centered on the Roll

Use the calculations below for horizontal and inclined mounting when the web is not centered on the roll.

The applied force at the load cell will be proportional to the distance between the tension force center and the load cell centerline, see figure.



Calculation procedure:

1. Horizontal or inclined mounting?
2. Calculate F_R and F_{RT} , see [Section C.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#).
3. Use the following equations:

$$F_R \text{ for single load cell} = F_R \times \frac{L-b}{L}$$

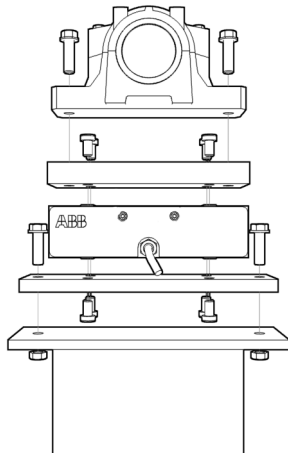
$$F_{RT} \text{ for single load cell} = F_{RT} \times \frac{L-a}{L}$$

$$F_{Rtot} \text{ for single load cell} = F_R \text{ for single load cell} + F_{RT} \text{ for single load cell}$$

where:

- L = Distance between load cell centerline and the opposite bearing centerline
- a = Distance between tare force center and load cell centerline
- b = Distance between tension force center and load cell centerline

C.7 Mounting the Load Cells

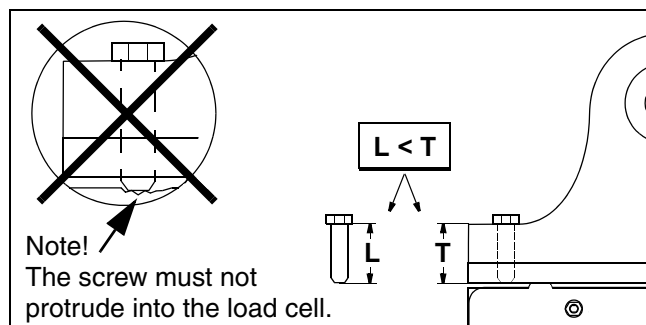


The instructions below apply to a typical mounting arrangement. Variations may be allowed, provided that they comply with the requirements in [Section C.4, Installation Requirements](#).

1. Clean the foundation and other mounting surfaces.
2. Fit the lower adapter plate to the load cell.
Tighten the screws (included in delivery) with a torque wrench to 24 Nm (18 ft.-lb).
3. Fit the load cell and the lower adapter plate to the foundation,
but do not fully tighten the screws.
4. Fit the upper adapter plate to the load cell.
Tighten the screws (included in delivery) with a torque wrench to 24 Nm (18 ft.-lb).
5. Fit the bearing housing and the roll to the upper adapter plate,
but do not fully tighten the screws.

CAUTION

When mounting bearings or other adjacent details to adapter plates, screws must not protrude into the load cell. The load cell can be damaged by an excessive applied force.



6. Adjust the load cells according to the installation requirements.
Tighten the foundation screws.
7. Adjust the roll according to the installation requirements.
Tighten the screws in the upper adapter plate.

C.7.1 Routing the Load Cell Cable

The cable must be supported with clamps and routed to prevent force shunting through the cable.

C.7.2 Connecting the Load Cell Extension Cable

See [Section C.10, Mounting Instruction, Cable Connector, 3BSE019064, Rev. A](#).

C.8 Technical Data

PFTL 301E					Unit
Nominal load					
Nominal load in measuring direction, F_{nom} For h = 135 mm (5.3 inch)	0.1 (22)	0.2 (45)	0.5 (112)	1.0 (225)	kN (lbs)
Permitted transverse load within the accuracy, F_{Vnom}	0.3 (67)	0.6 (135)	1.5 (337)	3.0 (675)	
Permitted axial load within the accuracy, F_{Anom} For h = 135 mm (5.3 inch)	0.5 (112)	0.5 (112)	1.0 (225)	1.0 (225)	
Extended load in measuring direction with accuracy class, bidirectional measurement $\pm 2\%$, F_{ext}	0.15 (33)	0.3 (67)	0.75 (169)	1.5 (337)	
Overload capacity					
Max. load in measuring direction without permanent change of data, $F_{max}^{(1)}$. For h = 135 mm (5.3 inch)	0.3 (67)	0.6 (135)	1.5 (337)	3.0 (674)	kN (lbs)
Max. load in transverse direction without permanent change of data, $F_{Vmax}^{(1)}$	0.5 (112)	1.0 (225)	2.5 (562)	5.0 (1125)	
Max. load in axial direction without permanent change of data, F_{Amax} . For h = 135 mm (5.3 inch)	0.5 (112)	0.5 (112)	1.0 (225)	1.0 (225)	
Spring constant	2 (11.3)	4 (22.6)	7 (39.7)	8 (44.6)	kN/mm (1000 lbs/inch)
Accuracy					
Accuracy class	± 1.0				%
Linearity deviation	$< \pm 0.5$				
Repeatability error	$< \pm 0.1$				
Hysteresis	$< \pm 0.3$				
Mechanical data					
Weight without adapter plates	approx. 2.5 (approx. 5.5)				kg (lbs)
Weight including adapter plates	approx. 5.4 (approx. 11.9)				
Length, width and height are given in Section C.11, Dimension Drawing, 3BSE019040D0094, Rev. B .					
Material					
Load cell	SS 2387 stainless steel, DIN X4CrNiMo 165. Corrosion resistance properties similar to AISI 304.				
Adapter plates	SS 1312, finished by black chromating. ASTM A 238-79 Grade C.				

(1) F_{max} and F_{Vmax} are allowed at the same time.

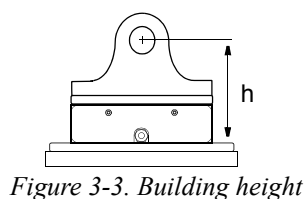


Figure 3-3. Building height

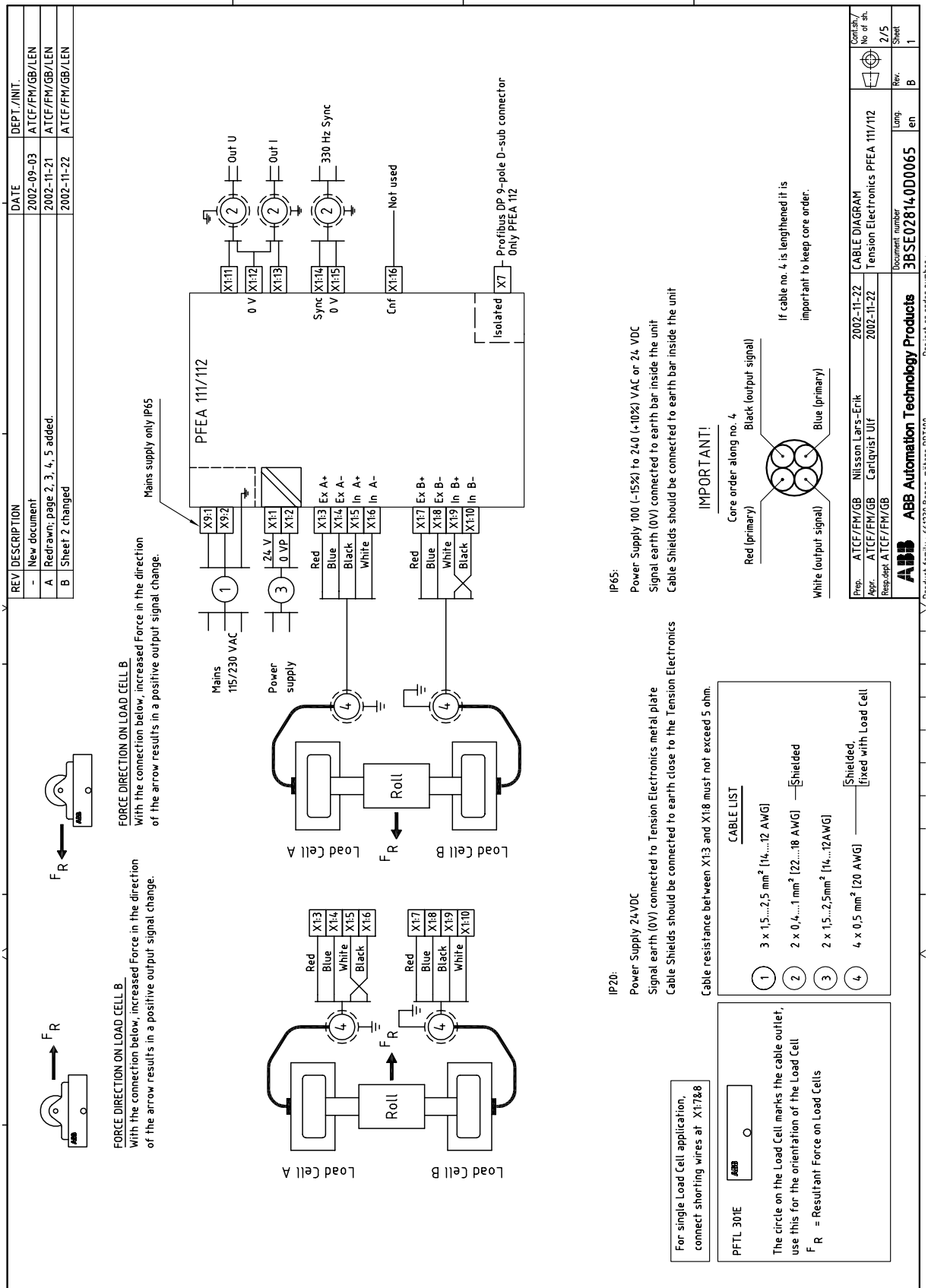
Table C-1. Environmental data for load cell PFTL 301E

PFTL 301E		Unit
Compensated temperature range	+20 - +60 (68 - 140)	°C (°F)
Zero point drift	< ± 150 (83)	ppm/K (ppm/F)
Sensitivity drift	< ± 250 (139)	
Working temperature range	-10 - +80 (14 - 176)	°C (°F)
Zero point drift	< ± 250 (139)	ppm/K (ppm/F)
Sensitivity drift	< ± 350 (194)	
Storage temperature range	-40 - +90 (-40 - 194)	°C (°F)
Degree of protection	IP 66 acc. to EN 60 529	

Table 3-2. Mounting screws

Type of screws	Strength class	Dimension	Tightening torque
Electro-zincplated steel screws, lubricated with oil or emulsion. Strength class according to ISO 898/1.	8.8	M8	24 Nm (18 ft-lb)

C.9 Cable Diagram 3BSE028140D0065, Page 1/5, Rev. B



C.10 Mounting Instruction, Cable Connector, 3BSE019064, Rev. A

REV	DESCRIPTION	DATE	DEPT./INIT.
-	New Document	1999.07.07	SEAPR / AGB / JRK
A	Core order along cable added.	00-02-25	SEAPR / AGB / JK

23

A

Screen

A - A

Blue

White

1

2

3

4

5

Black

Screen

Red

23

B

Screen

B - B

Blue

Black

White

Red

Screen

IMPORTANT!

Core order along cable

Blue

Black

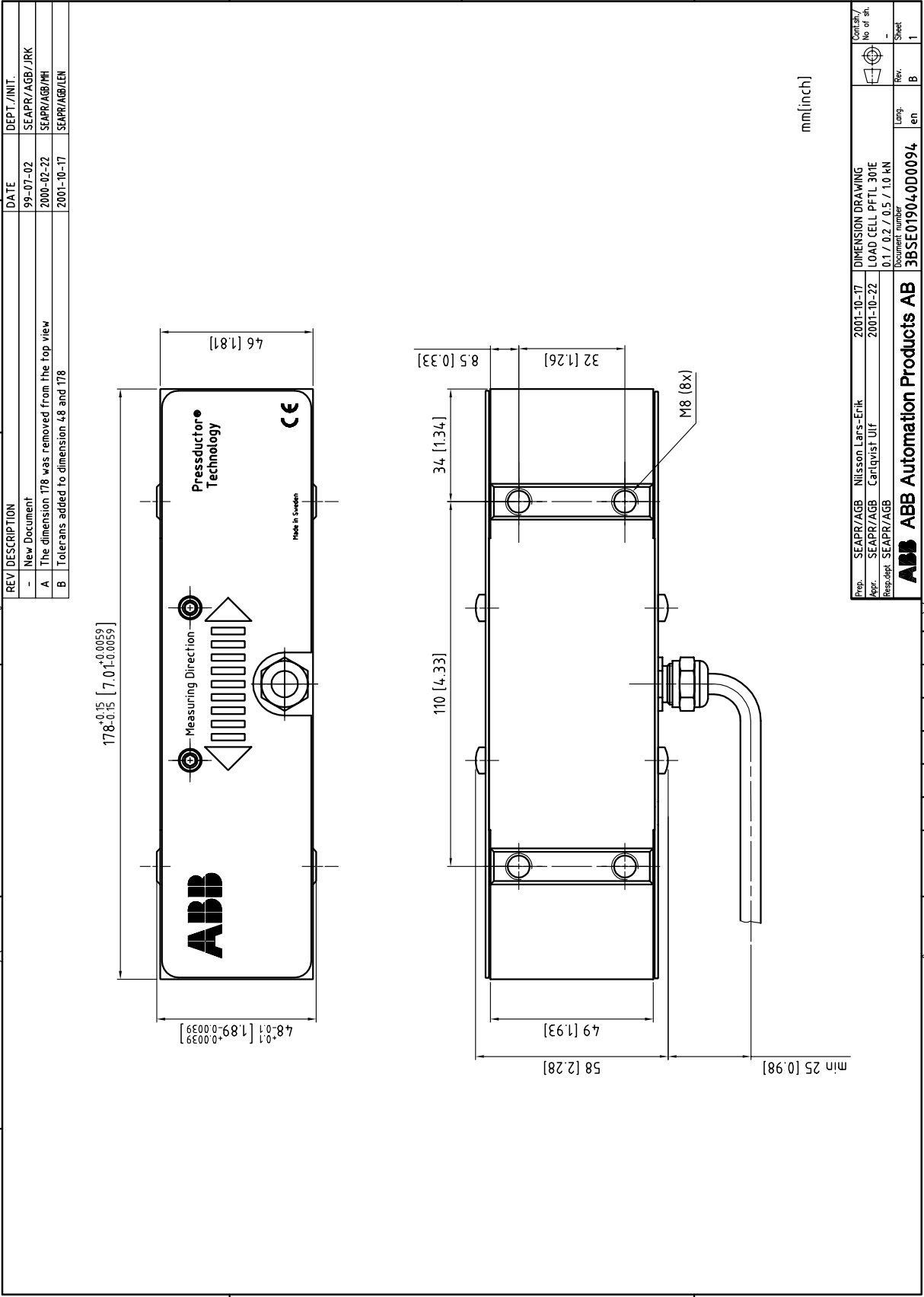
White

Red

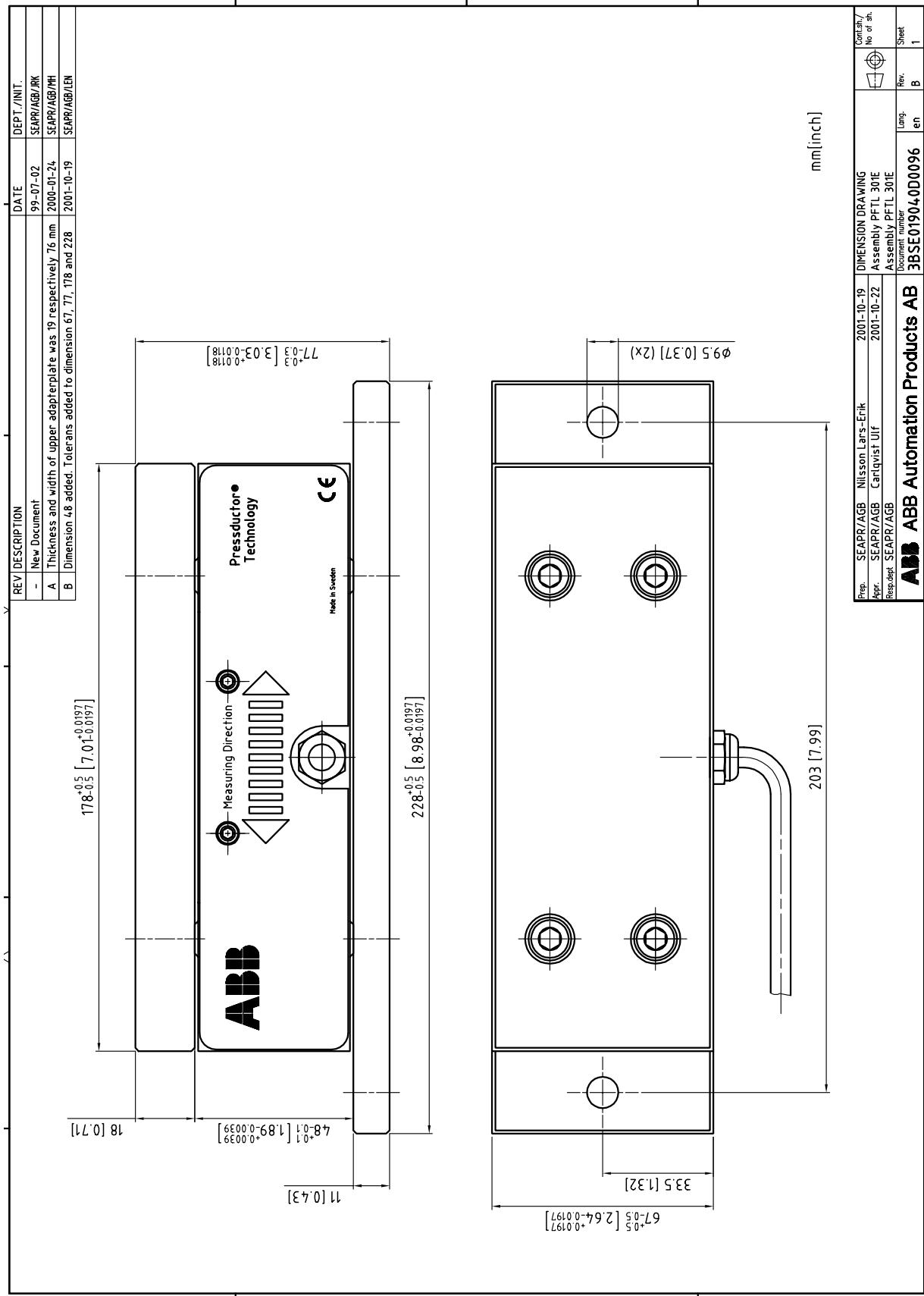
Prep.	SEAPR/AGB	Hugosson Mattias	2000-02-25	INSTRUCTION Mounting instr. for cable connector Monteringsinstruktion för kontakt		Cont.sh./ No of sh.	
Appr.	SEAPR/AGB	Carlqvist Ulf	2000-02-29				
Resp.dept	SEAPR/AGB						
ABB ABB Automation Products AB				Document number 3BSE019064	Lang. en	Rev. A	Sheet 1

Product family: 66192A Bases mätare DDT/MV/DBT Product order number:

C.11 Dimension Drawing, 3BSE019040D0094, Rev. B



C.12 Assembly Drawing, 3BSE019040D0096, Rev. B



Appendix D PFRL 101 - Designing the Load Cell Installation

D.1 About This Appendix

This appendix describes the procedure for designing the load cell installation.

The following sections are included:

- Basic application considerations
- Designing the load cell installation (step-by-step guide)
- Installation requirements
- Force and wrap gain calculation
 - Horizontal mounting
 - Inclined mounting
 - Single side measurement
- Mounting the load cells
- Technical data
- Drawings
 - Cable diagram(s)
 - Dimension drawing(s)

D.2 Basic Application Considerations

Each application has its own individual demands that have to be considered; though a few basic considerations tend to repeat themselves.

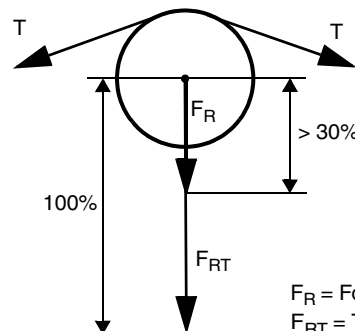
- What type of process is involved (papermaking, converting, etc.)?
Is the environment demanding (temperature, chemicals, etc.)?
- What is the tension measurement purpose; indication or closed loop control?
Are there any specific accuracy demands involved?
- What is the machine design like? Is there the possibility to modify the design,
in order to fit the most suitable load cell, or is the machine design fixed?
- What are the forces acting on the roll like (size and direction)?
Can they be altered by redesign?

If these questions are dealt with thoroughly, the installation is very likely to be successful. However, the extent to which measurement accuracy is needed, defines the requirements when designing a load cell installation.

D.3 Step-by-Step Guide for Designing the Load Cell Installation

The procedure below defines the main considerations involved in designing a load cell installation.

1. Check load cell data so that environmental demands are met.
2. Calculate forces; vertical, horizontal and axial (cross directional).
3. Size and orient load cell so that the guidelines below are met:
 - a. Try to achieve a measured value no less than 10% of web tension, in the load cell measurement direction!
 - b. Select load cell size so that it is loaded as close as possible to its nominal load! Do not dimension Force component of Tension in measuring direction, F_R , for less than 10% of the load cell nominal load!
 - c. If the span between maximum and minimum tension in the process is large, select load cell so that the maximum tension will be in the load cell extended range (when applicable)!
 - d. The measured force component of web tension is recommended to be at least 30% of tare force component (roll weight) acting in load cell measurement direction. The reason for this recommendation is load cell signal stability, especially when the system operates in a large temperature span.
This means that if $F_{RT} < 1/3$ of F_{nom} , F_R should be at least 10% of F_{nom} .
For larger F_{RT} , lowest F_R is recommended to be at least 30% of F_{RT} .



Rule 1: If $F_{RT} < 1/3$ of F_{nom}
 F_R should be at least 10% of F_{nom}

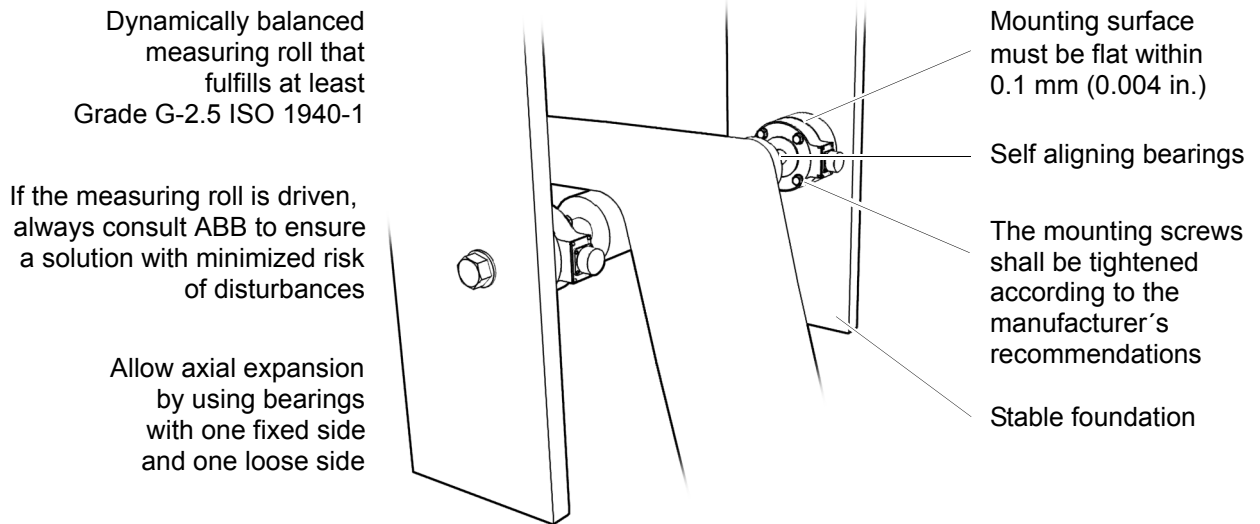
Rule 2: If $F_{RT} > 1/3$ of F_{nom}
 F_R is recommended to be at least 30% of F_{RT}

F_R = Force component of web tension in measuring direction
 F_{RT} = Tare force in measurement direction

- e. Check load cell data so that limits for building height, transverse and axial forces are not exceeded.
4. Design base frame and/or adapter plates.

D.4 Installation Requirements

To achieve the specified accuracy, the best possible reliability and long-term stability, install the load cells in accordance with the requirements below.



Alignment of the load cells

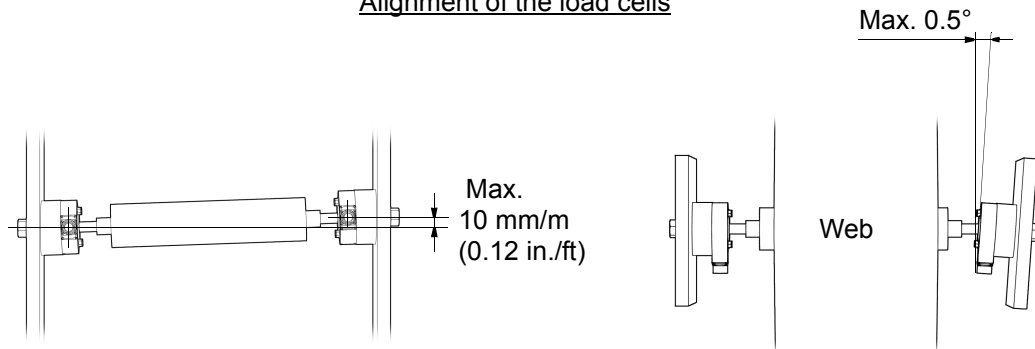
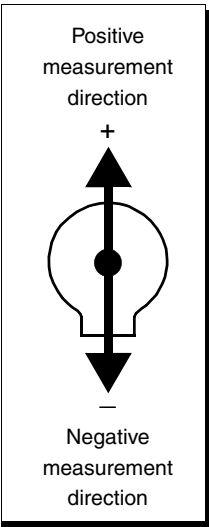


Figure D-1. Installation requirements

D.5 Load Cell Orientation Depending on Measurement Direction

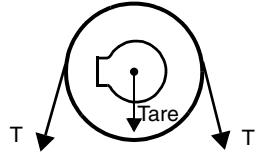


The radial load cell will only measure forces along the axis as shown in the figure to the left. Therefore the orientation of the measurement direction is important to the amount of signal output. To help understand how the orientation of the measurement direction affects the output see the figures below.

Orientation of measurement direction	Effects (Two load cells are assumed)
	The load cells measure $2 \times \text{Tension}$, but do not measure the roll weight (Tare).
	The load cells measure no Tension, but do measure the roll weight (Tare). Rotating the load cells counterclockwise will begin to gain in signal from the web tension and will eliminate the output due to roll weight (Tare). Maximum tension signal will occur at 90° of rotation.
	The load cells measure $1 \times \text{Tension}$, but do not measure the roll weight (Tare). Rotate the load cells 45° clockwise and the load cells sense $1.4 \times \text{Tension}$ and 70% of the roll weight.

D.6 Mounting Alternatives, Calculating Force and Calculating Wrap Gain

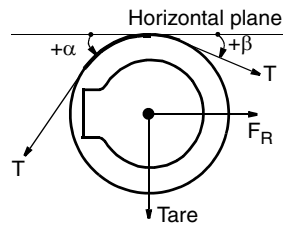
D.6.1 Horizontal Mounting



No horizontal web tension force applied to the load cell.

The PFRL 101 load cells can be mounted at any incline angle, 0 - 360°. However, it is recommended to minimize the influence of forces other than the tension to be measured. In most cases this means an orientation where the tare force (vertical) is perpendicular to the measured force (horizontal).

However, should the machine design require inclined mounting of the load cell or should the web path not give a sufficient horizontal force, see figure, the inclined mounting is permitted and the calculations are somewhat more complex, (see [Section D.6.2, Inclined Mounting](#)).



$$F_R = T \times (\cos \beta - \cos \alpha)$$

$$F_{RT} = 0 \text{ (Tare force is not measured)}$$

$$F_{Rtot} = F_R + F_{RT} = T \times (\cos \beta - \cos \alpha)$$

$$F_{Rtot} / \text{load cell} = F_{Rtot} / 2$$

$$T \text{ (Tension)} = \text{Wrap gain} \times F_R$$

$$\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T(\cos \beta - \cos \alpha)}$$

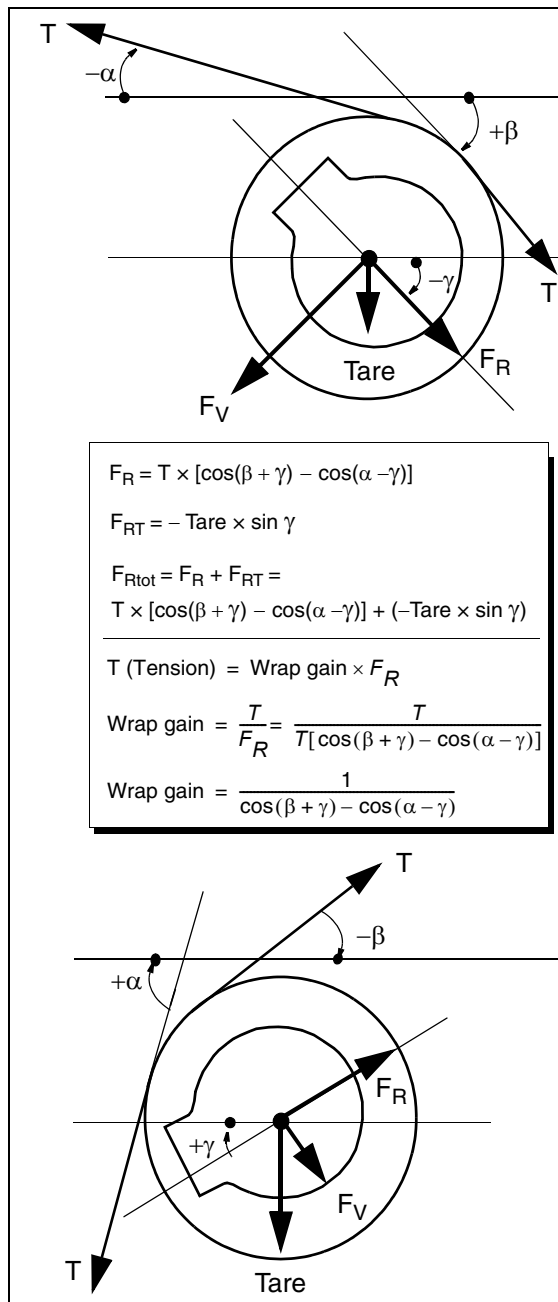
$$\text{Wrap gain} = \frac{1}{\cos \beta - \cos \alpha}$$

The load cell measures the horizontal forces. The load cell can measure in both directions. The vertical applied forces are not measured and do not influence the horizontal measurement. There is one source of horizontal forces, the force from the web tension (the tare weight has no force component in measuring direction). See force calculations in figure.

Divide the total vertical force F_{Rtot} by two to get the required capacity of each load cell.

Do not oversize an ABB load cell for overload purposes as the load cell has sufficient overload capacity.

D.6.2 Inclined Mounting



Sometimes it is necessary to mount the load cell on an incline due to mechanical design constraints of the machine or due to the need to have a proper force component applied to the load cell.

Inclined mounting adds a component of tare force and modifies the force components as shown.

NOTE

When calculating it is important that the angles are set into the equations with the correct signs relative to the horizontal plane.

D.7 Force Calculation for Measurement with a Single Load Cell

In some cases, it is sufficient to measure the tension with only a single load cell mounted at one end of the roll. The roll should nevertheless be supported at both ends.

D.7.1 The Most Common and Simple Solution

The most obvious and simple solution is horizontal mounting with the web evenly distributed and centered on the roll.

As long as the roll is supported at both ends, the calculations given in [Section D.6, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#) are valid. Note that the output signal is a summation.

NOTE

The accuracy of a single load cell measurement is highly dependent on how well the center of force can be determined. Since the cross-directional stress distribution generally is somewhat uneven, this is not easily done. The load cell will, however, produce a stable and repeatable measurement.

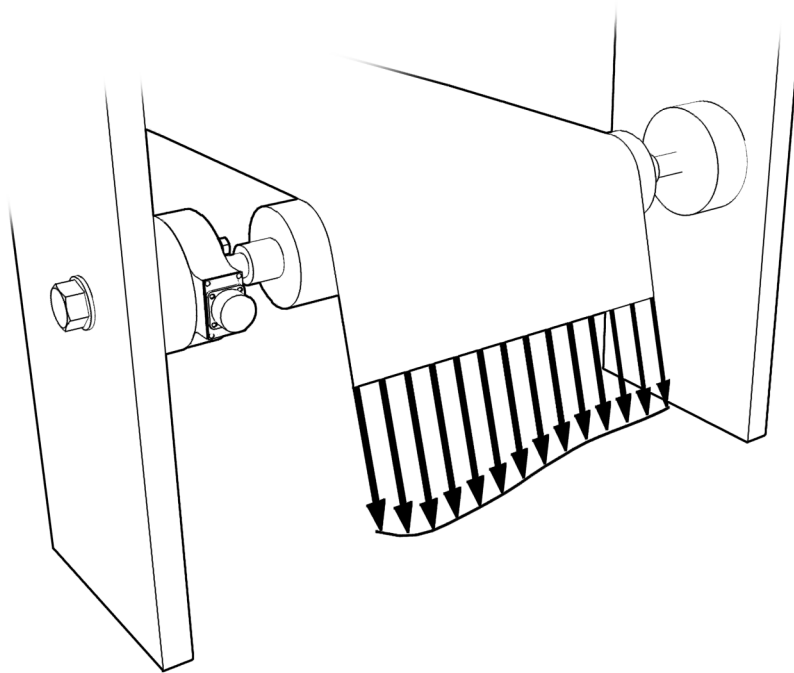
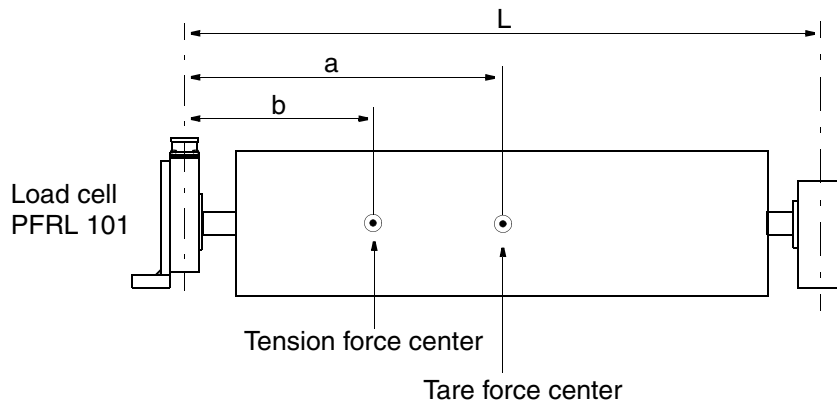


Figure D-2. Cross-directional stress distribution

D.7.2 Force Calculation when the Web is not Centered on the Roll

Use the calculations below for horizontal and inclined mounting when the web is not centered on the roll.

The applied force at the load cell will be proportional to the distance between the tension force center and the load cell centerline.



Calculation procedure:

1. Horizontal or inclined mounting?
2. Calculate F_R and F_{RT} , see [Section D.6, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#).
3. Use the following equations:
$$F_R \text{ for single load cell} = F_R \times \frac{L-b}{L}$$
$$F_{RT} \text{ for single load cell} = F_{RT} \times \frac{L-a}{L}$$
$$F_{Rtot} \text{ for single load cell} = F_R \text{ for single load cell} + F_{RT} \text{ for single load cell}$$

where:

- L = Distance between load cell centerline and the opposite bearing center line
a = Distance between tare force center and load cell centerline
b = Distance between tension force center and load cell centerline

D.8 Mounting the Load Cells

1. Mount the bearings in the load cells.

NOTE

Use tools and materials that not will damage the bearing or the load cell.

NOTE

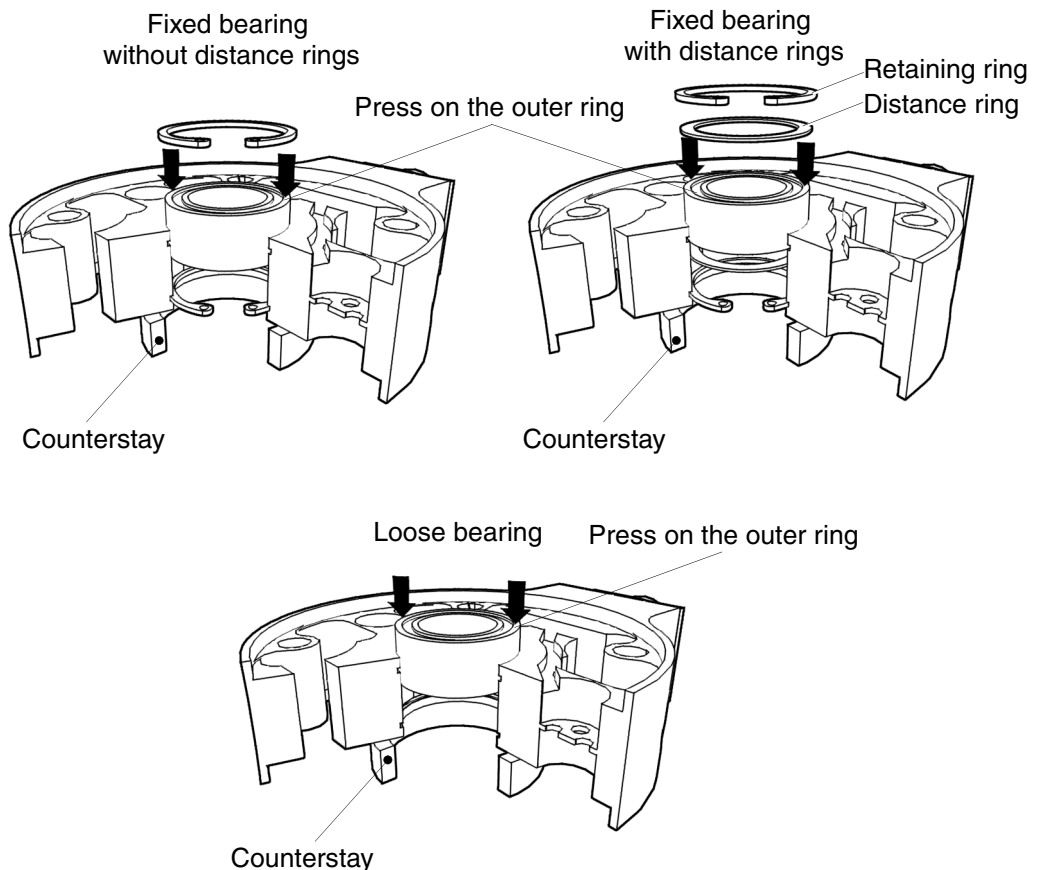
One of the bearings is locked in position with retaining rings while the other bearing is only pressed into the correct position to allow axial expansion.

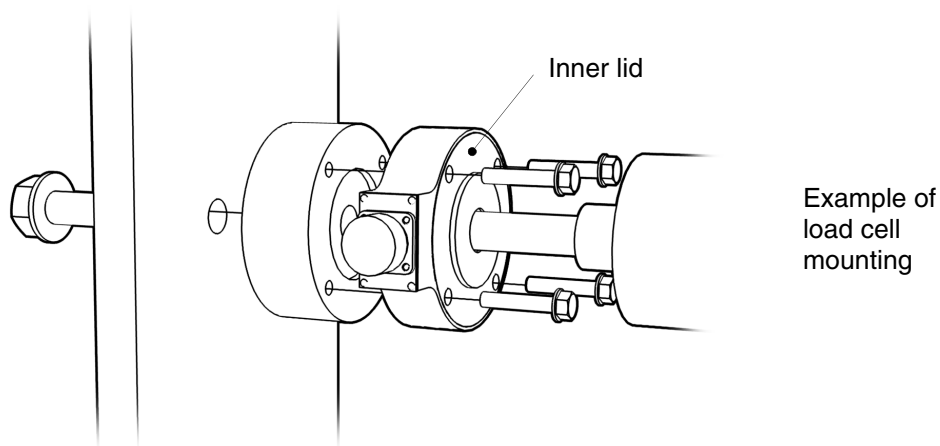
- a. Mount one of the retaining rings into the load cell.
- b. Arrange a counterstay as shown in the figure below.
- c. Press the bearing to correct position.

NOTE

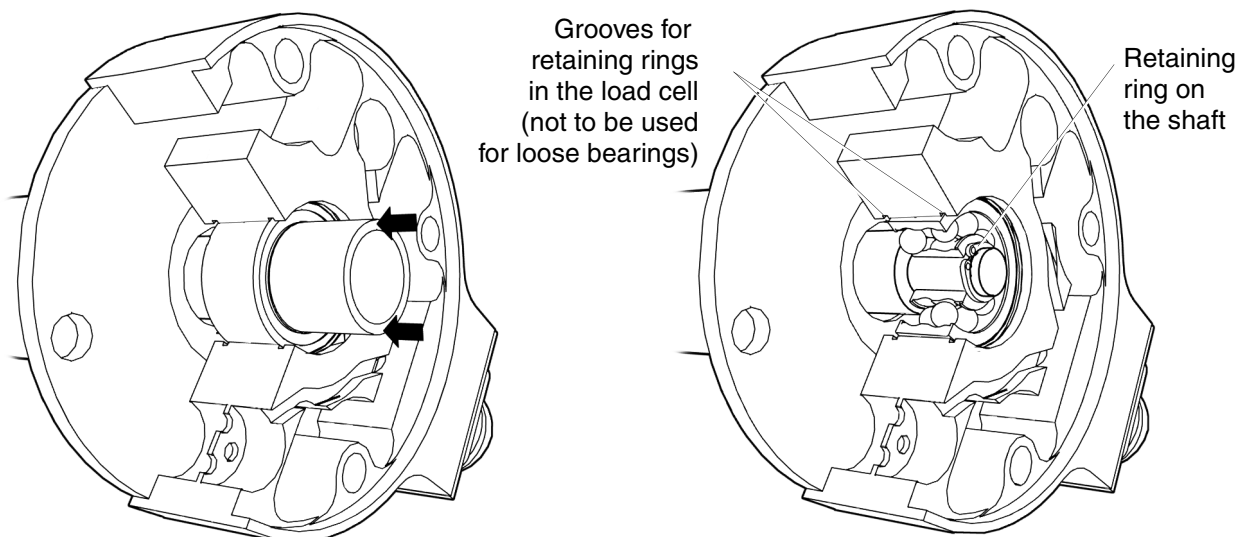
The bearing bed has only a slight interference fit, and, therefore, no heavy forces should be used.

- d. Mount the other retaining ring into the load cell.





2. Mount distance spacers and shaft sealings, if necessary.
3. Put the inner load cell lids in position, and also four mounting screws in their holes.
4. Press the load cells onto the shaft (press on the inner rings of the bearings only).



5. Mount the retaining rings for the bearings on the shaft.
Put the outer lids in position.
6. Position the measuring roll complete with the load cells into the correct position on the machine.

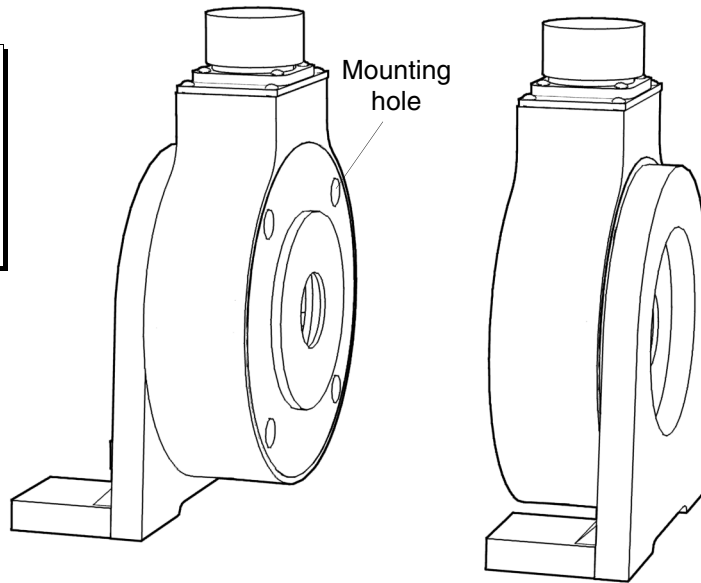
The load cell with the loose bearing is displaced towards the roll, in order to reduce the total length, so that the measuring roll with the load cells can be fitted in.

When the roll is in position pull the load cell with the loose bearing back to its proper position.
7. Fix each load cell by using the four mounting screws.
(tightening torque according to the manufacturer's recommendations)
8. Adjust the shaft sealings, if necessary.

D.8.1 Mounting with Brackets

The optional bracket is intended to facilitate mounting on horizontal surfaces.

To achieve the best measuring direction, turn the load cell to a suitable position before drilling the bracket.



Possible ways of mounting using brackets.

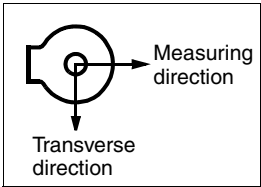
1. Mark the location of the mounting holes.
2. Drill the holes and cut threads, according to [Section D.15, Dimension Drawing, 3BSE010457, Rev. A](#).
3. Install according to the instructions in [Section D.8, Mounting the Load Cells](#).

D.8.2 Mounting Screws for the Load Cells

The load cell shall be mounted with screws according to [Table D-1](#).

NOTE

The screws shall be tightened according to the manufacturer’s recommendations.



Screws with strength class 8.8 are sufficient for normal applications without large transverse forces or overloads.

Screws with strength class 12.9 and a higher tightening torque are recommended in applications where large transverse forces or overloads can occur.

Prior to mounting, check that the mounting surfaces are clean and flat, e.g. free of burrs and other damage.

Table D-1. Mounting Screws

Load cell PFRL 101	Screw dimension
A	M8 (5/16 UNC)
B	M8 (5/16 UNC)
C	M10 (3/8 UNC)
D	M12 (1/2 UNC)

D.8.3 Routing the Load Cell Cable

The cable must be supported with clamps and routed to prevent force shunting through the cable.

D.9 Technical Data

Table D-2. Technical Data for Different Types of Load Cell PFRL 101

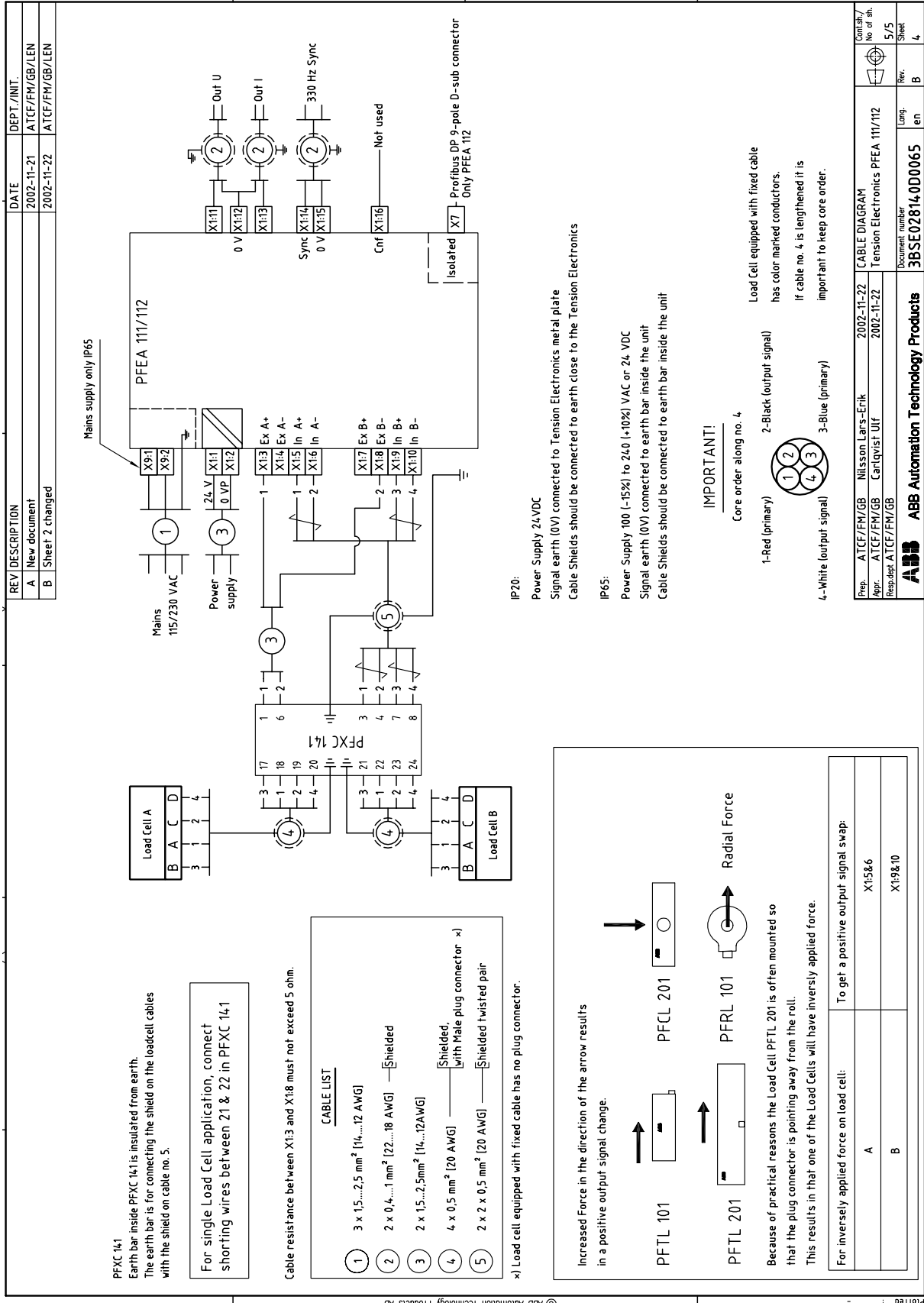
PFRL 101	Type	Data			Unit
Nominal load					
Nominal load, F _{nom}	A	0.5 (112)			kN (lbs)
	B	1 (225)			
	C	0.5 (112)	1 (225)	2 (450)	
	D	5 (1125)			
Accepted transverse load within the accuracy class, F _{Vnom}	A	2.5 (562)			
	B	3 (674)			
	C	1.25 (281)	2.5 (562)	5 (1125)	
	D	10 (2250)			
Accepted axial load within the accuracy class, F _{Anom}	A	2.5 (562)			
	B	5 (1125)			
	C	2.5 (562)	5 (1125)	10 (2250)	
	D	25 (5625)			
Overload capacity					
Maximal load in the direction of measurement without permanent change of data, F _{max}	A	2.5 (562)			kN (lbs)
	B	5 (1125)			
	C	2.5 (562)	5 (1125)	10 (2250)	
	D	25 (5625)			
Spring constant	A	50 (286)			kN/mm (1000 lbs/inch)
	B	100 (572)			
	C	50 (286)	100 (572)	200 (1143)	
	D	500 (2858)			
Mechanical data					
Weight	A	1.5 (3.3)			kg (lbs)
	B	2.0 (4.4)			
	C	5.0 (11)	5.0 (11)	5.0 (11)	
	D	8.5 (18.7)			

Table D-2. Technical Data for Different Types of Load Cell PFRL 101

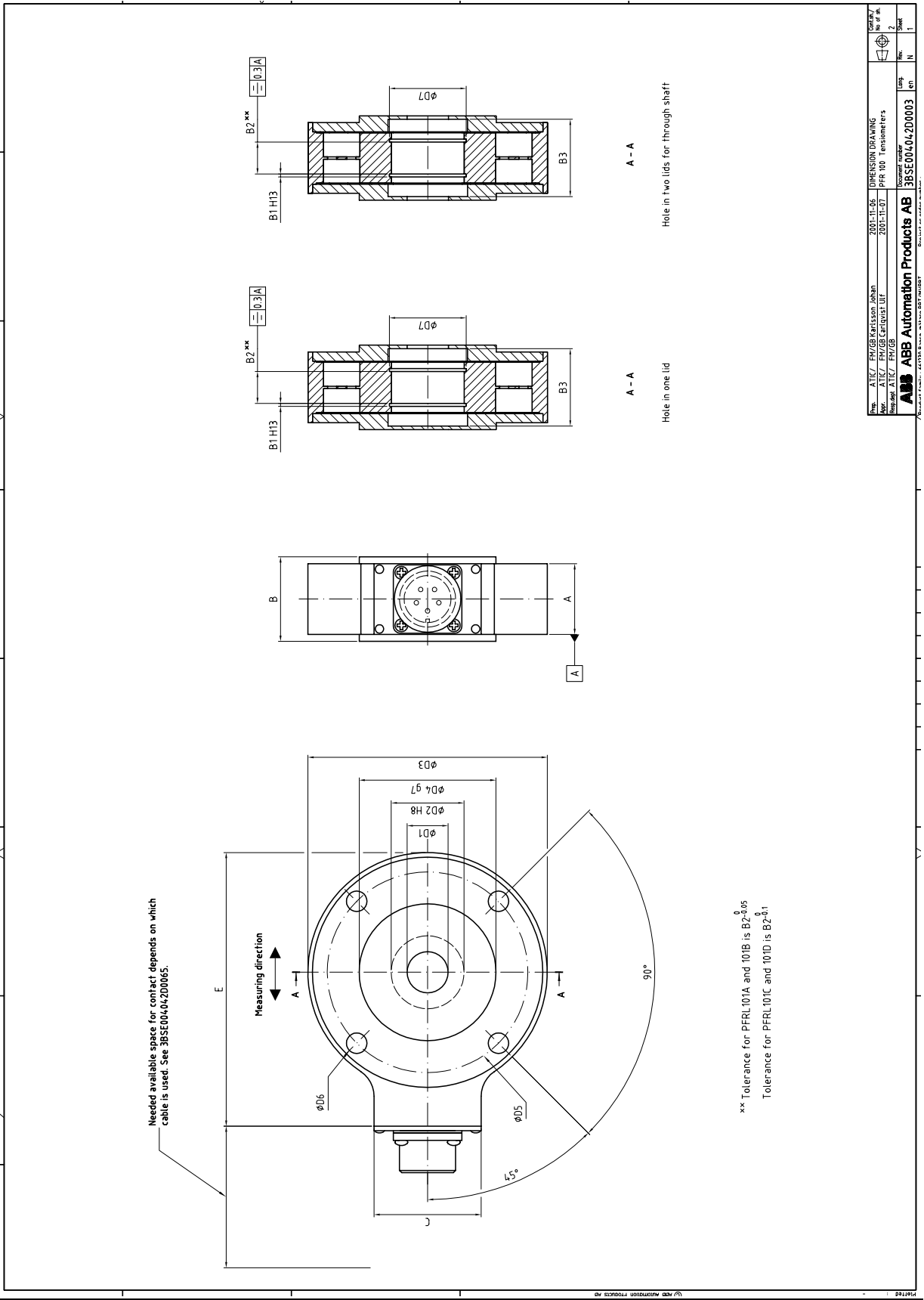
PFRL 101	Type	Data	Unit
Material	A B C D	SS 2387 stainless steel, DIN X4CrNiMo 16 5. Corrosion resistance properties similar to AISI 304.	
Accuracy			%
Accuracy class		± 0.5	
Repeatability error		$< \pm 0.1$	
Compensated temperature range		+20 - +80 (68 - 176)	°C (°F)
Zero point drift		150 (83)	ppm/K
Sensitivity drift		150 (83)	(ppm/F)
Working temperature range		-10 - +80 (14 - 176)	°C (°F)
Zero point drift		300 (167)	ppm/K
Sensitivity drift		300 (167)	(ppm/F)
Storage temperature range		-40 - +80 (-40 - 176)	°C (°F)



D.11 Cable Diagram, 3BSE028140D0065, Page 4/5, Rev. B



D.12 Dimension Drawing 3BSE004042D0003, page 1/2, Rev. N



D.13 Dimension Drawing, 3BSE004042D0003, page 2/2, Rev. N

REV DESCRIPTION										DEPT./INT.	DATE	REV DESCRIPTION	DEPT./INT.	DATE	REV DESCRIPTION	DEPT./INT.	DATE
- New document										SEISY/AGK/JK	94-05-20	J New version 3BSE002958R0004.R0014	SEISY/AGK/JK	98-01-22	J	SEISY/AGK/JK	98-01-22
A Dim. D7, B1 and B2 added										SEISY/AGK/JK	95-06-20	K Bearing tolerances changed H8 was G8	SEISY/AGK/JK	99-02-10	K	SEISY/AGK/JK	99-02-10
B Dim. A, B, for LC 2.0 and 5.0 adjusted										SEISY/AGK/JK	95-08-18	L Hole for Light Roll Pin removed	SEISY/AGK/JK	99-11-29	L	SEISY/AGK/JK	99-11-29
C New versions included in table. Format was A3										SEISY/AGK/JK	95-09-28	M Tolerance for A-dimension changed	SEISY/AGK/JK	99-11-29	M	SEISY/AGK/JK	99-11-29
D New versions incl. 3BSE002963R002 and 3BSE002968R002										SEISY/AGK/JK	96-03-06	N A-dimension for PFRL 10A and 10B changed	SEISY/AGK/JK	00-02-23	N	SEISY/AGK/JK	00-02-23
E # DIT for 3BSE002968R002 was G3										SEISY/AGK/JK	96-03-06	O A-dimension for PFRL 10A and 10B changed Tol. for øDL changed	SEISY/AGK/JK	00-02-23	O	SEISY/AGK/JK	00-02-23
F Bearing recommendation incl.										SEISY/AGK/JK	96-03-06	P Tolerance for B2-dimension changed	SEISY/AGK/JK	00-02-23	P	SEISY/AGK/JK	00-02-23
G Sheet 2 added E3 added.										SEISY/AGK/JK	96-03-06	Q Tolerance for B2-dimension changed	SEISY/AGK/JK	01-11-06	Q	SEISY/AGK/JK	01-11-06
H Versions note in two lids for through shaft add.										SEISY/AGK/JK	96-03-06	R 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	R	SEISY/AGK/JK	01-11-06
I Versions note in two lids for through shaft add. from list										SEISY/AGK/JK	97-02-24	S 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	S	SEISY/AGK/JK	01-11-06
J 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	T 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	T	SEISY/AGK/JK	01-11-06
K 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	U 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	U	SEISY/AGK/JK	01-11-06
L 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	V 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	V	SEISY/AGK/JK	01-11-06
M 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	W 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	W	SEISY/AGK/JK	01-11-06
N 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	X 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	X	SEISY/AGK/JK	01-11-06
O 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	Y 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	Y	SEISY/AGK/JK	01-11-06
P 3BSE002958R0006 and 3BSE002958R0006 removed from list										SEISY/AGK/JK	97-02-24	Z 3BSE002958R0006 and 3BSE002958R0006 included	SEISY/AGK/JK	01-11-06	Z	SEISY/AGK/JK	01-11-06
H New version PFRL10D-5,0kN ø125, PFRL10D-5,0kN - ø110 add.										SEISY/AGK/JK	97-02-24		SEISY/AGK/JK	97-02-24		SEISY/AGK/JK	97-02-24

Dimensions																			
Article number	x	Type designation	From kN	øD1	øD2	øD3	øD4	A	B	C	E	øD5	øD6	øD7	B1	B2	B3	Bearing recommendations	d = shaft diameter
3BSE002958R0001		PFRL 101A-0,5kN ø32 B2=14	0,5	18	32	105	60	32	37	47	120	88	9	33.7H12	1.3	14	34		
3BSE002958R0011		PFRL 101A-0,5kN ø35 B2=14	0,5	23	35	105	60	32	37	47	120	88	9	37 H12	1.6	14	34		
3BSE002958R0002		PFRL 101A-0,5kN ø40 B2=16	0,5	23	40	105	60	32	37	47	120	88	9	42.5H12	1.85	16	34		
3BSE002958R0012		PFRL 101A-0,5kN ø40 B2=16	0,5	23	35	105	60	32	37	47	120	88	9	37 H12	1.6	11	34		
3BSE002958R0003		PFRL 101A-0,5kN ø40 B2=11	0,5	23	40	105	60	32	37	47	120	88	9	42.5H12	1.85	16	34		
3BSE002958R0004		PFRL 101A-0,5kN ø40 B2=11	0,5	23	40	105	60	32	37	47	120	88	9	42.5H12	1.85	16	34		
3BSE002958R0014		PFRL 101B-1,0kN ø40 Application	0,5	23	40	125	60	32	37	47	140	106	9	42.5H12	1.85	16	34		
3BSE002958R0001		PFRL 101B-1,0kN ø40	1,0	23	40	125	60	32	37	47	140	106	9	42.5H12	1.85	16	34		
3BSE002958R0011		PFRL 101B-1,0kN ø47	1,0	30	47	125	60	32	37	47	140	106	9	49.5H12	1.85	18	34		
3BSE002958R0014		PFRL 101B-1,0kN ø52	1,0	33	52	125	60	32	37	47	140	106	9	55H12	2.15	18	34		
3BSE002958R0005		PFRL 101B-1,0kN ø40 Application	1,0	23	40	125	60	32	37	47	140	106	9	42.5H12	1.85	16	34		
3BSE002958R0015		PFRL 101B-1,0kN ø40 Application	1,0	23	40	125	60	32	37	47	140	106	9	42.5H12	1.85	16	34		
3BSE002963R0502		PFRL 101C-0,5kN	0,5	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002963R0512		PFRL 101C-0,5kN	0,5	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002963R1002		PFRL 101C-1,0kN	1,0	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002963R1012		PFRL 101C-1,0kN	1,0	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002963R0002		PFRL 101C-2,0kN	2,0	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002963R0012		PFRL 101C-2,0kN	2,0	56	80	175	100	44	50	47	190	152	11	83.5H12	2.65	23	46		
3BSE002968R0002		PFRL 101D-5,0kN ø110	5,0	77	110	225	130	50	56	47	240	200	14	114H13	4.15	28	52		
3BSE002968R0012		PFRL 101D-5,0kN ø125	5,0	77	125	225	130	50	56	47	240	200	14	129H13	4.15	31	52		
3BSE002968R0003		PFRL 101D-5,0kN ø125	5,0	77	125	225	130	50	56	47	240	200	14	129H13	4.15	31	52		

× Hole in one lid 3BSE00____R____0____

Hole in two lids for through shaft 3BSE00____R____1____

Units: mm

ABB Automation Products AB

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

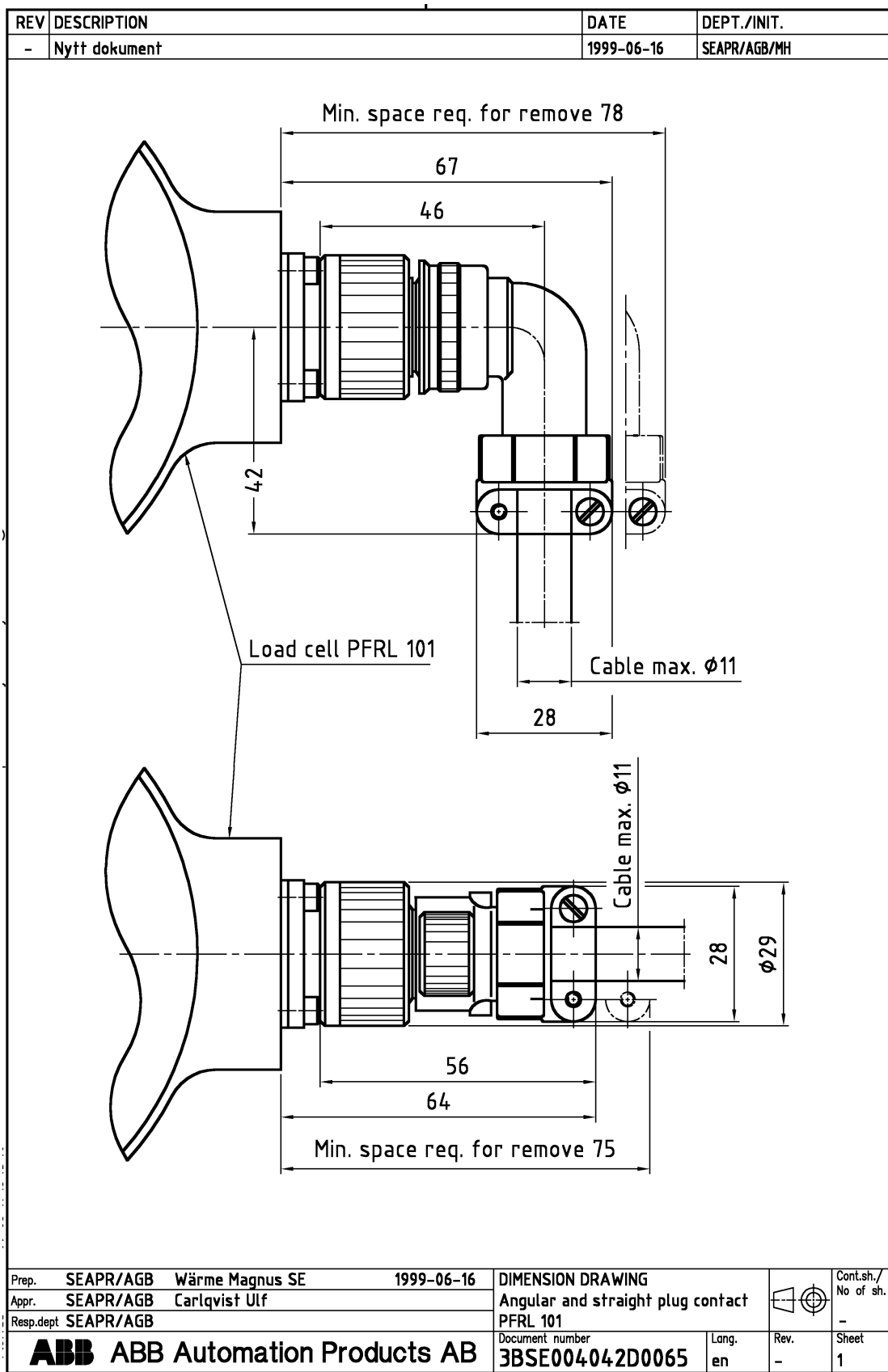
Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan

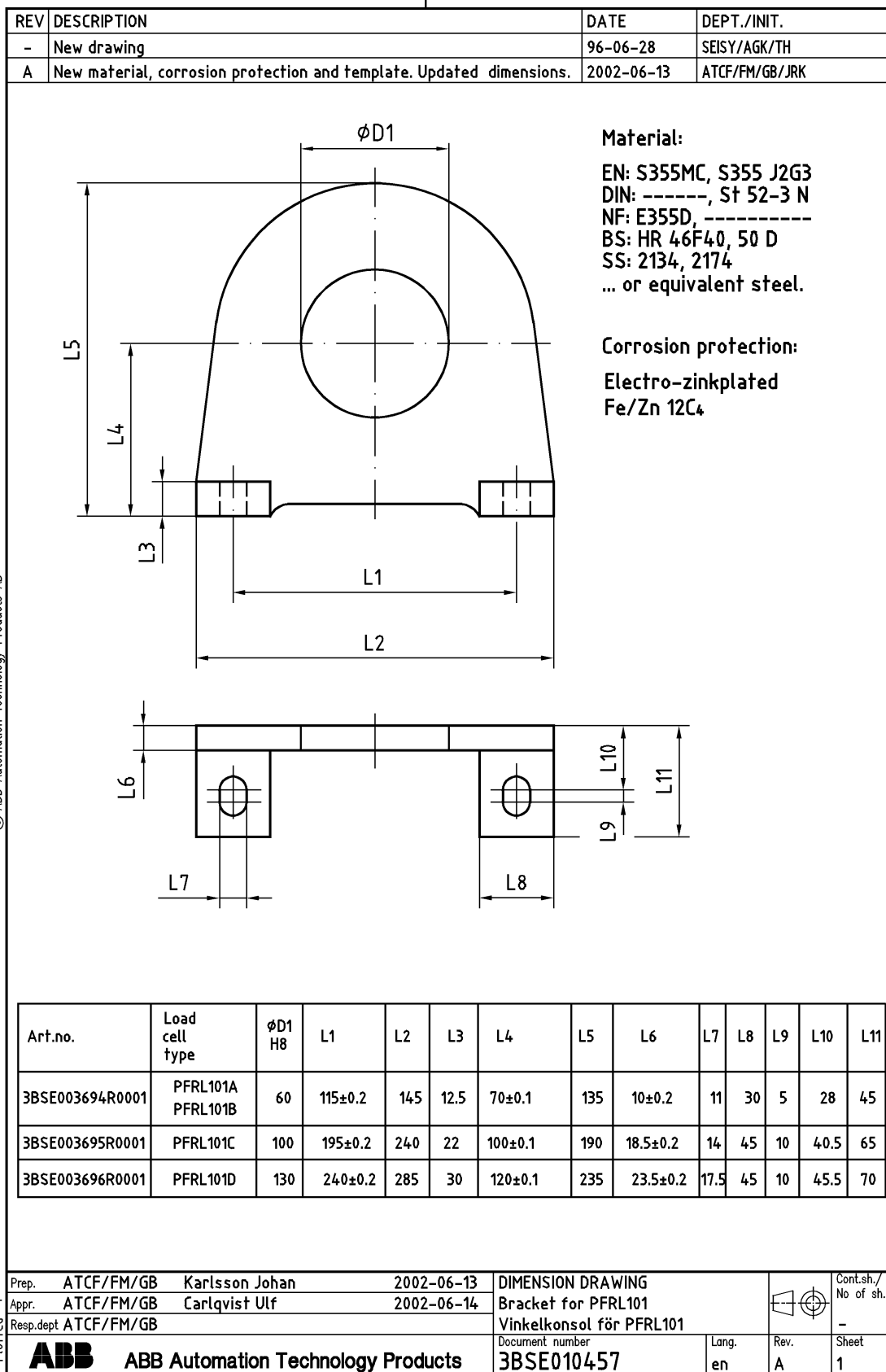
Rev: A1/E1 / 02/08 K. Eriksson, Johan

Rev: A1/E1 / 02/08 K. Eriksson, Johan</

D.14 Dimension Drawing, 3BSE004042D0065, Rev. -



D.15 Dimension Drawing, 3BSE010457, Rev. A



Appendix E PFTL 101 - Designing the Load Cell Installation

E.1 About This Appendix

This appendix describes the procedure for designing the load cell installation.

The following sections are included:

- Basic application considerations
- Designing the load cell installation (step-by-step guide)
- Installation requirements
- Force and wrap gain calculation
 - Horizontal mounting
 - Inclined mounting
 - Single side measurement
- Mounting the load cells
- Technical data
- Drawings
 - Cable diagram(s)
 - Dimension drawing(s)
 - Assembly drawing(s)

E.2 Basic Application Considerations

Each application has its own individual demands that have to be considered; though a few basic considerations tend to repeat themselves.

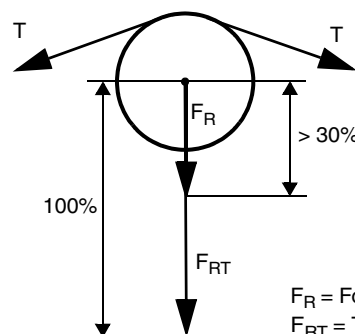
- What type of process is involved (papermaking, converting, etc.)?
Is the environment demanding (temperature, chemicals, etc.)?
- What is the tension measurement purpose; indication or closed loop control?
Are there any specific accuracy demands involved?
- What is the machine design like? Is there the possibility to modify the design,
in order to fit the most suitable load cell, or is the machine design fixed?
- What are the forces acting on the roll like (size and direction)?
Can they be altered by redesign?

If these questions are dealt with thoroughly, the installation is very likely to be successful. However, the extent to which measurement accuracy is needed, defines the requirements when designing a load cell installation.

E.3 Step-by-Step Guide for Designing the Load Cell Installation

The procedure below defines the main considerations involved in designing a load cell installation.

1. Check load cell data so that environmental demands are met.
2. Calculate forces; vertical, horizontal and axial (cross directional).
3. Size and orient load cell so that the guidelines below are met:
 - a. Try to achieve as large portion as possible no less than 10% of web tension, in the load cell measurement direction!
 - b. Select load cell size so that it is loaded as close as possible to its nominal load! Do not dimension Force component of Tension in measuring direction, F_R , for less than 10% of the load cell nominal load!
 - c. If the span between maximum and minimum tension in the process is large, select load cell so that the maximum tension will be in the load cell extended range (when applicable)!
 - d. The measured force component of web tension is recommended to be at least 30% of tare force component (roll weight) acting in load cell measurement direction. The reason for this recommendation is load cell signal stability, especially when the system operates in a large temperature span.
This means that if $F_{RT} < 1/3$ of F_{nom} , F_R should be at least 10% of F_{nom} .
For larger F_{RT} , lowest F_R is recommended to be at least 30% of F_{RT} .



Rule 1: If $F_{RT} < 1/3$ of F_{nom}
 F_R should be at least 10% of F_{nom}

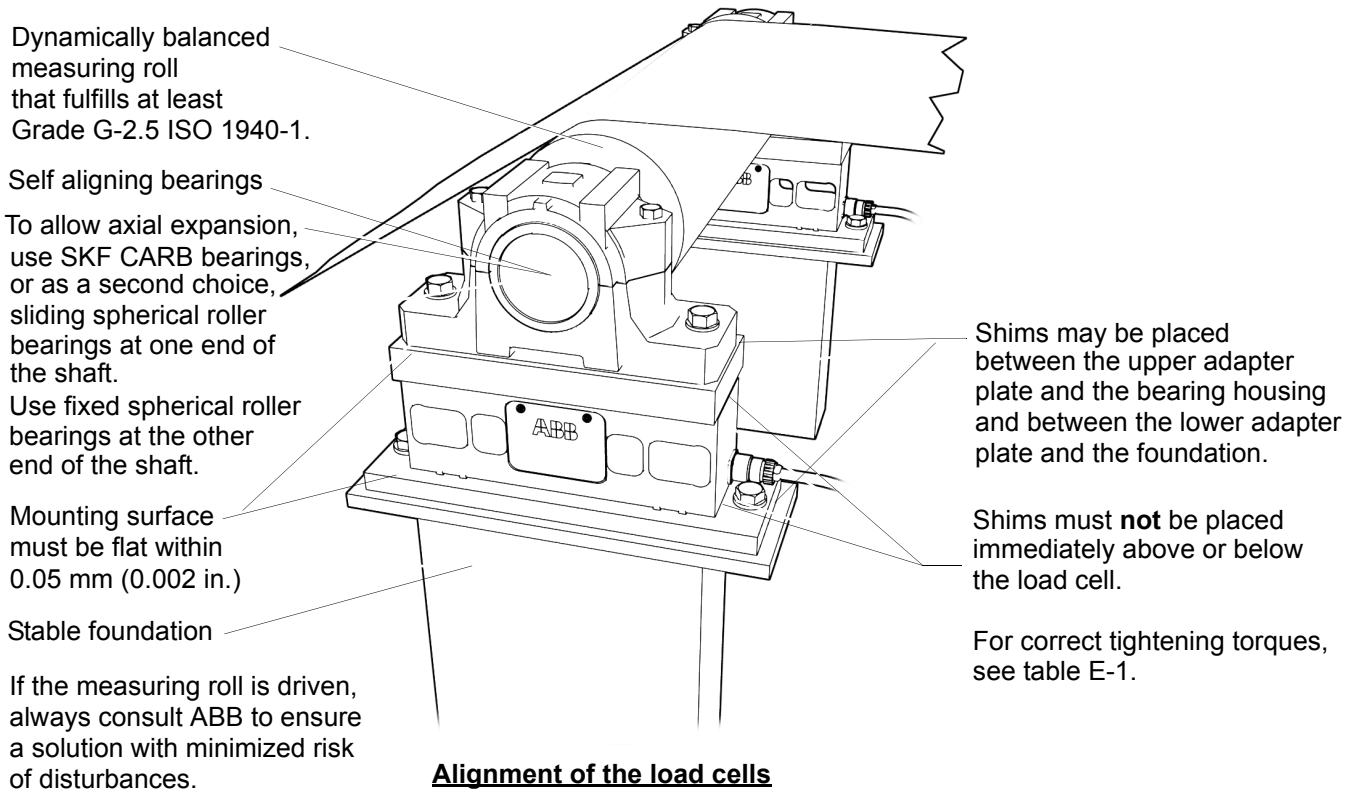
Rule 2: If $F_{RT} > 1/3$ of F_{nom}
 F_R is recommended to be at least 30% of F_{RT}

F_R = Force component of web tension in measuring direction
 F_{RT} = Tare force in measurement direction

- e. Check load cell data so that limits for building height, transverse and axial forces are not exceeded.
4. Design base frame and/or adapter plates.

E.4 Installation Requirements

To achieve the specified accuracy, the best possible reliability and long-term stability, install the load cells in accordance with the requirements below.



Alignment of the load cells

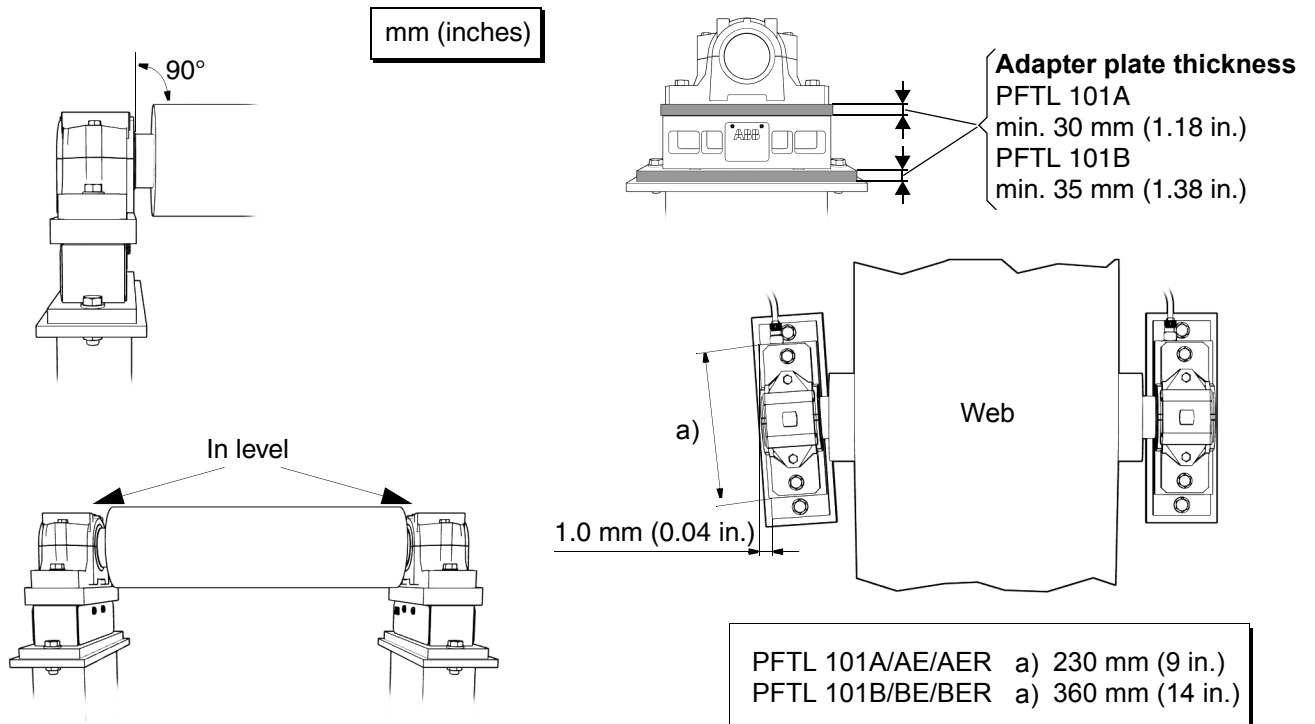
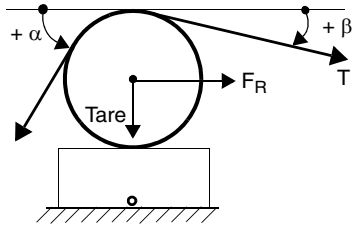


Figure E-1. Installation Requirements

E.5 Mounting Alternatives, Calculating Force and Calculating Wrap Gain

E.5.1 Horizontal Mounting



In most cases, horizontal mounting is the most obvious and simplest solution. The load cell should thus be mounted horizontally when possible.

$$F_R = T \times (\cos \beta - \cos \alpha)$$

$$F_{RT} = 0 \text{ (Tare force is not measured)}$$

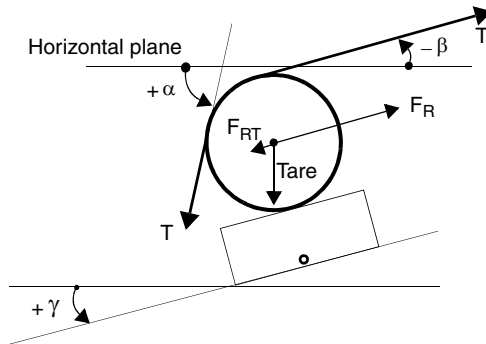
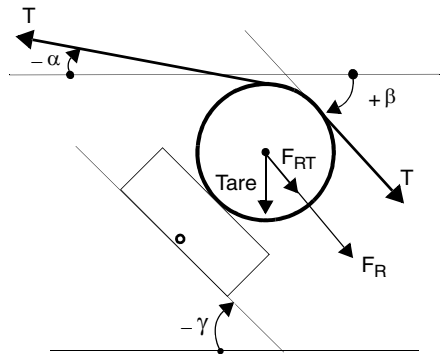
$$F_{Rtot} = F_R + F_{RT} = T \times (\cos \beta - \cos \alpha)$$

$$T \text{ (Tension)} = \text{Wrap gain} \times F_R$$

$$\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T(\cos \beta - \cos \alpha)}$$

$$\text{Wrap gain} = \frac{1}{\cos \beta - \cos \alpha}$$

E.5.2 Inclined Mounting



Sometimes it is necessary to mount the load cell on an incline due to mechanical design constraints of the machine or the need to have a sufficient force component applied to the load cell.

Inclined mounting adds a component of tare force in the measuring direction and modifies the force components as shown.

NOTE

When calculating, it is important that the angles are set into the equations with the correct signs in relation to the horizontal plane.

$$F_R = T \times [\cos(\beta + \gamma) - \cos(\alpha - \gamma)]$$

$$F_{RT} = -Tare \times \sin \gamma$$

$$F_{Rtot} = F_R + F_{RT} =$$

$$T \times [\cos(\beta + \gamma) - \cos(\alpha - \gamma)] + (-Tare \times \sin \gamma)$$

$$T \text{ (Tension)} = \text{Wrap gain} \times F_R$$

$$\text{Wrap gain} = \frac{T}{F_R} = \frac{T}{T[\cos(\beta + \gamma) - \cos(\alpha - \gamma)]}$$

$$\text{Wrap gain} = \frac{1}{\cos(\beta + \gamma) - \cos(\alpha - \gamma)}$$

E.6 Force Calculation for Measurement with a Single Load Cell

In some cases, it is sufficient to measure the tension with only a single load cell mounted at one end of the roll. The roll should nevertheless be supported at both ends.

E.6.1 The Most Common and Simple Solution

The most obvious and simple solution is horizontal mounting with the web evenly distributed and centered on the roll.

As long as the roll is supported at both ends, the same calculations given in [Section E.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#) are valid.

NOTE

The accuracy of a single load cell measurement is highly dependent on how well the center of force can be determined. Since the cross-directional stress distribution generally is somewhat uneven, this is not easily done. The load cell will, however, produce a stable and repeatable measurement.

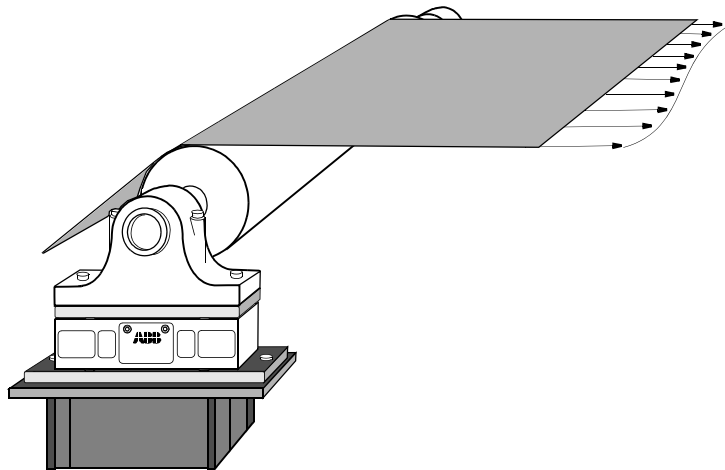
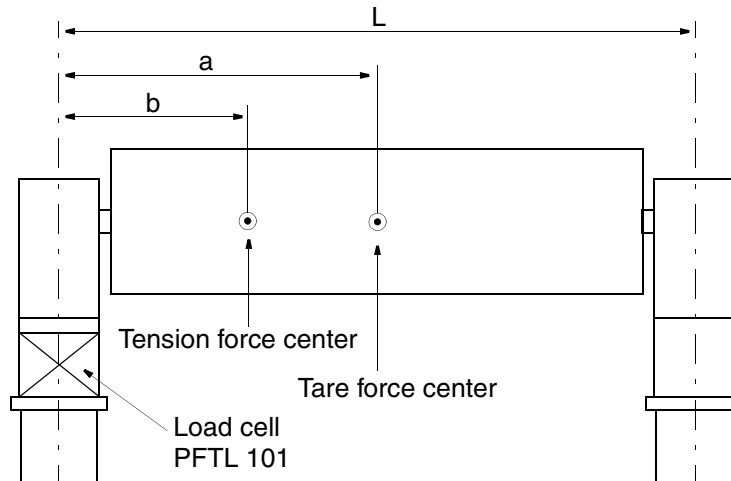


Figure E-2. Cross-directional stress distribution

E.6.2 Force Calculation when the Web is not Centered on the Roll

Use the calculations below for horizontal and inclined mounting when the web is not centered on the roll.

The applied force at the load cell will be proportional to the distance between the tension force center and the load cell centerline, see figure.



Calculation procedure:

1. Horizontal or inclined mounting?
2. Calculate F_R and F_{RT} , see [Section E.5, Mounting Alternatives, Calculating Force and Calculating Wrap Gain](#).
3. Use the following equations:

$$F_R \text{ for single load cell} = F_R \times \frac{L-b}{L}$$

$$F_{RT} \text{ for single load cell} = F_{RT} \times \frac{L-a}{L}$$

$$F_{Rtot} \text{ for single load cell} = F_R \text{ for single load cell} + F_{RT} \text{ for single load cell}$$

where:

- L = Distance between load cell centerline and the opposite bearing centerline
a = Distance between tare force center and load cell centerline
b = Distance between tension force center and load cell centerline

E.7 Mounting the Load Cells

The instructions below apply to a typical mounting arrangement. Variations are allowed, provided that the requirements of [Section E.4, Installation Requirements](#) are complied with.

If it is necessary to use tubular dowel pins to secure the position of the load cell, see instructions in [Figure E-3](#).

1. Clean the foundation and other mounting surfaces.
2. Fit the lower adapter plate to the load cell.
Tighten the screws to the torque stated in [Table E-1](#).
3. Fit the load cell and the lower adapter plate to the foundation,
but do not fully tighten the screws.
4. Fit the upper adapter plate to the load cell.
Tighten the screws to the torque stated in [Table E-1](#).
5. Fit the bearing housing and the roll to the upper adapter plate,
but do not fully tighten the screws.

CAUTION

During this operation it is possible to over load the load cells if the operation is done not careful enough, especially if the roll is heavy. The most critical load cells are naturally the PFTL 101A-0.5 kN and PFTL 101B-2 kN. Applications with inclined mounting are most critical.

6. Adjust the load cells so that they are in parallel with each other and in line with the axial direction of the roll. Tighten the foundation screws, see [Table E-1](#).
7. Adjust the roll so that it is at right angles to the longitudinal direction of the load cells. Tighten the screws in the upper adapter plate, see [Table E-1](#).

Table E-1. Tightening Torques for Load Cell PFTL 101

Alternative	Type of screws	Strength class	Type of lubrication	Dimension	Tightening torque [Nm] $\pm$ 5%
1 (Recommended)	Alloyed steel screws Strength class according to ISO 898/1	12.9	Oil	M12	136 Nm
				M16	333 Nm
				M20	649 Nm
2 (Recommended)	Alloyed steel screws Strength class according to ISO 898/1	12.9	MoS ₂	M12	117 Nm
				M16	286 Nm
				M20	558 Nm
3	Stainless steel (A2-80) or acid resistant steel (A4-80), Strength class according to ISO 3506	A2-80	Wax	M12	76 Nm
		or		M16	187 Nm
		A4-80		M20	364 Nm
4	Stainless steel (A2-80) or acid resistant steel (A4-80), Strength class according to ISO 3506	A2-80	Oil	M12	65 Nm
		or	or	M16	161 Nm
		A4-80	emulsion	M20	313 Nm

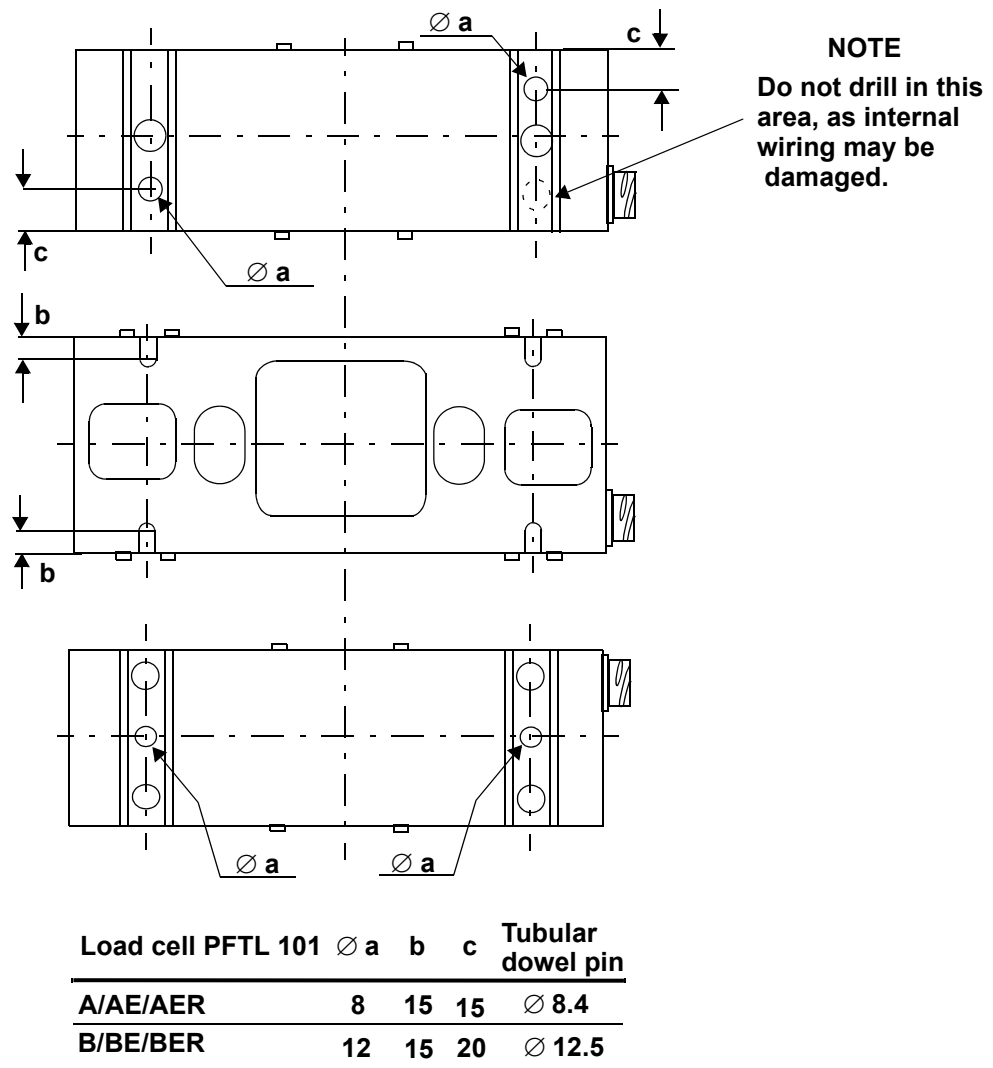


Figure E-3. Drilling Dowel Pin Holes

E.7.1 Routing the Load Cell Cable

The cable must be supported with clamps and routed to prevent force shunting through the cable.

E.8 Technical Data

Table E-2. Technical Data for Different Types of Load Cell PFTL 101

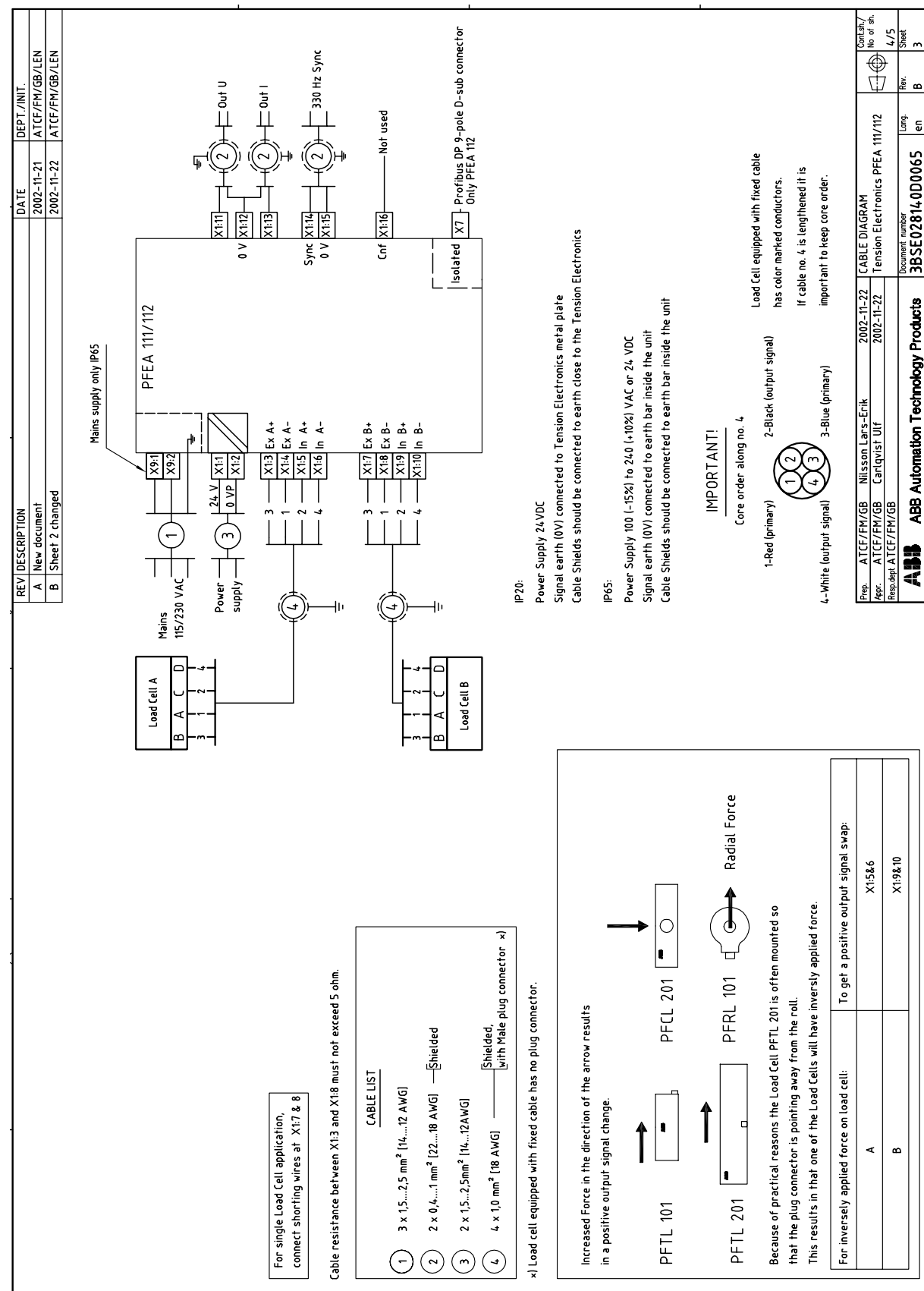
PFTL 101	Type	Data				Unit	
Nominal load							
Nominal load in measuring direction, F _{nom}	A/AE/AER	0.5 (112)	1.0 (225)	2.0 (450)			
	B/BE/BER	2.0 (450)				5.0 (1120)	10.0 (2250)
Permitted transverse load within the accuracy, F _{Vnom}	A/AE/AER	5 (1120)	10 (2250)	10 (2250)			
	B/BE/BER	30 (6740)				30 (6740)	30 (6740)
Permitted axial load within the accuracy, F _{Anom}	A/AE/AER	2 (450)	5 (1120)	5 (1120)			
	B/BE/BER	5 (1120)				10 (2250)	10 (2250)
Overload capacity							
Max. load in measuring direction without permanent change of data, F _{max}	A/AE/AER	2.5 (562)	5 (1120)	10 (2250)			
	B/BE/BER	10 (2250)				25 (5620)	50 (11200)
Spring constant	A/AE/AER	32 (183)	65 (372)	130 (744)			
	B/BE/BER	130 (744)				325 (1860)	650 (3718)
Mechanical data							
Length	A/AE/AER	230 (9)	230 (9)	230 (9)			
	B/BE/BER	360 (14)				360 (14)	360 (14)
Width	A/AE/AER	84 (3.3)	84 (3.3)	84 (3.3)			
	B/BE/BER	104 (4)				104 (4)	104 (4)
Height	A/AE/AER	125 (5)	125 (5)	125 (5)			
	B/BE/BER	125 (5)				125 (5)	125 (5)

Table E-2. Technical Data for Different Types of Load Cell PFTL 101

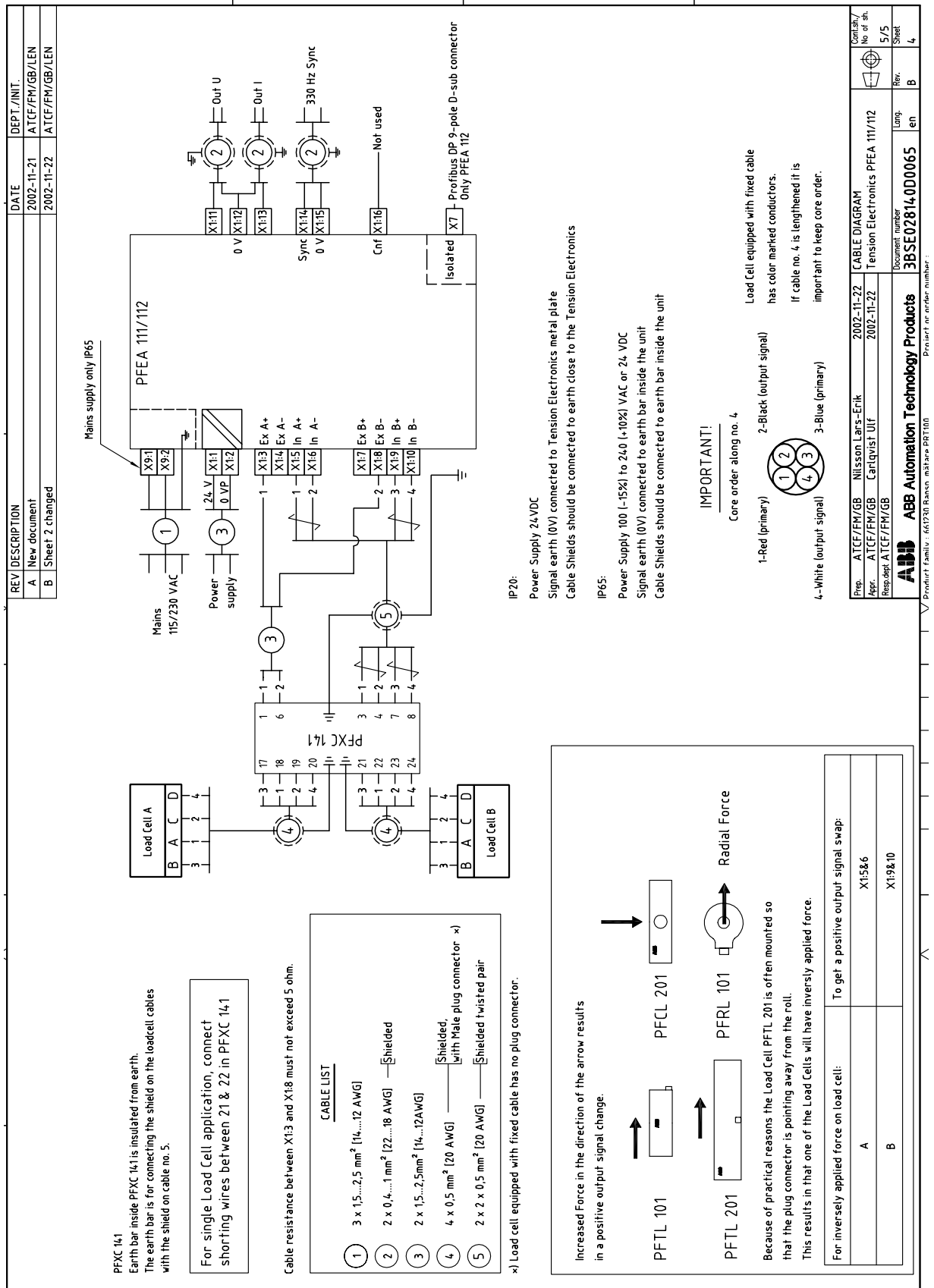
PFTL 101	Type	Data				Unit
Weight	A/AE/AER	9 (20)	9 (20)	10 (22)		kg (lbs)
	B/BE/BER			20 (44)	21 (46) 21 (46) 23 (51)	
Material	A/AE/B/BE	Stainless steel: SS 2383 DIN 17440 X12CrMoS17 Werkstoffnr 1.4104 AISI 430F				
	AER/BER	Acid resistant steel: SS 2348 DIN 17440 X2CrNiMo17 13 2 Werkstoffnr 1.4404 AISI 316L				
Accuracy	A/AE/AER B/BE/BER					%
Accuracy class		± 0.5				
Linearity deviation		± 0.3				
Repeatability error		< ± 0.05				
Hysteresis		<0.2				
Compensated temperature range		+20 - +80 (68 - 176)				°C (°F)
Zero point drift		30 / 80 ⁽¹⁾ (17 / 44 ⁽¹⁾)				ppm/K (ppm/F)
Sensitivity drift		150 (83)				
Working temperature range		−10 - +105 (14 - 221)				°C (°F)
Zero point drift		50 / 100 ⁽¹⁾ / (28 / 56 ⁽¹⁾)				ppm/K (ppm/F)
Sensitivity drift		250 (139)				
Storage temperature range		−40 - +105 (−40 - +105)				°C (°F)
Degree of protection	A/B	IP 65		According. to EN 60 529		
	AE/BE	IP 66				
	AER/BER	IP 66/67				

(1) PFTL 101AER -0.5 kN/ -1.0 kN

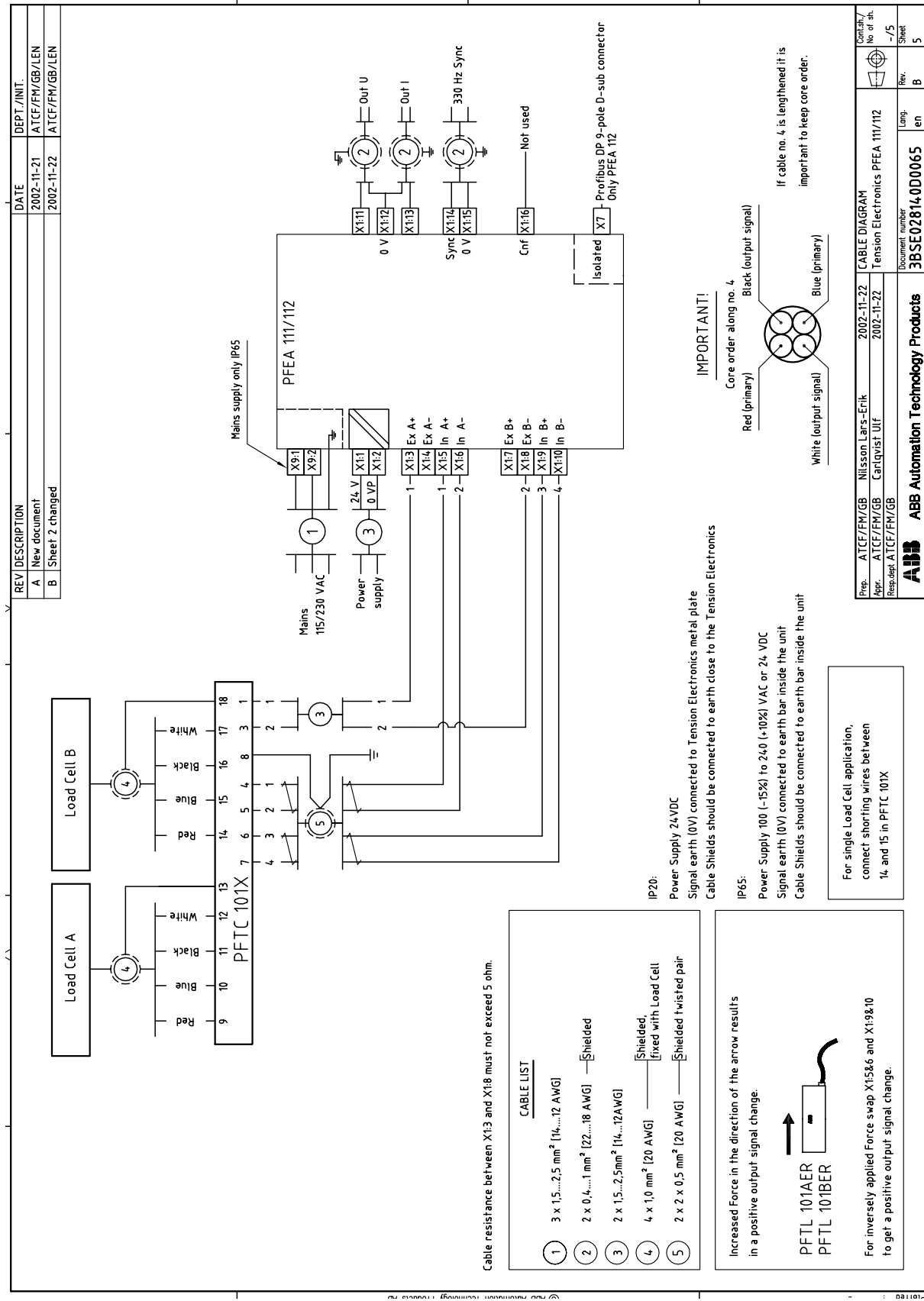
E.9 Cable Diagram 3BSE028140D0065, Page 3/5, Rev. B



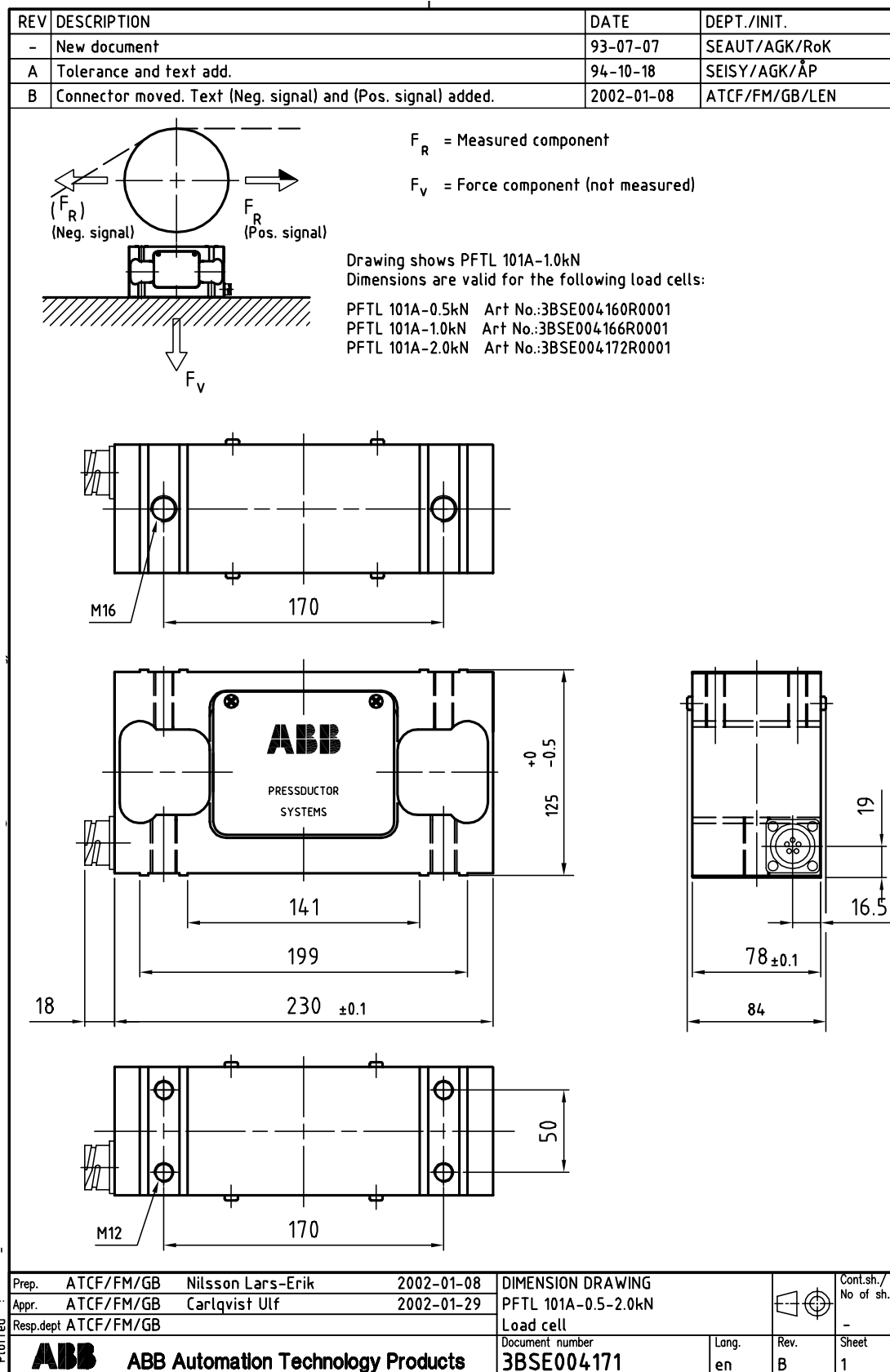
E.10 Cable Diagram 3BSE028140D0065, Page 4/5, Rev. B



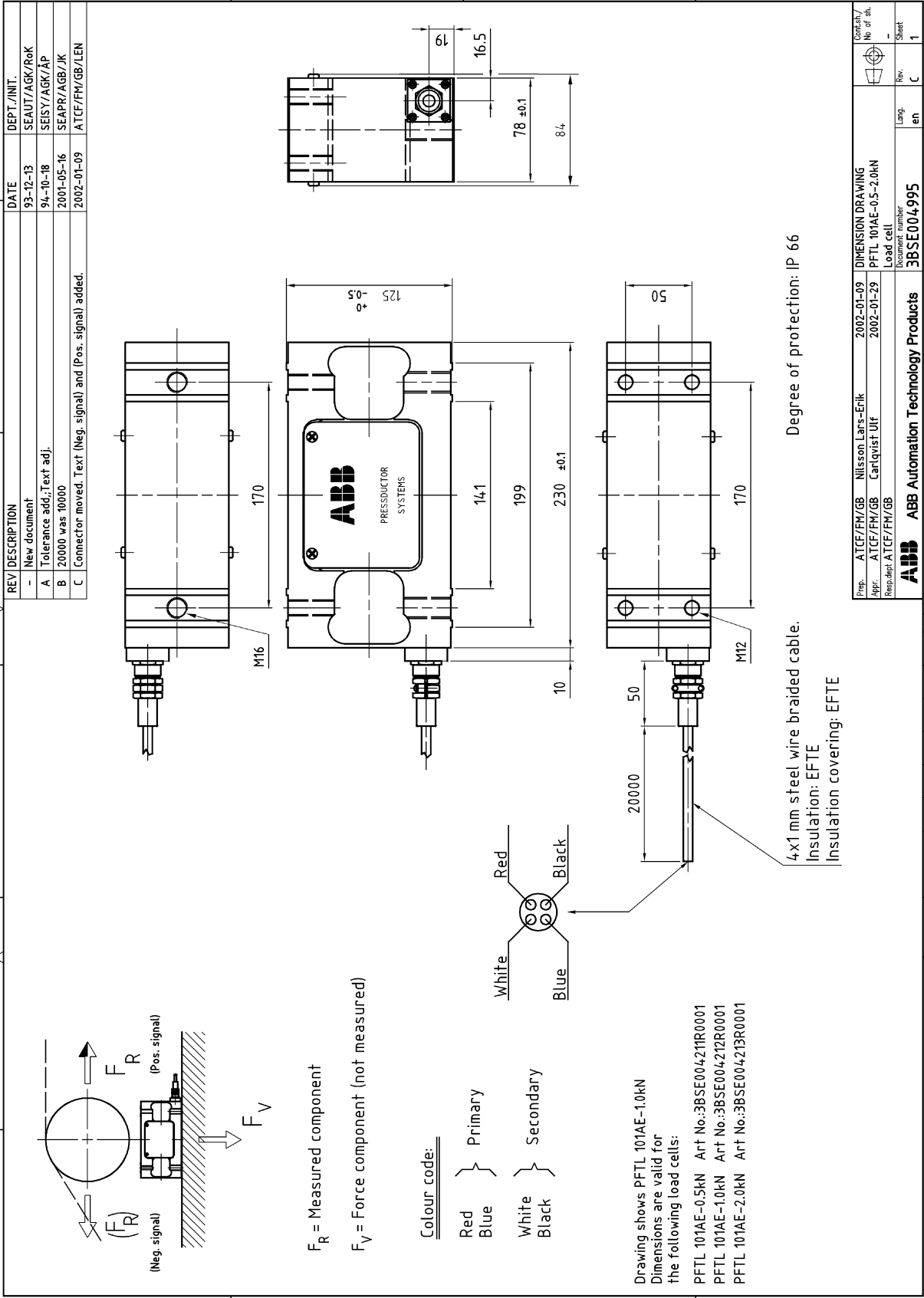
E.11 Cable Diagram 3BSE028140D0065, Page 5/5, Rev. B



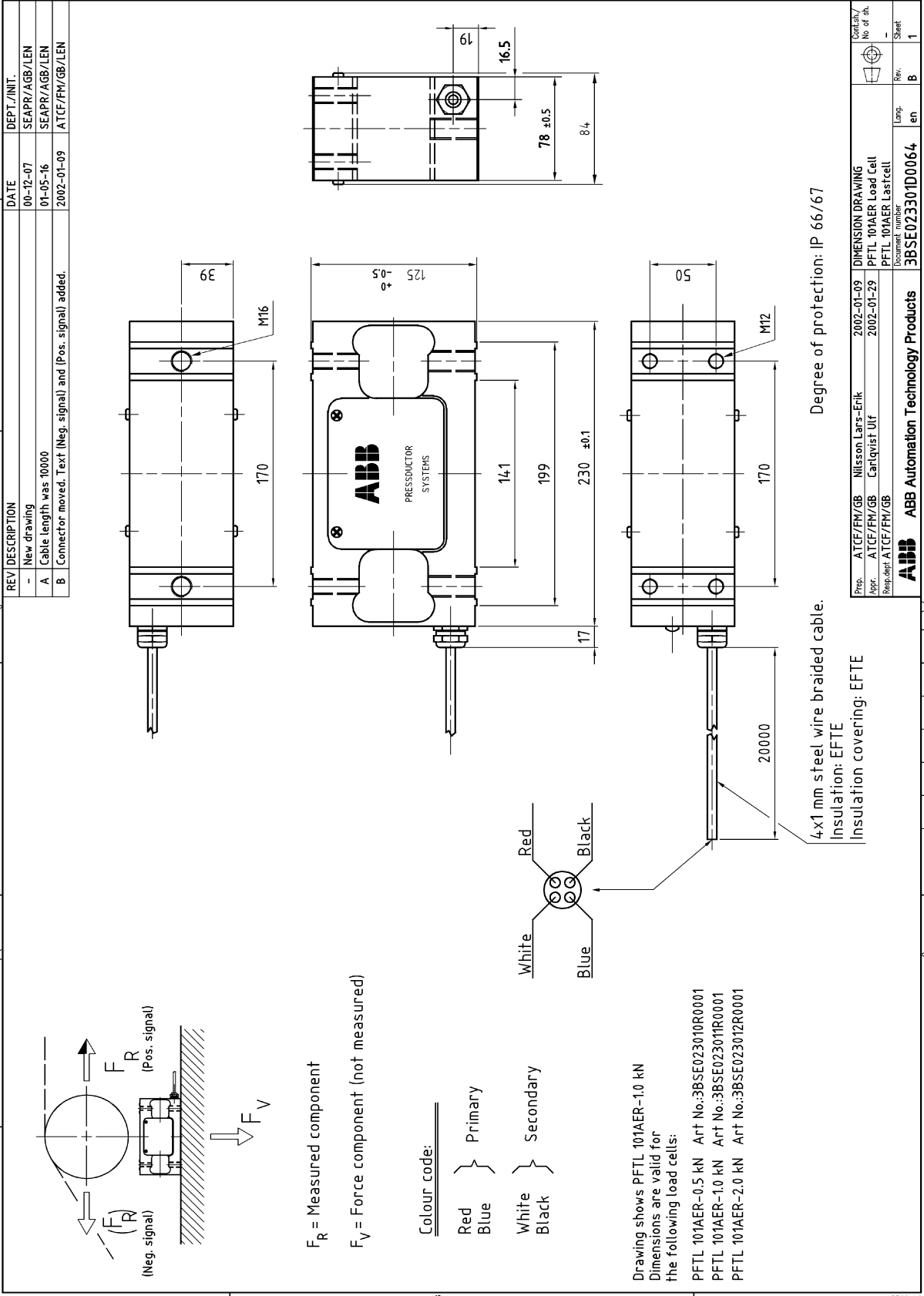
E.12 Dimension Drawing, 3BSE004171, Rev. B



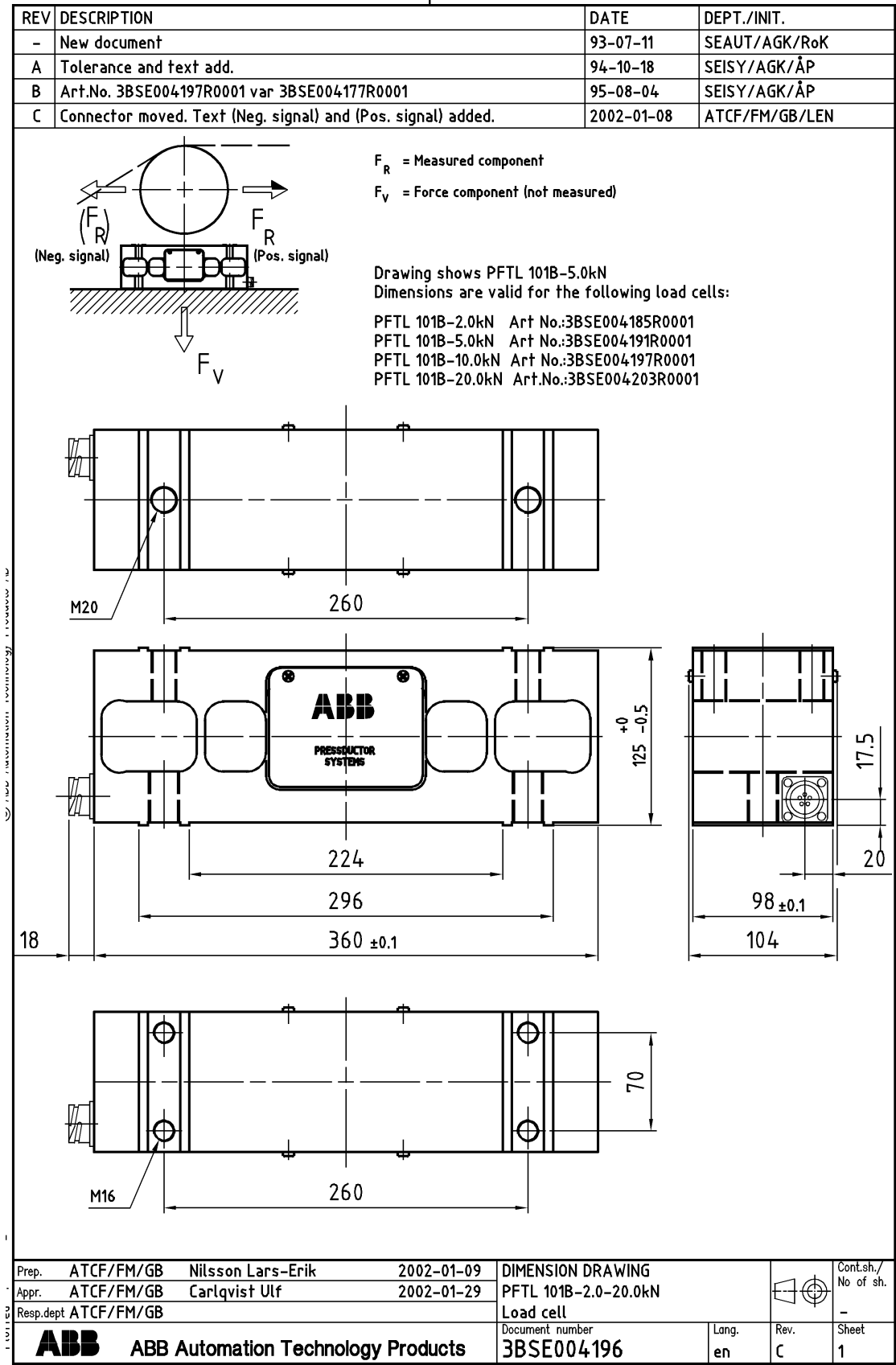
E.13 Dimension Drawing, 3BSE004995, Rev. C



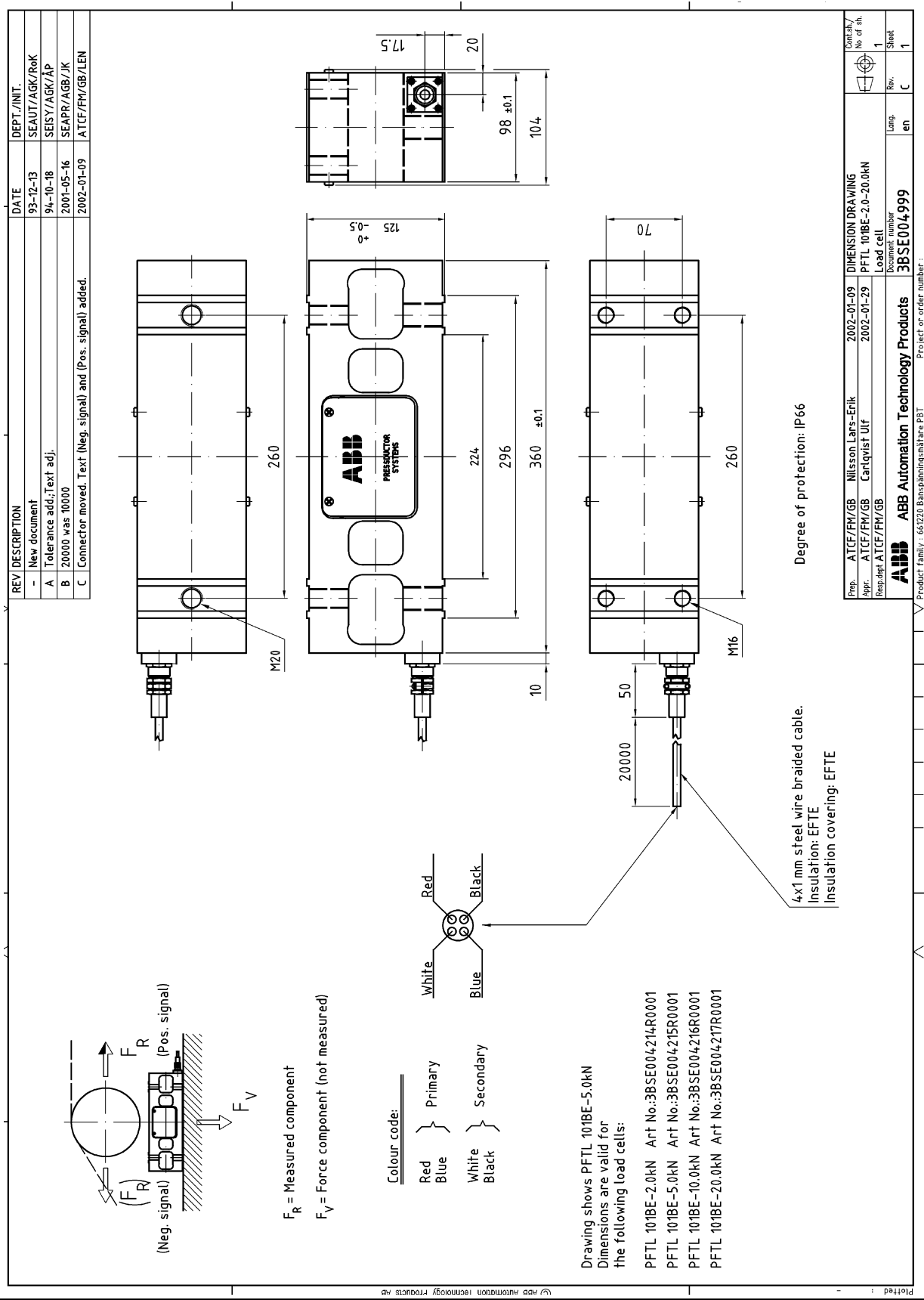
E.14 Dimension Drawing, 3BSE023301D0064, Rev. B



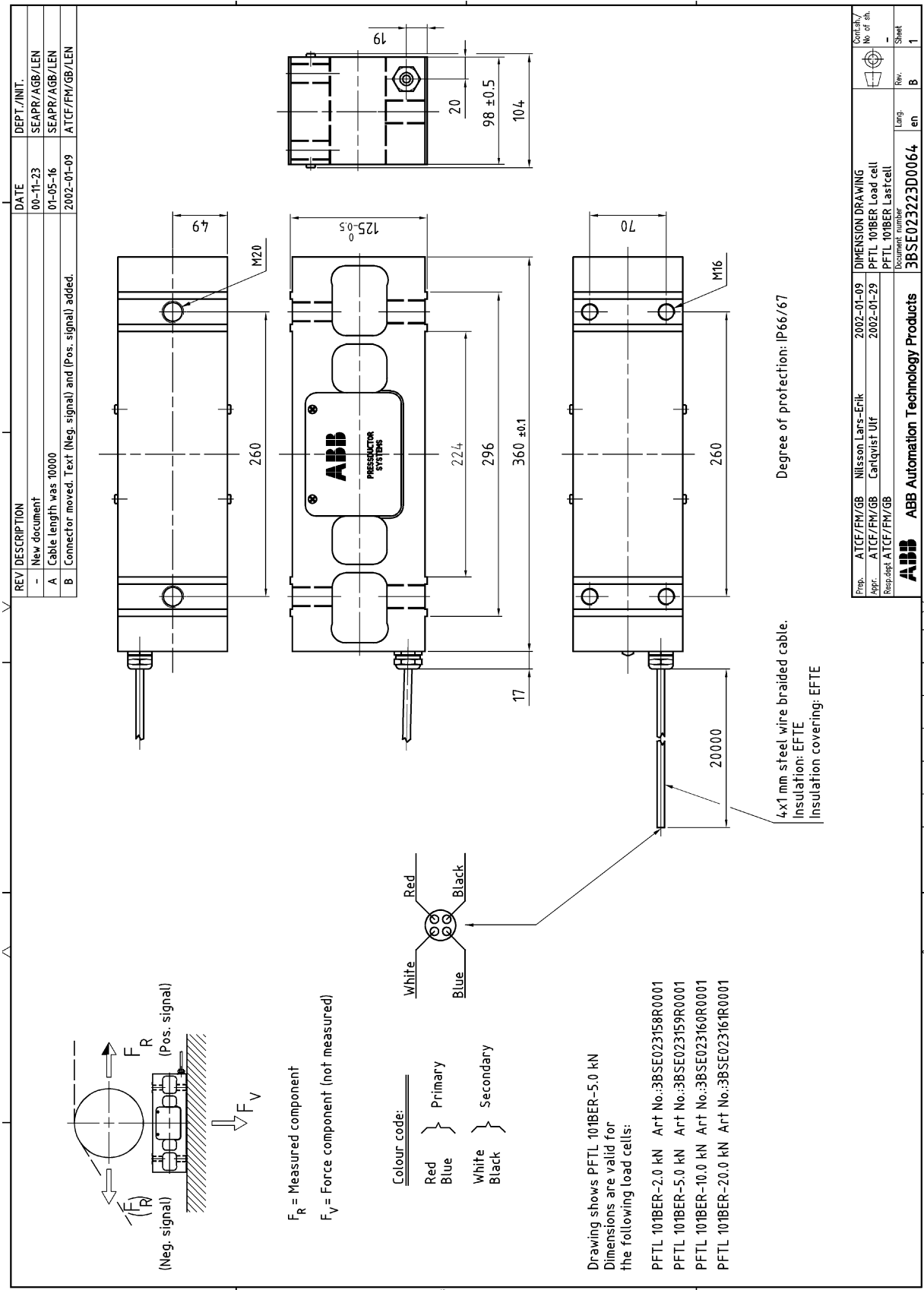
E.15 Dimension Drawing, 3BSE004196, Rev. C



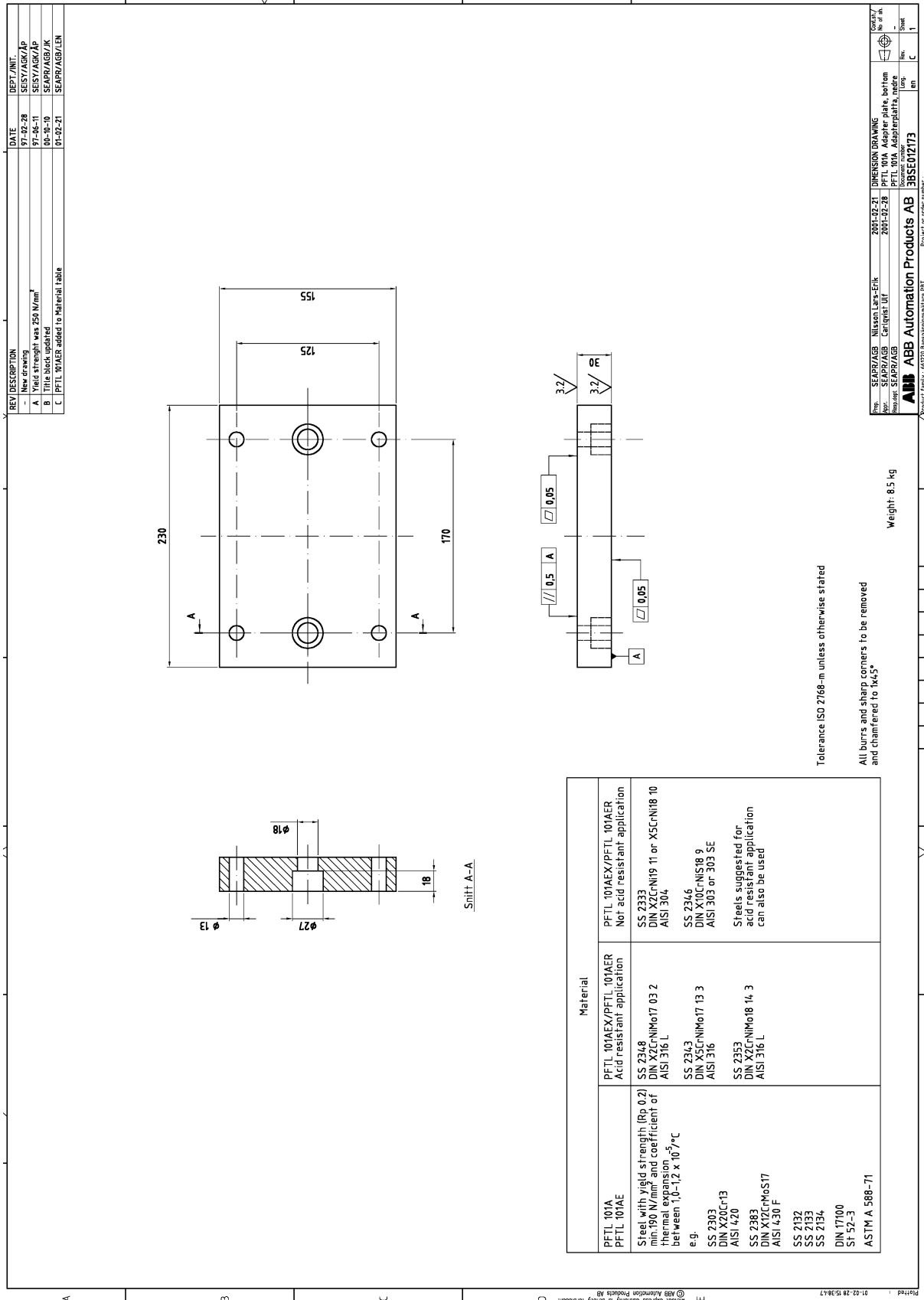
E.16 Dimension Drawing, 3BSE004999, Rev. C



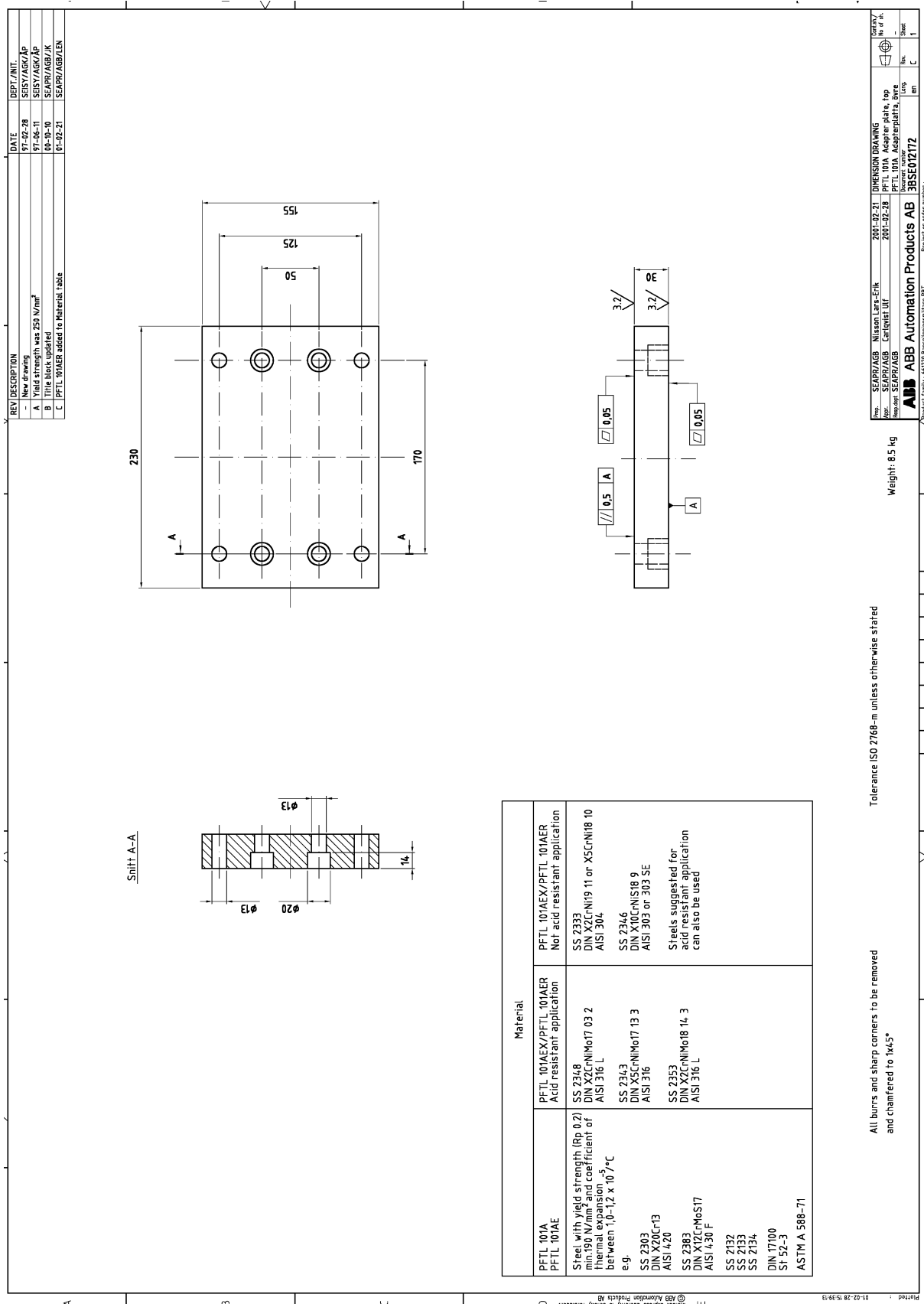
E.17 Dimension Drawing, 3BSE023223D0064, Rev. B



E.18 Dimension Drawing, 3BSE012173, Rev. C



E.19 Dimension Drawing, 3BSE012172, Rev. C



E.21 Dimension Drawing, 3BSE012170, Rev. C

[illegible]

Appendix F Actual Data and Settings at Commissioning

F.1 Document the Commissioning in this Form

Fill in actual data and settings to document the commissioning.

Data and Settings	PFEA111	PFEA112	Unit
Display language			
Display unit			N, kN, kg, lbs
Object type (Load cells per roll)	Standard roll (2 load cells)	Standard roll (2 load cells)	
	Single side (1 load cell)	Single side (1 load cell)	
Load cell nominal load			kN, lbs
Set wrap gain			
- Wrap gain*			
Voltage output			
- Step response time			ms
- High Tension			N, kN, kg, lbs
- High Output			V
- Low Tension			N, kN, kg, lbs
- Low Output			V
- High Limit			V
- Low Limit			V
Current output			
- Step response time			ms
- High Tension			N, kN, kg, lbs
- High Output			mA
- Low Tension			N, kN, kg, lbs
- Low Output			mA
- High Limit			mA
- Low Limit			mA
PROFIBUS			
- Address	-		

* If HangWeight has been used at commissioning, go to menu "EnterWrapGain", read the Wrap gain value calculated by the electronics and fill in this Wrap gain value in the table.



ABB Automation Technologies AB
Force Measurement
SE-721 59 Västerås, Sweden
Telephone: +46 (0) 21 34 20 00
Telefax: +46 (0) 21 34 00 05
Internet: www.abb.com/pressductor

3BSE029380R0001 Rev B

2004-08